

Econ 221 Human Progress

Numbered Transcript

ECON 221 • Dr. Marian L. Tupy

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Lecture 1: What is Human Progress?

At a glance — Tupy introduces what may be the first course on human progress and defines what he means by it.

- 1.1** Well, hello and welcome to what may well be the world's first course on human progress. Over the next few lectures, we will, first of all, ask: what is human progress? Second, we will ask ourselves: is there evidence for human progress? In other words, can we actually prove that things in the world are getting better? And if so, to what extent? If it is true that the world is getting better, that there has been progress in human affairs, we will ask ourselves: why are so many people pessimistic? Why is there so much negativity in the world?
- 1.2** I will talk about "progressophobia," that some people just don't want to admit that there is progress going on. I'll be talking about negativity biases and so forth. Then we will try to discern what is behind human progress. In other words, how did it come about? What are the theories for human progress and what are the drivers for human progress? Can we distill some lessons from world history as to what makes the world a better place?
- 1.3** Then finally, we'll be talking about challenges and criticisms of human progress. As you'll see, my definition of human progress is a very specific one, and other people may disagree. And so, what I think actually makes the world a better place, some people may think makes the world actually a worse place. So we'll be talking about those criticisms, and we'll be looking also at the future of human progress.
- 1.4** So first of all, I want to say a few things about myself, not because I'm a narcissist, although some people might disagree, but perhaps I do have a little bit of a perspective. I was born in communist Czechoslovakia, and obviously, I experienced my childhood without any political and economic freedom. The country was very poor, and we were desperate to get out. Eventually, we were able to get out as a family. My parents, because they're medical doctors, ended up in South Africa, working in South African hospitals. And after that, I went to Great Britain, where I went to university, and eventually, I ended up in the United States.
- 1.5** So I went through four, or rather, I experienced four different cultures, four very different standards of living. Eastern Europe under communism was very poor. South Africa at that point was actually somewhat more developed. I then went to Britain, which is obviously a first-world country. And eventually, I ended up in the United States, which is an extremely rich country, irrespective of what a lot of Americans feel. By global standards and by historical standards, America is a very, very rich country. And so, I guess I started to think about what makes countries rich, what makes countries poor.

- 1.6 Human Progress is a website which I started in 2013. It is part of the Cato Institute in Washington, DC. But anybody who wants to learn more about Human Progress can go to humanprogress.org. So that's what I do for a living. I'm both the founder and editor of humanprogress.org, and I'm also a senior fellow at the Cato Institute. And that's where we do our thinking. That's where we do our writing. That's where we are trying to figure out what makes countries rich, what makes countries poor. Why are some places in the world where millions of people want to move? And why are there some places in the world where millions of people want to escape?
- 1.7 The second picture there is actually from the capital city of my native country. I'm an American now, but back then I was still a citizen of Czechoslovakia. In 1968, the Soviet Union invaded my native country of Czechoslovakia to suppress whatever economic and political freedom were beginning to emerge during what was called the Prague Spring. And so, experience in communism obviously made me very much appreciative of being able to move from countries without political freedom to countries with political freedom.
- 1.8 The third picture is a typical African village story. When I lived in Africa, I lived there for 10 years. I would go to places which are very prosperous, but also which are very poor. And that made me appreciate the importance of economic freedom, the importance of economic growth. Here in the United States, we are very rich, but a lot of people in Africa are still very poor. And finally, there is my employer, the Cato Institute.
- 1.9 OK, so I guess the main point of the opening was to explain to you why I am very passionate about the subject of human progress, and why perhaps after 20 years of thinking about what makes countries rich, what makes countries poor, and 15 years of writing about it, I may be at least somewhat qualified to talk to you about it. So I guess the first obvious question must be: what is human progress? So the dictionaries of this world will tell you that human progress is an advancement to a further or higher stage, or to a further or higher stage successively—growth, development, progress, moving towards somewhere, moving upwards, usually.
- 1.10 But to what end? To what end? Progress can be good, can be neutral, it can be bad. If somebody tells you your cancer is progressing, that's not a good thing. OK, so we are progressing, but to what end? Progress is not a goal by itself. What counts as genuine progress is a movement toward a set of predetermined ultimate values. We have to decide what it is that we want to progress toward. So the definition of human progress rests on what smart people, people smarter than I am, talk about axiology. They call it axiology, our theory of value.
- 1.11 Now, to a humanist—and my definition of progress will be a humanistic one, hence the word human progress—so to a humanist, which is a current in philosophy, a certain way of thinking about the world and about people in it, to a humanist, progress means improvement in human well-being. Improvement in human well-being. Can we agree on what those improvements might be? Well, Steven Pinker, a Harvard University

psychologist, whose book, "Enlightenment Now," is still, I think, the best book on the subject—it came out in 2018, and I highly recommend it to you—defines progress in the following way.

- 1.12 "Most people agree that life is better than death. Health is better than sickness. Sustenance is better than hunger. Wealth is better than poverty. Peace is better than war. Safety is better than danger. Freedom is better than tyranny. Equal rights are better than bigotry and discrimination. Literacy is better than illiteracy. Knowledge is better than ignorance. Intelligence is better than dull-wittedness. Happiness is better than misery. Opportunities to enjoy family, friends, culture, and nature are better than drudgery and monotony. All these can be measured. And if they have increased over time, that is progress."
- 1.13 So that is what I mean when I talk about improvements in human well-being. Do we live longer, as opposed to dying younger? Are we more satiated, rather than hungry? Do we have access to more information, rather than living in ignorance? Can we speak our minds, rather than be oppressed by a tyrant and so on and so forth? Now, not everybody agrees with Steven Pinker's definition. There are alternative visions of progress.
- 1.14 There's the theocentric or salvationist mode of progress. People like Augustine and Aquinas would have told us that conformity to divine will is more important than material gains. Material gains are secondary than fulfilling, for example, a certain theological vision of what humans ought to be like. So, welfare only counts if it contributes to salvation, for example. So the theocratic view privileges religious law over comfort, for example.
- 1.15 We also have the ascetic or monastic view of progress. These are people like Saint Benedict, who believed in spiritual purification. Human progress is not about getting more stuff. It is about having a soul that is pure from corruption.
- 1.16 We have a traditionalist viewpoint, which is that human progress is actually a return to a previous order. It is a return to a better world of tradition, rather than starting something new. So here we have often a view that, previously, the world was understandable, it was predictable, it was hierarchical, everybody knew their place. Now everything is in flux. So we have this traditionalist viewpoint, or traditionalist view of human progress.
- 1.17 We also have the romantic or anti-industrial view of human progress. Here, the idea is that capitalism and industry, they are making everything cheap, yes, but they're making everything the same. Perhaps human progress would be better if we had crafts and if we were all authentic and if everything was crafted for us. If you go to a fancy sandwich place, they will say, "It is crafted for you." What are they saying to you? They're saying, "It's not mass-produced. It is produced just for you." So, we have this romantic, anti-

industrial view of human progress. It talks about restricting mechanization and returning to the craftful way of doing things.

- 1.18** There's cultural pessimism or cyclical history's view of progress, which is to say that civilizations rise and fall in an endless cycle. And if you think about linear progress, if you think that the world is eventually going to end up in some good place, well, you are lying to yourself, right? You are being completely hubristic.
- 1.19** There is the biocentric or ecocentric view of progress, which is to say that what's important is the integrity of the ecosystems; that human gains, if they clash with environmental protection, that's actually not progress, it's a regress.
- 1.20** We have anti-civilizational attitude to progress, or primitivist attitude to progress, which says that the way to progress is actually to destroy a large-scale civilization. Large-scale civilization corrupts us. If we can return to the small, if we can return to the local, that is better.
- 1.21** We have the post-development, or anti-development, view of progress, whereby if we can protect local lifestyles, say, the way that some Indians live in the Amazon, that is better than subjecting people who live in that primitive state to what we would call civilization, but they may call actually just a destruction of their way of life. So development is a problem. It's the original way of doing things which is better. So we want to halt mega-projects. We want to preserve the smallness.
- 1.22** We have the Frankfurt School or the critical theory of people like Horkheimer, Adorno, and Marcuse, who talk about emancipation from domination. We have been dominated by media. We have been dominated by, for example, advertisement agencies. We are not really free. So Frankfurt School says that we have to restrain technology. We have to restrain mass culture.
- 1.23** We have the postmodern skepticism. People like Foucault, who appears a lot of times in Peterson Academy lectures. Postmodernists say there is no single tale of us. You cannot tell a single story about human progress because any kind of mega-story is actually quite oppressive. We must avoid universal metrics. So when I tell you in Lecture 2 that life expectancy around the world has increased from 35 years to 73 years, for example, that's irrelevant. They don't recognize metrics.
- 1.24** We have the virtue ethical attitude, which talks about the importance of civic virtue over comfort. We see that a lot in certain countries. Don't prioritize how much stuff you can buy in the shops. The question is: are you a good citizen? Do you pay homage to the flag? Do you respect the Constitution? Those are the hallmarks of a good citizen, rather than somebody who cares about how many types of deodorants there are in the stores.
- 1.25** The Heideggerian critique of technology, again by German philosopher Heidegger, is going back to the idea that technology is bad, that we need to protect nature and so

forth. Another version of this anti-technological and pro-environmental view of looking at progress would be degrowth, people who believe that it's not enough if we simply stop growing in order to protect the environment. We should actually shrink our economic footprint. We should actually shrink our economy. So the American economy is, what, \$30 trillion dollars a year? And people who are against development would say, "Just stop. We have enough." Degrowthers would say, "We should do with less. Maybe we should have a \$20-trillion dollar economy or a \$10-trillion dollar economy." People like Kohei Saito, a Japanese Marxist, is a big advancer of these ideas.

- 1.26 You have Nietzsche, our old friend Friedrich Nietzsche, from Germany, who believes that this whole business of human progress, where we become much more egalitarian and tolerant of each other, is nonsense. The "Übermensch," the Superman, should be able to determine what happens to the rest of humanity. So the idea that women now have equal rights, and gay people can marry and whatever else, that's crazy. That's crazy talk. That's egalitarianism. That's toleration. For Nietzsche, what's important is the assertion of the will of the Superman, of the Übermensch, on the rest.
- 1.27 We have the Stoic criticism of human progress, or the Stoic approach to human progress. People like Marcus Aurelius, for example, or Epictetus, who say that real progress really comes from inner mastery, tranquility. In other words, instead of being concerned about switching your old-model car for a new-model car, just accept the old car, or maybe not have a car at all. The fewer ambitions you have, the fewer desires you have, the better for you. You're a Stoic, right?
- 1.28 And we also have communitarian conservatism, people like Alasdair MacIntyre or Edmund Burke, who talk about human progress in terms of social cohesion. What really matters is that there is tradition, that the society hangs together, and what we want to avoid is disruptive innovations.
- 1.29 What Steven Pinker presented to you is a humanistic definition of human progress. He talks about human flourishing. The more options you have, the better you feel about your life; the better off you are, so much the better. But there are alternative viewpoints. Not everybody agrees that human flourishing should be the objective of progress, and I've just given you a bunch of alternative visions of what makes for a good society.
- 1.30 So let's read through some of the more prominent dissents, criticisms, of human progress. Here is our old friend Jean-Jacques Rousseau, the originator of Romanticism in France in the 18th century. He says, "The progress of the sciences and the arts has added nothing to our true happiness, and that our souls have become corrupted to the extent that our sciences and our arts have advanced toward perfection."
- 1.31 We have Carl Becker, in "Progress and Power," who writes after the First World War, so he had some reason to be skeptical about human progress. He writes, "The present moment, therefore, when the fact of progress is disputed and the doctrine discredited,

seems to me proper time to ask the question: What, if anything, may be said on behalf of the human race? May we still, in whatever different fashion, believe in the progress of mankind?"

- 1.32** And we have Jared Diamond, the famous author, celebrated author, of a popular book called "Guns, Germs, and Steel." "I do not assume," he says, "that industrialized states are 'better' than hunter-gathering tribes, or that the abandonment of the hunter-gathering lifestyle for iron-based statehood represents 'progress,' or that it has led to an increase in human happiness." OK? A hunter-gatherer, somewhere in the Amazon, just as happy as an American in Phoenix or Scottsdale, and it makes absolutely no difference.
- 1.33** So, can we agree on the basics? I'm going to read another quote from Steven Pinker's "Enlightenment Now." And here's what he says: "Humanistic values," the ones to which I subscribe as well, "leave out religious, romantic, aristocratic virtues like salvation, grace, sacredness, heroism, honor, glory, and authenticity." Remember the previous slide? We were talking about the criticism of human progress from the romantic, aristocratic, heroic, romantic perspectives. So Pinker acknowledges that.
- 1.34** He says, "Humanistic values leave out all of those religious, romantic, and aristocratic virtues, like salvation, grace, sacredness, heroism, honor, glory, and authenticity. But most would agree that it is a necessary start," the humanism. "It's easy to extol transcendent values in the abstract." I want to be a hero. I want to be a Superman. "But most people prioritize life, health, safety, literacy, sustenance, and stimulation for the obvious reason that these goods are the prerequisite for everything else." Health, life, safety, literacy, sustenance are the prerequisite for everything else.
- 1.35** "If you are reading this," he writes in his book, "you are not dead, you are not starving, you are not destitute, you are not moribund, you are not terrified, you are not enslaved or illiterate, which means that you are in no position to turn your nose at these values — nor to deny that other people should share your good fortune." What he's saying is that you are still free to pursue ascetic fulfillment, or that you should devote your life to spiritual purification, that you should spend your life in contemplation of the higher things, figuring out what's good and what's bad. That may well be worthwhile.
- 1.36** But the start, the basic that you have to have in order to do that, you have to be alive. You cannot be dead at the age of 30. You have to be free to be able to read books and contemplate and write about it, rather than being a woman who has no rights and who is prevented from being schooled. You have to be able to have your stomach full, because it's much easier to contemplate higher things in life on a full stomach than on an empty one, when all you can think about is where my meal is going to come from.
- 1.37** Now, Steven Pinker is not the only one who represents this humanistic attitude to human progress. Not the only one who talks about it, writes about it. There are many others. There is Matt Ridley, Jason Crawford, Joel Mokyr, who just received a Nobel

Prize in economics this year. There is Johan Norberg, Deirdre McCloskey, and David Deutsch. And in their works, such as "Bourgeois Dignity," or the Bourgeois Trilogy by Deirdre McCloskey, "Culture of Growth" by Joel Mokyr, "Open: The Story of Human Progress" by Johan Norberg, and "The Rational Optimist" by Matt Ridley, "The Beginnings of Infinity" by David Deutsch, and "The Techno-Humanist Manifesto" by Jason Crawford—they basically echo many of these motifs, or many of these ideas that we have discussed with regard to Steven Pinker's work.

- 1.38** Now, there are different types of progress. There is the technological progress. When you are replacing, for example, horses with steam power, you are making progress in the technological arena. There are institutional types of progress. When we are replacing, for example, absolute monarchy for constitutional monarchy, in other words, the absolute monarchy of Henry VIII, for the constitutional monarchy of Charles III, we are making progress.
- 1.39** When we are replacing Athenian democracy, where the majority can, by one vote, condemn Socrates to death, for liberal democracy, where individual rights are protected, we are making institutional progress. We are also making legal progress. There is legal type of progress. Instead of arbitrary imprisonment, for example, you have now access to habeas corpus; you have access to due process.
- 1.40** We have scientific progress. For example, the movement from Aristotelian to Newtonian physics, that's scientific progress. Aristotle believed that birds flew and rocks fell to the ground because it was in their nature. Newton understood that it was because of gravity. So eventually, Aristotelian physics has to give way to Newtonian physics. That's what we call scientific progress.
- 1.41** We have medical progress. People used to believe in miasma, that miasma was what made people sick. The air was filled with tiny, impossible-to-see droplets of disease that would flow through the air and would infect your lungs and would give you some terrible disease. But then, in the 1860s and 1870s and 1880s, we came up, or other people much smarter than me, came up with the idea of a germ theory of disease. Now we know that there are these tiny, little, invisible pathogens which do that. It's not air, per se. It is these pathogens.
- 1.42** We have cultural progress. Yesterday, I flew here from Washington, DC. But instead of praying for a flying carpet, I bought a ticket on an airplane. So we have, essentially, instead of superstition or believing in a genie out of a bottle who will fulfill your wishes, we buy airline tickets, and we hope that the plane will stay in the sky.
- 1.43** And we have ethical progress. People in the old days, including Nietzsche, in the 19th century, believed that "might makes right." If you have power, you can force your fellow human beings to do anything you want. But now we believe in golden rule, or the categorical imperative. We have developed this new ethic, which says, "Don't do

unto others as you wouldn't want them to do unto you." There's some ethical improvement as well.

- 1.44** "By progress," says one of America's most prominent economists, Tyler Cowen, and Patrick Collison, who is a very respected and highly successful entrepreneur in Silicon Valley, "by 'progress,' we mean the combination of economic, technological, scientific, cultural, and organizational advancements that has transformed our lives and raised standards of living over the past couple of centuries." It's the combination of all of these types of progress that contribute to making one whole human progress.
- 1.45** Now, that doesn't mean that all of these types of progress move in lockstep, reminds us Jason Crawford, from the Roots of Progress, a philosopher in, I believe he's in California. What he means is that all aspects of progress support each other. If you have a better understanding of science, maybe you'll have a better understanding of technology. If you have better technology, maybe you'll be able to measure science better.
- 1.46** If you have political freedom where you can express your views, and when you can write those views down and share them with your friends, that too contributes to progress, because you are able to maybe share your scientific or political discoveries. So it doesn't mean that all aspects of progress move in lockstep. They don't. They support each other, and over the long run, they depend on each other.
- 1.47** It may even be possible that certain different types of progress are actually diverging, or maybe come into opposition with one another. For example, smartphone is an excellent invention. It allows you access to GPS. It allows you to watch videos. It allows you to take pictures. It's an alarm clock. It's all sorts of things. But maybe, just maybe, it is not such a good idea for kids under a certain age. What we are seeing in the United States is that, in a lot of schools, and indeed, abroad as well, in a lot of schools, smartphones are now being banned in schools for kids, so they can focus on their classes.
- 1.48** Types of progress may even occasionally, or different types of innovations, may actually go and undermine progress. Then you have to adjust to it. You have to adapt to it. So it is certainly possible that you can diverge, and certain forms of progress may be in opposition to one another. We'll discuss that a little bit later. Now we'll talk about the idea of progress. Where does it come from?
- 1.49** So progress is not just about numbers— measuring how long people live, how many calories they have access to. It is also an interpretation of history, according to which different forms of progress have, by and large, been happening. I'll just repeat that. Progress is an interpretation of history, according to which different forms of progress have, by and large, been happening. Interpretation of history.
- 1.50** Max Roser from Oxford University has this saying. "The world is awful. The world is much better. The world can be much better." All of those three statements are true.

The world was much worse. It is much better. It could be much better. But there is a timeline connecting these three statements together. What we are really talking about is interpretation of history.

- 1.51 So, going back to Jason Crawford, he says, in this sense, the interpretation-of-history sense, in this sense, the study of progress is the intersection of axiology and history. Given a standard of value, which in our case is human flourishing, are things getting better? So, the idea of progress is a normative statement. In other words, it is good to have it. By saying that we like human progress, we are saying, we are making a normative statement, "It is a good thing that we have human progress."
- 1.52 And it is also an empirical claim that human life can improve over time through cumulative knowledge, better institutions, and deliberate actions. So interpretation of history, it says that life has been getting better, but it's also a normative statement, saying, "And it is a good thing that it was getting better, because it ought to have gotten better."
- 1.53 So, in the lectures that follow, I will be focusing on four key elements of human progress. One is the material human progress, which is largely driven, but not solely, by economic factors. I'll be looking at moral and ethical progress. I'll be talking about institutional and political and legal progress. That's the third one. And the fourth will be an epistemic progress, our understanding of science and technology.
- 1.54 And so, when it comes to this idea of progress as an interpretation of history, can we really say that, whether you want to begin with the Stone Age tools 2.4 million years ago, or whether you want to start in ancient Athens, can we really say that humanity has been progressing? That's the idea of progress. Can we actually say that? And this is why a debate between two scholars, long dead, actually matters. One was J. B. Bury, and the second one was Robert Nisbet. What they have argued about is: where does the idea of progress come from? Was it always there, or is it a modern invention? Is it a secular idea, or is it a religious idea? So here they are, J. B. Bury and his 1922 book, "The Idea of Progress," and then Robert Nisbet's "The History of the Idea of Progress," which came out in the 1980s.
- 1.55 Now, Bury's main theme, main idea in the book, is that the idea of progress, that life can actually get better, is a modern invention that really starts in the 17th and 18th centuries, primarily with the scientific revolution and the Enlightenment. He claims that it is new, that at some point in time, people decided, based on evidence, that life could actually improve, whereas before they didn't believe in that.
- 1.56 He says that the ancients, for example, believed in the cycles. Society improves, then it deteriorates, destroys itself, then it's renewed. So the ancients, ancient Greeks and ancient Romans, would have a cyclical understanding of history. So the ancients are cyclical. Yes, you can have certain improvements, but then there's a collapse, renewal, improvement, collapse, eternally, like that.

- 1.57 Pythagoreans, for example, were extreme believers in this view. In their view, everything that happens in the world always repeats itself. So, for example, if, in this world, you have the Second World War breaking out by Hitler's invasion of Poland, then after the destruction of the world, there will be a new world which will also have a Second World War that starts with Hitler's invasion of Poland. Nothing ever changes.
- 1.58 Now, that's quite different from other ancients, like, for example, the ancient Greek writer, Hesiod, who believed in perpetual decline. His idea, in "Works and Days," which was written about 700 or 800 BC, is the idea of perpetual decline. You start with the Golden Age, the age of the gods. Then there's the Silver Age, then there's the Bronze Age, and eventually the Iron Age. He believed that he lived in the Iron Age, when there was a lot of violence, when there was a lot of destruction. But he believed that in the Golden Age, the age of gold, there was no violence; there was no want or hunger and whatever else. So Hesiod introduces us to the idea of a perpetual decline.
- 1.59 So Bury's argument is that the ancients were either believers in the cyclical nature of history or in perpetual decline. And indeed, in some ways, you can see how Christian religion mirrors the ancient Greek idea of a perpetual decline from the Garden of Eden. The Garden of Eden is a place where there is no pain, there is no hunger, there is no violence. And when Adam and Eve do what they do, eventually you see a decline, postlapsarian decline, from the Garden of Eden until the present. So this idea of continual decline is something that Bury claims was very much present in the ancient world.
- 1.60 Now, the medievals, people in the medieval world, look at things a little bit differently. They stop seeing human evolution as a perpetual decline from the Garden of Eden. They now see a possibility of an improvement leading to salvation, but it is purely spiritual salvation. The medievals don't believe that, in anno Domini 2025, we are going to be riding in chauffeur-less cabs, and we'll have access to strawberries in winter and whatever else. To them, the idea of improvement is a purely secular, salvationist idea. It is not an improvement in the human condition. It is the notion that, at the end, there is going to be the day of judgment, and humanity will be saved.
- 1.61 Now, Bury then goes on to say that, because of scientific revolution and the discovery of the empirical method, about which we'll be talking a lot in this particular course, there is a rejection of classical authority and of scholasticism; in other words, that, at some point in Europe, 15th, 16th, 17th centuries, we reject this notion that all of human wisdom is contained in ancient texts. We start believing that it is possible to come up with new knowledge. And once you have new knowledge, then you can build new knowledge on top of other new knowledge, until you end up in a place where you can actually have progress, and you can start believing in it.
- 1.62 So, because of some early shoots of scientific and technological progress that he could observe, such as, for example, the compass, the usage of compass, gunpowder, and printing, all of which had been invented in China hundreds of years before, but then

were imported into Europe and much improved, Bury now says, "The ancients have been overcome." The conversation between the ancients and the moderns is over. We no longer have to believe that all of the information that we need for survival and flourishing is contained in the old books. We now have new knowledge coming in that can make the world a better place.

- 1.63** But more importantly, it is at this stage, says Bury, that this notion of a linear progression towards salvation becomes secularized. It is no longer divine providence which guides humanity towards salvation. It is now human action— discovery of the telescope, discovery of the microscope, discovery of gravity, discovery of printing press —that imbues the human being with the ability, imbues the human being with the agency, to make the world a better place.
- 1.64** So, in Bury's account, we still maintain this linear progression towards something good. We are not talking about decline like Hesiod. We are not talking about secular understanding of history. We are now talking about a movement, linear movement, towards something good. But unlike the medievals, who believed that divine providence would have to cause all of that, we now believe that humans can accomplish human progress. It is humans who now have the agency to bring about this better world.
- 1.65** In Bury's narrative, it is Francis Bacon who is very important, because Bacon is the one who, in the 16th century, introduces a new scientific method. Basically, Bacon says that it is experimentation that produces new knowledge. He rejects the Greek idea of deduction. He loves induction. You collect evidence, you experiment at the bottom, and then if everything works together, you build upon it a new theory. You don't deduce theory from the top down; you create it from the bottom up. So, it is Francis Bacon who proposes that this experimentation and organized research can produce new knowledge.
- 1.66** But very importantly, as a Christian, Francis Bacon still justifies human progress theologically, curing disease, extending life, improving arts. These are the works to honor God. So Bacon announces a research program, if you will, but not yet does he tell us that it is the human beings who are going to be in the driving seat of human progress. It's still divine providence; it is still—Bacon's idea of research and empiricism is still imbued within the moral theological framework. Bacon's view of human progress is still rooted in theology that J. B. Bury credits Bernard de Fontenelle, a French philosopher, for being the first true humanist progressive, so to speak.
- 1.67** It is Fontenelle, according to Bury, who is the first to state and to secularize the idea of progress. Progress is no longer dependent on the whims of a deity of divine providence; it is in the hands of human beings. So, knowledge and society, according to Fontenelle, advance cumulatively. We surpass ancients, and the future can be better because of human effort. He secularizes the ideas which Bacon still was very much beholden to, theology and religious belief. So here we have, according to Bury, a shift from a

research program, as it was outlined by somebody like Bacon, to a historical narrative driven by human agency, as outlined by Fontenelle. So that's Bury's attitude.

- 1.68** Robert Nisbet disagrees. Robert Nisbet doesn't believe that the idea of progress is a new idea. He doesn't believe that it appears suddenly in the 17th and 18th centuries. He believes that it matured over centuries. He believes that the concepts, such as the idea of linear time as opposed to cyclical nature of history, the meaning of history, human perfectibility, he believes, existed in religious and classical forms a long time before the scientific revolution and before the Enlightenment.
- 1.69** He points, Robert Nisbet, points to some excerpts from Greek and Roman literature, where he, for example, points out that some of the early Greek philosophers, they have actually made fun of their gods. They wanted to base their thinking on reason, rather than the commands of Zeus or Hera or Apollo. Xenophanes mocked the very idea of an anthropomorphic god. Sophists, and then later Hellenic scholars, treated knowledge and technique as teachable and improvable. So, they believed that you are not just stuck with one technology, for example, one way of doing things, that you can improve on it. From this, Nisbet derives the idea that there had been shoots of the idea of progress all the way back in antiquity.
- 1.70** And he believes that, from antiquity, modern historians of progress derive two very important lessons. One is criticism of authority, that just because something is written down in a book, like books by Aristotle or Plato, doesn't mean that it is correct forever. It can change. So Nisbet tells us, from the Greeks, we get criticism of authority, and secondly, confidence in human reason.
- 1.71** Then we move on to the Christian era. Here, Nisbet tells us, Christianity supplies us, meaning humanity, with the architecture for a directional history. It taught us a beginning, a purposeful middle, and an end. So history becomes a narrative with stages of promise and fulfilment, leading to something good at the end. And according to the Christian way of looking at things, human history is imbued with meaning. It is not simply an ancient Greek cyclical idea of history. So there's a meaning, there's a linearity to history, as we were talking about medievals, and it has a meaning. So that's what Christians contribute to thinking about progress, from Nisbet's perspective.
- 1.72** And then eventually, just like Bury, Nisbet moves on to the Renaissance and the scientific revolution, the voyages of discovery and the Enlightenment; whereas before, you have human progress being guided by divine hand, by divine providence, at some point, humans figure out the laws of nature. They become much more agentic. They can now be in charge of their own future.
- 1.73** So, by the time we have the Enlightenment, Nisbet argues, we can— and I suppose Bury would agree—we can now replace the idea of a linear spiritual progress, which is something that the Christian era was mostly about, to one, we have secular progress, which is measured by science, by arts, by public welfare, and so forth. So finally, in

Nisbet's view, the modern secular movements inherit religious form. The idea of progress keeps the structures of expectation, universal scope, and moral purpose, while relocating them from divine providence to human reason and action.

- 1.74** So what are the drivers of human progress? Again, in Lecture 2 and Lecture 3, we'll be talking about evidence for progress, but assuming, just for the sake of argument, that there has been progress, what are the drivers of progress? Well, there are a lot of different thinkers who think very differently about what has driven human progress forward. If you have Francis Bacon, about whom we spoke, Francis Bacon believed that it was the growth of useful knowledge, new tools, patents, experimental output. René Descartes, the French philosopher, believed that it was methodological reasoning, applied science, engineering feats, but methodological reasoning.
- 1.75** Thomas Hobbes, the British philosopher, believed that it was the move from insecurity. "Man is another man's wolf." The world was at war, and then it became more peaceful. So Hobbes talks about the move from insecurity to peace. Fontenelle, cumulative knowledge. We already discussed him. David Hume, that it was actually commerce that has driven progress forward; that people who trade together have a much greater appreciation together. If you have something to sell me that I want, I don't care whether you're a woman, I don't care whether you are black or white, I don't care about anything. I just want to have a good relationship with you, so that we can trade.
- 1.76** Immanuel Kant, for example, places human progress at the feet of moral and political maturation. Condorcet talks about the advance of science and rights. You have Hegel, who believes that human progress is driven by unfolding of human freedom. The more freedom you have, the more progress humanity can experiment. You have Saint-Simon, who believes in social reorganization. In other words, industrialization will bring about human progress. You have Auguste Comte, that it is really scientific education that drives humanity forward.
- 1.77** You have John Stuart Mill, who believes that it is advancement of human liberty. You have Karl Marx, who believes that by moving from agriculture to capitalism to communism, that's what will bring about human progress. You've got Herbert Spencer, an English philosopher, talking about increasing differentiation and social complexity. Basically, he believes that, as society becomes more complex, you are developing vaccines, I'm developing driverless cars, that's what drives human progress.
- 1.78** We have Max Weber, who believes in things like, for example, rationalization, rational bureaucracy. We have Joseph Schumpeter, who believes in innovation. We'll be talking about innovation quite a lot in the future. So he believes that human progress is driven by innovation. We have Friedrich von Hayek, who believes that—he was an Austrian economist—who believes that what's important in terms of driving progress is actually decentralized decision-making, where new ideas can come up from the bottom up, rather than centralized decision-making from the top down.

- 1.79 There is Karl Popper, who talks about conjectures and refutations, that human progress is guided by the idea that I propose an idea and you shoot it down; and it's a constant false correction mechanism that drives society forward. We have Douglass North, the Nobel Prize-winning economist, who puts human progress at the feet of institutional development. We have Joel Mokyr, the Nobel Prize-winning economist, who talks about diffusion, creation and diffusion of useful knowledge in open cultures. Open cultures create an environment in which new ideas can spread.
- 1.80 We have Deirdre McCloskey, my colleague at the Cato Institute, who talks about dignity and liberty for innovators; that in the old days, innovators were persecuted. Innovation was looked down upon because it challenged society; it challenged the status quo. And once innovators were treated with a certain degree of dignity, or were even admired, like Elon Musk is admired by a lot of people today, that that is what has driven humanity forward. And finally, we have Steven Pinker, who places it, human progress, at the feet of humanism, science, reason, and so forth.
- 1.81 Let us now talk about what human progress is not. First of all, human progress is not utopia. Progress doesn't mean that everything is working out for everyone, everywhere, at all times. That would be a miracle, not progress. So progress is not a miracle, it is problem-solving. Progress is not a miracle, it is problem-solving. Progress lessens but does not eliminate problems. Solutions to big problems very often lead to new problems. We used to be hungry once, and then we figured out how to grow more food, and then we became obese. Then we had a problem that we are now trying to solve through Ozempic and many other things.
- 1.82 So we had a massive problem. Tens of millions of people were dying all over the world every year because they didn't have enough food. Then we figured out how to make a lot of food, and we all put on weight, and now we are trying to overcome that problem. Who knows? Maybe Ozempic has side effects, and we will have to deal with those again and again and again. So, progress lessens but does not eliminate problems. The solution of big problems often leads to smaller ones.
- 1.83 Progress is jagged, not straight. Very often, progress means two steps forward, one step back. Crime is a very good example. At the time of Michelangelo, 45 out of 100,000 Italians could expect to be murdered in their lifetimes. Today, in Italy, that number is 0.3. So, from 45 per 100,000, to 0.3 per 100,000. But that doesn't mean that Italy didn't have spikes in murder rates in between. America is a much less violent place than it was when sheriffs and gangsters were shooting each other out west. But we did have a spike recently in crime during 2020 and 2021 and 2022, and now that's coming down. So we are much less violent, but there are often two steps forward, one step back. There are sometimes reversions.
- 1.84 Progress, I would argue, is not optional. It's not optional if humanity wants to survive. Pathogens evolve. Climates vary. We hear a lot about climate change in terms of global warming. But there was the Little Ice Age, which begins in 14th century, that we will be

talking about, in the northern hemisphere. And that brings about collapses in agriculture productivity and millions of deaths. So sometimes the world is warming, sometimes it is cooling. But what we do know is that it's always changing. And what you want to be is a civilization that is capable of handling those problems.

- 1.85** So when I say that human progress is not optional, what I mean by that is that, if we stop progressing technologically and scientifically, eventually we will run up against a problem that we cannot solve. And if we cannot solve it, then it's all over. Progress, very importantly, is not just GDP. People often pooh-poo GDP per capita, and they say, "All you care about is money." But as we will see in Lecture 2 and Lecture 3, GDP is highly correlated with everything else.
- 1.86** Countries which have higher incomes have higher life expectancy. They have better education. They have more rights for women, etc. So it's not just about GDP. It is about much more than that. Progress is not Western-only. Scientific ideas, like gravity, yes, Newton was an Englishman, but it applies universally. Nor did progress happen only in the West, as we will see in further lectures. Often, there were periods when most of human progress happened in the Middle East. Sometimes it happened in Asia.
- 1.87** So human progress is a story that is almost universal, and the lessons that we learn and the new knowledge that we come up with, that is also a universal story because, if there are certain lessons that work, certain ideas that work, then they should be applicable, no matter whether you live in Europe or Asia or anywhere else.
- 1.88** Progress cannot be centrally designed. This is another very important thing. Starting with the French Revolution and later in communism, there were people in charge who had a vision of progress, a classless society perhaps, which the French tried to create by killing their aristocrats, and the Soviets by killing their capitalists. They had a vision of a classless society as an idea of progress. But that turned out to be a very bad idea. Those ideas turned out to be very bad.
- 1.89** Those were very bad times for humanity because they were centrally planned. You cannot plan where you want to go. It is a complex—humanity is a complex system that requires a lot of experimentation that is local experimentation, that happens at a small scale. So, complex systems improve through iteration, not by having some sort of blueprint into which you are trying to herd all of humanity.
- 1.90** Progress also doesn't excuse current problems. I'm going to be arguing that there has been a lot of progress in the world. But that doesn't mean, I'm certainly not arguing that there are no problems to be resolved. There are plenty of problems. There will always be problems to be resolved. But the fact that there are problems doesn't mean that we didn't have more problems and bigger problems before.
- 1.91** And finally, and I'm coming to the conclusion of this lecture, progress is not inevitable. There have been times in the past, especially in the 19th century, when people saw so much progress, when they saw so much increase in wealth and science and technology,

they convinced themselves that there is such a thing as an inevitability of human progress, that there was History, with a capital H, some sort of a force which was guiding humanity forward.

- 1.92 Saint-Simon believed that science was going to be the mechanism by which progress would be unavoidable. Hegel had his dialectic. Condorcet and Turgot were talking about cumulative knowledge as a guiding force toward inevitable progress. Herbert Spencer was talking about specialization and social evolution. For Marx and Engels, it was economic determinism. For Adam Ferguson, it was the stadial model. We progressed from the Stone Age to agriculture, then to industry, and then to something else. So he believed that that was unavoidable.
- 1.93 Even Immanuel Kant dabbled a little bit in historical inevitability, by claiming that human history unfolds a plan in nature that brings about more freedom and order. He claimed that history of the human race can be regarded as the realization of the hidden plan of nature. Now, that didn't happen. Progress was not unavoidable. Progress is not unavoidable or inevitable. And we know that, because there have been periods in human history when a sophisticated civilization came crashing down.
- 1.94 Perfect example of that would be the Western Roman Empire, which falls in the fifth century. After the fall of the Western Roman Empire, we have a decline in political and legal institutions. The imperial administration and professional bureaucracy disappear. Unified Roman law, courts, taxes fragment. Cities shrink and lose administrative and commercial roles. Roads, bridges, aqueducts, harbors, public works receive little maintenance. Sometimes they collapse.
- 1.95 Literacy falls. Schools and libraries close. Large-scale engineering and monumental structures, which the Romans were so good at building, cease. Many specializations, crafts, and standardized methods of production disappear or survive only in small localities. Mediterranean trade network shrinks, and the use of coinage, the payment of taxes, falls. So there are times when a society, which may seem like it's very progressive, like it's at its pinnacle of human accomplishment, which the Romans were, can collapse, and human progress ceases.
- 1.96 Now, why do I talk about this? It's because I think that a belief in some sort of immutable, unavoidable historical forces can be very dangerous. I believe that progress is happening, but I don't believe in its unavoidability. Here, I'm guided by the ideas of Karl Popper, an English philosopher, who talked about historicism. Historicism is a belief that history unfolds by necessary laws and stages that allow unconditional prophecies about the societal future. So if history has iron laws, if we know where we are going, then rulers can have the mandate to coerce society to achieve those unavoidable goals.
- 1.97 We already talked about the French Revolution and communism. If you already know what the future is like, what the future—how magnificent it will be, if only you follow a

certain path, then you can start treating everybody who disagrees with you, and everybody who criticizes you, as an enemy. And you begin to use people as means rather than ends. You suppress dissent and you suppress criticism. And it's exactly the opposite of the kind of trial-and-error idea that people like Hayek, for example, would be talking about, or Hume, or even Bacon.

- 1.98** You want to experiment on small scales and develop from it; you build up from it, rather than top down. Once you start treating people as means rather than ends, then you depart from the humanistic concept of human progress. Progress which is not guided by humanism, because it ignores the well-being of an individual human being, is not progress. That's essentially what I'm saying.
- 1.99** I already talked about how a belief in unavoidability of some sort of a future that somebody has imagined, be it a French revolutionary or a communist Soviet apparatchik, how it suppresses criticism and learning. So, if I'm an anti-historicist, if I don't believe that that's the right way to go, how can I believe in human progress? How can I believe that there is still such a thing as human progress?
- 1.100** I think that the way to reconcile these two is by saying that, I look at human progress, the trends in life expectancy, the trends in education, calorie consumption, whatever else, as conditional trends, not as inevitable laws. I'm not saying that what has been happening in the last 200 years or so has to happen in the future. So I'm looking at them as conditional trends, not inevitable laws.
- 1.101** I'm not prophesying human progress; I'm talking about probabilities. I'm treating progress, longevity and knowledge, prosperity, as empirical regularities that can stall and that can reverse. They are not fated endpoints. I also insist that, instead of grand plans, which the French revolutionaries had or the Soviet communists, we should embrace experimentation, piecemeal approach. A teams versus B teams, trying to figure out what works better, rather than big plans. Feedback loops to ensure that when something goes wrong, we learn about it. We don't suppress dissent. That's how you learn in a complex system.
- 1.102** We also need a lot of institutional preconditions. Progress is contingent on, for example, free inquiry, on criticism, on market prices and constitutional checks. And in order to treat human beings as ends rather than means, we also have to have institutional preconditions. We have to maintain free inquiry. We have to maintain our ability to criticize. We have to keep market prices. We have to have constitutional checks. We have to ensure that nobody can force us into a future that we don't desire, essentially.
- 1.103** So, can I be pro-progress without believing that progress is unavoidable? And I think I can. I think that you can be probabilistic. I think that you can embrace the contingent nature of progress. I think that you can do so, and that you're most likely to succeed if

you preserve the institutions, such as error correction, that make progress and humanism compatible for the future.

- 1.104** I've been fortunate enough to travel to countries that are not as free as America. How do you even start to convince them that this might be the way, when there's greed, when women don't have as many rights, all sorts of things, power? How do you go about, what does Cato do? Do you go to different countries? How do you get through to the minds of people, to convince them that human progress is the best outcome?
- 1.105** Well, as we will see, actually, in Lecture 3, I believe, or maybe in Lecture 4, people in countries where there has been a lot of progress happening, mostly Third World countries, or developing nations, as we now call them, tend to be much more optimistic about the future than people in the West, precisely because they have seen so much change happening in the last 20 or 40 or 50 years.
- 1.106** So, if you are a 50-year-old man or a 60-year-old man in China, you would have seen your country rise up from the very bottom of humanity, maybe \$300 or \$500 a year, to a country where the income per capita is maybe \$15,000 a year. So you've seen tremendous amount of progress, from the depths of communism to a highly successful mixed economy society. Same goes for India. People tend to be very optimistic because they have seen decades of very high economic growth rates.
- 1.107** So it's not actually very difficult to convince people that life can be better in poor countries, because they can either see successful poor countries or they can see the United States and say, "Hey, I want to be like that." The problem is actually convincing people like us that life can be better, because we are already at the pinnacle of economic success. I know that people like to complain in the United States about how difficult life is; but in terms of large countries, we are by far the richest. There are small countries around the world which are even richer than the United States; but I think that we are suffering a little bit from a deficit of imagination, thinking about how much richer the country could be. So I hope that answered your question.
- 1.108** In a perfect world, I agree with you. I do think that it is easier to, in countries that have more freedom, India, perfect example. But—
- 1.109** Well, there are countries that are stuck, no question about it. I'd be much less optimistic about human progress if I didn't see in the last, say, since the Second World War, if we haven't seen huge success stories: Hong Kong, Taiwan, South Korea, Chile. They were all very, very poor countries, but today they are very rich within a couple of generations. So, we do see countries escaping from poverty into riches. It's obvious that human progress is possible because many countries have become incredibly prosperous.
- 1.110** But what we'll be talking about in the subsequent lectures, we'll be talking about the conditions which are required for growth and prosperity to happen, and those preconditions are not there for everyone. They could be, but they are not. South Korea

and North Korea is a very good example: same ethnicity, same culture, certainly when they separated— when they were separated forcibly, I should say—very similar geography, similar natural resources—North Korea was actually richer at the time of the separation of the two countries than South Korea. But today, South Korea is fabulously rich, and North Korea is dirt poor.

1.111 So, if it's not ethnicity or culture that holds you back, what is it? We'll be talking about the importance of institutions, for example. But the bottom line is that, if you are willing to follow the recipe—there is a recipe for human progress—and if you are willing to follow it, then there's no country that needs to remain poor forever. Thank you very much.

Lecture 2: Evidence of Progress

At a glance — The evidence of progress across major measures of human well-being.

- 2.1** In the first lecture, I tried to define what I mean by human progress, and I explained that it is human progress from a humanistic perspective, which means that it takes into account, or it focuses on, human flourishing. It asks questions like: is education improving? Are people living longer? Do they have more food? And if all of those things are true, then human progress is actually progressing.
- 2.2** In this lecture, we will actually look at the evidence. It is going to be a lecture full of charts from a reputable source. This is Our World in Data. It's an outfit that exists within Oxford University, run by Max Roser, and it is really a gold standard in terms of statistics that are being used by people who are trying to measure the human condition. Let us start. We'll be looking at economic growth. We'll be looking at income per capita and how it correlates with things like, for example, better education, more access to food, etc.
- 2.3** As I explained in the first lecture, human progress is not just about GDP, but GDP is very important. Gross domestic product, or income, is very important because the richer the society, the more good things follow. If you have more money, you can buy a better sewage system. If you have more money, you can buy better universities. If you have more money, you can get better hospitals, etc. We'll also be looking at income inequality. We'll be looking at poverty.
- 2.4** We'll be looking at social indicators of human well-being, which is to say, infant, child, and maternal mortality, diseases, vaccines, education and literacy, access to electricity, happiness, hunger, life expectancy, population, urbanization, water and sanitation. We'll be looking at democracy and human rights, terror, violence, homicides, weapons, and war. We'll also look at the environment, because even though human progress is human-centric, and I'll be the first to admit it, living in a clean environment and having access to beautiful nature is very much a part of having a good life. And finally, we will look at energy and space.
- 2.5** So let's start with wealth and GDP. The first extraordinary thing that you see, this is called the hockey stick of great enrichment, coined by my colleague at Cato, Deirdre McCloskey. Now, the important thing to understand is that you could take the horizontal line, the horizontal axis, all the way back to millions of years ago, and it would be just as flat—which is to say that, in terms of the wealth produced on the planet, we were poor, poor, poor for a very long time. Then about 200 years ago, what you see suddenly is that hockey stick, meaning the long shaft, straight shaft, changing into the upward blade. And we are at the top of that blade.
- 2.6** Today, global economy is roughly 161-162 trillion dollars, whereas before, you could measure it in terms of billions. So, very important to understand, the world has never

been richer. If you do a logarithmic scale, which is to say, a slightly different way of visualizing it, what you see is that there is actually some progress happening over the last 1,000 years. Nothing like what we see over the last 200 years, but it is not as though there haven't been episodes where economic growth hasn't improved somewhat. And in the last 1,000 years, we do see, for example, between the year 1000 and 1500, we do see the global economic output rising to about, from negligible amounts to about half a trillion. But once again, it is the economic growth in the last 200 years that we want to focus on, and we will do so later.

- 2.7** Here is how global economic output is differentiated by regions. As you can see, Asia and the Western world really account for most of the global economic output. Most of the wealth produced in the world is produced in Asia and in Western countries. By Western Europe, you all know what Western Europe is. Western offshoots—because we'll be talking a lot about Western offshoots—here we are talking about the United States and Canada and Australia and New Zealand and so forth. But the bottom line from this chart, you can see that most of the wealth is produced in Asia and in the West.
- 2.8** What I've talked about until now was global economic output. Now we are looking at GDP per capita. So you take the global economic output, you divide it by the number of people who live in the world or in individual countries, you get GDP per capita—global, sorry, gross domestic product per capita or, in other words, income. Western offshoots are in the lead, with United States, Canada, Australia, and New Zealand basically being the richest people in the world, followed by Western Europe, followed by Asia, then Latin America, all the way down to sub-Saharan Africa, which continues to be the world's poorest continent.
- 2.9** But once again, notice how extraordinarily recent the growth in GDP per capita is. Now, very often what you will see is that there is maybe just one country that we take back to year one. That's the country that we have data for. But the way that economic historians think about it is that, since everybody was poor before, if we can show that, I don't know, England or Italy had a GDP per capita of a few dollars per person per day in the year one, we can deduce from it that, in the rest of the world, the situation was very similar, because everybody was equally poor. That is why we sometimes have this one line going all the way back to year one, and the other line starting later because that's when the data kicks in. But we are simply making an estimate, a guesstimate, a guess, that everybody was basically poor in the pre-industrial era.
- 2.10** The last 200 years, suddenly a massive jump in income per capita. Here is just a more detailed view of the same charts beginning 200 years ago. You can see the massive gap that opens between Western Europe and Western offshoots and the rest of the world. That's essentially the essence of industrialization, that you have industrialization beginning in Western Europe and then spreading to North America, to Australia, and so forth. These countries just grow like crazy. Suddenly, they become much richer. The

countries and the regions and the continents that lag behind, that start industrializing late, there's a huge gap that opens, that then needs to shrink. That is why sub-Saharan Africa, for example, but even Latin America, is still very much behind.

- 2.11** Very often it is said, workers produce X amount of wealth, but they're not being compensated enough; that productivity, which is to say how much workers produce and their income, are somehow decoupled. But as you can see from this chart, there is actually a very high correlation between productivity and income. A different way to understand it is that the more productive a worker is, the more he's compensated, or she.
- 2.12** If you are an agricultural worker, a peasant, armed with just a sickle or a scythe, living in a developing country or a very poor country, you're going to be—maybe you're a Cambodian agricultural laborer. You don't have access to good technology, you're maybe producing a few dollars an hour, your productivity equals your income. But if you're in the United States, you may still be an agricultural laborer, but now you are sitting in an air-conditioned tractor or combine harvester, and you are producing hundreds of dollars rather than \$5 per hour of labor.
- 2.13** Here you can see how, basically, the more growth the economy experiences, the higher the GDP per capita. In other words, this chart just simply shows you the relationship between total economic output of a country and per-capita income. As you can see, on the horizontal line, you've got GDP per capita. On the vertical line, you've got the wealth of the country, and it's a positive relationship. The richer the country, the more stuff it produces, the higher the incomes.
- 2.14** I've explained to you that our GDP per capita is tied to our productivity. The worker who produces more stuff gets more compensation. The more productive you are as a laborer, the more you can produce for your company, the higher your reward. Why should you care? Why is GDP so important?
- 2.15** Well, because it's correlated with all sorts of good things in life. If you look at this particular chart, what you see is income per capita on the X axis, the horizontal line, and you can see life expectancy in terms of years on the Y axis. There's a positive correlation. The higher the income, the higher the life expectancy. Child mortality—there we see the relationship inverse. The higher the income on the X axis, the lower the child mortality on the Y axis. Maternal health, once again, positive correlation: the richer the country, the fewer mothers die at birth. Medical doctors: positive relationship; the richer the country, the more medical doctors.
- 2.16** Access to improved drinking water—there, the relationship is actually a little more complex, because what it shows you is that, actually, even people who are making very little money, even countries which are very poor, have surprisingly great access to clean drinking water. Nonetheless, it is still true that the positive relationship holds. Access to electricity: positive relationship. Tourist departures: again, a positive relationship, if a

little bit more messy, which is to say, the richer the country, the more people can afford to go and take a trip to Bali.

- 2.17** Annual working hours—this will come maybe as a surprise to some of you, or certainly to some of our viewers—the richer the country, the more productive the workers, the less they work, fewer hours of labor. The amount of labor that an American worker performs today is much lower than it was in the 1950s and much lower than it was in the 1850s. So, there's a relationship here that we'll be talking about a little bit later, but the relationship is clearly negative. The richer the country, the fewer hours people work. It is not true that we are being overworked.
- 2.18** Average years of schooling: positive relationship. Literacy rate: positive relationship. Learning outcomes— which is to say, test scores on math, reading, and science— positive relationship. And life satisfaction. Very often, you hear critics say things like, "OK, so maybe you are rich, but you are unhappy." Not true. When people actually talk about their life satisfaction, we have surveys all over the world; the richer the people, the more satisfied they are with their lives.
- 2.19** Human Development Index is organized by the United Nations, and it looks at education, life expectancy, and GDP per capita. Again, very highly correlated with GDP per capita. Basically, the richer you are, the higher you are on the index. Life expectancy: positive relationship with GDP per capita, a little messy, if you look at it, and that's because it is possible for even poor countries to experience very high gains in life expectancy. How would that happen?
- 2.20** Well, let's say that antibiotics is developed in the West by Fleming in 1928, and then it becomes widespread in production in the 1940s, and it becomes very, very cheap. You don't have to do the same discovery in Burkina Faso or in the Congo. At some point, these drugs become dirt cheap, and they can be exported or even given to you. And so, even poor countries can experience very high improvements in their life expectancies.
- 2.21** Nonetheless, it is still true that life expectancy and GDP per capita are highly correlated, because maybe cheap drugs solved your problem with malaria or bacterial infection. But if you end up with cancer, you still want to be in a super-rich country because you get access to the latest medical treatment, and that is why the relationship is still positive. Child mortality, GDP per capita, we've already looked at. Years of schooling— this is just a more detailed view than we looked at before.
- 2.22** Global hunger—not surprisingly, the richer the country, the fewer people who are hungry. Hidden hunger—now, hidden hunger is a little more involved measure. It really averages three deficiencies, which is to say, stunting, anemia, vitamin A deficiency. Here, even when it comes to what is called the hidden hunger index, we can still see that there is a negative correlation with GDP per capita: the fewer people, the richer the economy, fewer people go hungry.

- 2.23** Share of children in employment—in the next lecture, we'll be looking at long-term trends in employment. This is just a correlation with GDP per capita. As you can see, in countries which are very poor, many more children as a share of the population have to work than in countries which are very rich. Here is a slide showing you the share of children in employment. As you can see here, once again, we see a negative correlation: the richer the people, the fewer children have to work.
- 2.24** Literacy rate—once again, literacy rates increase with income. Medical doctors, we already talked about before, but this is a much more detailed view that shows you how richer societies can afford more doctors per 1,000 people. And median income or consumption—basically, what it says is that the richer the country and the richer the people, the more they consume. They have more money left over after taxes to be able to buy all the things that they desire.
- 2.25** And once again, here we have a detailed view of what happened to working hours in three Western nations. So, in the United States, back in 1870, you could expect to work over 3,000 hours a year. Today, we are down to about 1,800 or so, 1,800 hours a year. In France and Germany, in 1870, people worked well over 3,000 hours a year. Now, they are down to about 1,500. The richer the people, the less they work. Why? Because they don't have to.
- 2.26** I would like to ask you to pay attention to this particular slide because it shows you self-reported happiness and GDP per capita. You very often hear people say that just because you are wealthier, you are not necessarily happier, but that's not really true. What we see around the world is that people who are richer also tend to self-report greater happiness. The way to understand this chart is basically what happened in self-reported happiness between 1985 and 2022.
- 2.27** So, in all the cases where you have the arrow basically pointing upwards, what this shows you is, over the 37 years, all the countries that have moved from one income level to a higher income level have also experienced higher self-reported happiness. I think that's quite an important finding. Here is a different way of showing you the same thing. Whereas the previous chart showed you what happened over time, which is why you had these long lines, this is just a snapshot in time; it tells you how people felt about their life satisfaction back in 2024. What it shows you, once again, is that the richer countries self-reported higher life satisfaction.
- 2.28** Here is a different way of showing the relationship between happiness and income. The position of the arrow shows the average life satisfaction reported by the population of a country, and the slope of the arrow shows the gradient between the income of individuals and their reported life satisfaction within each country. It is not just that rich countries tend to be happier, but it is also that richer people within countries tend to be happier. If you are at this level of wealth and you are reporting this level of happiness, if you're increasing your level of richness, you are probably also going to report higher levels of happiness. This is the way to look at it.

- 2.29** We looked at life satisfaction. We looked at happiness. These are two slightly different measures. One is looking at a momentary feeling of whether you're happy or not. The other one is looking at how you feel about your overall life, more of a long-term measure of satisfaction. They both show the same thing, which is that richer people tend to be more happy and more satisfied with their lives. And once again, this relationship is being shown just in a different visual way, but once again, an incredibly powerful, positive relationship.
- 2.30** Countries with higher GDP per capita produce more electricity. Countries with higher GDP per capita also have greater access to electricity. Rich countries have very few people dying from being poisoned by unsafe water; in poor countries, that's not the case. Improved sanitation—that would be things like building sewers, for example, so that you don't have to live next to the refuse and stuff like that—so once again, the richer the country, the more of these projects you can build, the more of these infrastructure projects you can build.
- 2.31** Air pollution—here's a very interesting thing that tends to happen as you industrialize. It's called the environmental Kuznets curve, which is to say that when countries first start industrializing, they also experience degradation of the environment. When Britain started industrializing, everybody was burning coal, and the air became very, very filthy. You can see here that suspended particulate matter shot up after 1700. Then it reached its peak, and now it has collapsed to a point where London has some of the cleanest air in the world.
- 2.32** So, you first have to become rich, and you inflict some harm on the environment as well as on your fellow human beings. But at some point, you become very rich, and you're going to start using some of that wealth in order to limit environmental damage and eventually create a good, clean environment. Now, some people might say, "Why not simply start off by having a clean environment? Why not simply stick with a clean environment? We had the clean environment when we were hunter-gatherers."
- 2.33** Well, that is true, but we were also dirt poor. As you can see, a lot of babies and a lot of mothers died in childbirth and so on and so forth. Now we can have both; we can have a clean environment, and we can be rich. But we do go through that hump. The reason why I picked Delhi there is because India is basically undergoing the same process. India is going through a process where it is industrializing, modernizing. It's experiencing a lot of pollution, but that too will drop off, just like in China, it has dropped off.
- 2.34** Very often, when you hear about poverty in the United States, what we're really talking about is relative poverty. Very few people in America experience poverty in the way that people in Africa, say, experience poverty. When we talk about absolute poverty, we talk about somebody living in Burundi, for example. But in the United States, we have relative poverty, which is that, nationally, we simply decide what the poverty level should be. It's usually a multiple of the national, or maybe it's 30% of the average income, or something like that. If you fall below it, you are at a relative poverty level.

- 2.35** But what's important is that the richer the country, the higher those relative levels are. The richer the country, the more is expected that you'll be given in order not to be poor anymore. So relative poverty in the Democratic Republic of the Congo will be very similar to absolute poverty; but the relative poverty in, say, Brazil is still at a much lower level than what it would be in the United States.
- 2.36** Extreme poverty is different from relative poverty. That means living on \$1, \$2, or \$3 per person per day. Here you can obviously see how, when you get to a position like Switzerland or the United States, we don't really have absolute poverty. Everybody can be taken care of. Nobody has to work on \$2 or \$3 per person per day.
- 2.37** Another thing which is very important is that the richer the country, the better off the poorest people are. We talked about relative poverty. But if you just take the bottom decile, the bottom 10% of the population, and you ask: how much wealth do these people have, or how much income do they have? Well, if you are in a rich country like the United States, the bottom 10% will have wealth that people in Malawi or Niger or Burundi can only dream about. So even the poor people in rich countries are much better off.
- 2.38** Here we have income or consumption of the poorest 10% in 2025. Basically, as you can see, a poor person in Luxembourg, which just happens to be there, has income or consumption of about, I don't know what it is, \$50 per person per day; whereas by the time you get to the Democratic Republic of Congo, it may be \$1 or \$2 per person per day. These are the poorest people in developed countries.
- 2.39** Another aspect of human progress is moving from jobs that are dangerous, that are difficult, and that will destroy your body, into jobs that don't. And the move from agriculture is actually important. In other words, countries which are very poor tend to employ a lot of people in agriculture. People who are employed in services, which are the cushiest jobs that we currently have, are usually people who live in rich countries. So there, you can see how the share of the population which is involved in agriculture is actually declining with wealth.
- 2.40** That is actually a good thing. We want machines to do the tough agricultural jobs, dig ditches, and so forth. We want human beings to do the easier and less dangerous jobs. When you start industrializing, what tends to happen is that more and more people start working in the industry. But at some point, there is a drop-off, and the number of people working in manufacturing declines.
- 2.41** One of the very important conversations that we are having in the United States right now is what is happening to manufacturing employment, and it has been falling; it has been falling for decades. But what they don't tell you is that manufacturing employment is also shrinking in many other countries. It is even shrinking in China, and it's certainly shrinking in other Western nations. Why? Well, because people then go

into the service sector. You have a decrease in population working in the manufacturing sector, but increase in services.

- 2.42** And so, this is the United States. This is what happens to our distribution of jobs between 1839 and 2016. As you can see, basically, what used to be an agricultural nation, essentially, back in the early 1800s, over 40% of Americans were farmers. Now it's down to 1 or 2%. Industry has peaked, especially after the Second World War, but now it's in decline. But the services show you the inexorable progress of services, which are also the most well-compensated and the least dangerous jobs. I would argue that this is actually a sign of progress.
- 2.43** The richer the country, or the richer the citizens, the more energy they use. Very often in the West, because we are so rich and so concerned about the environment, energy use is very often seen as a bad thing. You are using too much energy. But energy is a very important component of human progress. If you live in a very hot place, you want to be able to switch on the air conditioner and cool it down. If you live in a very cold place, maybe Sweden or Iceland, you want to be able to warm your dwelling.
- 2.44** And consequently, what tends to happen is that there is a very high correlation between income per capita and energy use. The key is not to limit energy use. The key is to start producing energy in such a way that it's not destroying the planet. Maybe having more nuclear fission or nuclear fusion, maybe switching from coal to gas, which produces much less CO₂ into the atmosphere. But the point is that we want to have more machines making our lives better. We want to be able to cool or warm us whenever we need. That is what the people in Burundi or Somalia are dreaming about having, and we have it. That's also a sign of progress.
- 2.45** But of course, human progress does occasionally create some problems. I already talked about how we solve one problem, and then we often end up with another one. And so, energy use is very much part of human progress. But as a result of energy use, we also end with more CO₂ emissions in the atmosphere. The very important question that a lot of smart people are talking about is: how do we get away from contributing more CO₂ into the atmosphere? Can we have new energy sources, which will basically allow us to have as much energy without more CO₂?
- 2.46** Health care spending and GDP per capita: highly correlated. And public spending, social spending, as a share of GDP is very highly correlated. This is what's called Wagner's Law. And what it shows you is, basically, the richer the country, the more we spend on welfare programs—I would argue that up to a point where we cannot afford it. I mean, the United States is running trillions of dollars of annual deficits, and we have to pay a lot of money in terms of our debt obligations. Something like \$1 trillion that we raise from the taxpayers every year has to go to paying off our debt. And that's partly, or largely, driven by social welfare spending.

- 2.47** So this is a double-edged sword. What it shows you is that we can take very good care of our citizens when we are rich. We are giving them a lot of money in welfare support. So that's human progress. You take care of your fellow human beings, but you may reach a point where you're spending so much because you don't know how to stop, where you're actually going to bankrupt your country. And you have to have a sense of when to pull back.
- 2.48** Income inequality—here you have a chart of what is called the Gini coefficient. The Gini coefficient basically means that if you have a Gini coefficient of one, it means that one person owns all the wealth in the country. If the Gini coefficient is zero, it means that wealth is perfectly evenly distributed. Here you see the income distribution around the world in 1800 compared with 2015. And in every chart, you see a line called the poverty line, which is the international absolute poverty line. And what you see is that, in 1800, most of the world was before that poverty line. Most of the world was dirt poor.
- 2.49** Then by 1975, you have these two humps. It's like a camel with two humps. Most of the developed countries, the Western world, is far beyond that international poverty line, right? Whereas Asia and Africa are very much still on the left-hand side of the line. But by 2015, basically, most of the world is now beyond that poverty line. And what that really means is that global income distribution has very much shifted in the direction of greater prosperity. Most people have become prosperous, and that huge inequality, which was created by 1975, has basically disappeared.
- 2.50** And that is what we are seeing in this chart. This is a chart that comes from Branko Milanovic, who is a world authority on income inequality. What do we see here? We see here income inequality over the last 200 years or so. When the Western world starts industrializing, a massive gap opens between the West and the rest. The Gini coefficient basically says that, if one person or one country owns all the wealth in the world, then that will be at 1 or 100%, whereas if income around the world is equally distributed, that will be at zero.
- 2.51** Now, what we see here is a massive increase in global inequality after the West starts industrializing. A gap opens between the West and the rest. Then, after the new millennium, especially with the rise of Asia, countries like China and India, we see for the first time in 200 years a decrease in global inequality. Basically, developing countries, China and India especially, are growing at such a fast rate that, even though we are growing, they are growing at such a fast rate that the global income inequality between countries is decreasing.
- 2.52** Let's look at poverty. This is extreme poverty or absolute poverty around the world over the last 200 years. Basic rule of thumb, in the early 1800s, nine out of 10 people lived in absolute poverty. Today, one out of 10 people live in absolute poverty. A way to read this chart is that the green are the billions of people who did not live in absolute poverty. And back in 1820, it was a tiny sliver of the population, whereas now it's most

of the world. So these are absolute numbers. There are 8 billion people in the world, and this is how they are distributed. Tiny sliver in absolute poverty, most of them not.

- 2.53** This is the share of population living in extreme poverty. Once again, in 1820, roughly 80% of people lived in absolute poverty. Today, it's down to about 10% and vice versa. This is just a different way of looking at it. This one, going back to 1990, but it basically shows the same relationship. Whether you're looking at the share of the population or the number of people who are living in extreme poverty, they are collapsing.
- 2.54** Now, very often when I talk about decline in absolute poverty, somebody will say, "Oh, but you're only talking about people living on \$2 or \$3 per person per day. Everybody else is still suffering." No. Whether you take \$1 or \$3 per person per day, or \$3 and \$5 per person per day, or \$5 and \$7 per person per day, or \$10, or \$20, or \$30 per person per day, fewer and fewer people are living at that level. If you look at this particular chart, what you see here, no matter how you measure real poverty, be it on \$30 or \$3, fewer and fewer people are living at that level and are moving to the higher levels.
- 2.55** And that's what's important. It's not just, people are moving from \$3 and then they are all hanging around at \$4 per person per day. No, everybody is moving to higher and higher levels of income, adjusted for inflation. This is important. We always use adjusted for inflation because \$3 a hundred years ago meant something different than what it is now. So we always have to adjust for inflation. So, essentially, we got to a point in global economic development where absolute poverty is now mostly an African problem. Asia, Europe, Americas, even Latin America, very little absolute poverty to speak about. But of course, we do have a problem in sub-Saharan Africa.
- 2.56** Here is long-term development, broken down by regions. I showed you the global situation. Now, I'm showing you the situation in different regions. As you can see, basically, in the Western world, we stopped talking about absolute poverty somewhere in the 20th century, middle of the 20th century. Then we were caught up with by the Middle East, and now we are still waiting for sub-Saharan Africa, where absolute poverty is still a problem for about 40% of the population.
- 2.57** Let's talk about infant and child and maternal mortality. This chart goes back to the early 1950s and shows you infant mortality rates by regions. And as you can see, up to 14% of babies in South America—these are babies who are younger than one year, so infants are between ages of zero and one—13%, 14% of infants in South America would expect to die in infancy. Right now, it's down to about 2%.
- 2.58** The same story could be told for Europe and North America. Asia experienced a massive drop, and even Africa. Obviously, in Africa, it being a very poor continent, we still have far too many infants dying. But consider the slope. In a scope of 70 years, we have seen a massive decline in infant mortality all over the world. And the steeper the slope, the quicker that region is catching up with the rest.

- 2.59** By the way, in case you ever wonder, why do I start sometimes in 1990, why do I start in 1953, or why do I start sometimes in 1820, I always start where the data takes me. In other words, if the data starts in 1820, that's what I will show to you, because I want to show you the longest possible period. But sometimes that's impossible. Sometimes, for the world, for example, infant mortality rate, we only have data for the last 70 years, so that's what we use. Otherwise, we'd be showing you some other charts.
- 2.60** Child mortality rate, so children before the age of five—why do I have Sweden going to 1751? Because we have data for Sweden going to 1751. And because the rest of the world was also very poor in the 18th century, we can say that the situation in the rest of the world was just as bad, possibly worse, than it was in Sweden. So, because Sweden is such an organized country and they have this data, we can show you what the rest of the world looked like in a positive scenario. Most likely, the situation in other countries was much worse than it was in, what it was in Sweden.
- 2.61** But what does this tell you? It tells you that up to 50% on certain years, up to 50% of children before the age of five would die. Right now, in Sweden, child mortality rate is almost nonexistent, and the rest of the world is catching up with Sweden. Again, look at the slope: the more dramatic the slope, the faster the continent or the region is catching up with the rest.
- 2.62** Child mortality by income level by country is commonsensical. Basically, what it tells you is that, if you want to avoid child mortality, get rich. This is long-run history of child mortality going back all the way to hunter-gathering societies thousands of years ago. These are children before the age of 15. Now, whether you're looking at ancient Rome, whether you're looking at ancient Egypt, you're looking at ancient hunter-gathering societies thousands of years ago, child mortality before the age of 15 was about 50%. Two hundred years ago, we see this sudden massive drop, whereby today, child mortality rate before the age of 15 is about 4%, so from 50% to 4%, and in very developed Western nations, such as Iceland, Finland, Norway, et cetera, it's 0.3%.
- 2.63** So, an extraordinary level of progress. I'm hoping I'm winning you over to the idea that there is actually progress going on. Mothers died like flies giving birth. This is, once again, I'm introducing here, I believe it's Sweden. Well, let's just stick with Sweden and possibly even Finland. Once again, we have the data going back to 1751 because we have it. It shows you that, up to 1,400 mothers per 100,000 live births could die in childbirth. Now, once again, it's close to zero. You can see how incredibly dangerous being a mother meant, only 300 years ago.
- 2.64** Now, as I will show you a little bit later, life expectancy is increasing. And what we would expect to see when life expectancy increases is that cancer death rates should be rising. But in fact, the opposite is happening. Once you adjust for age, what we see is, actually, people dying due to cancer are decreasing in the United States. Once again, we are such a rich society. We have access to such excellent technologies that, even though our people keep on living longer and longer, and the longer you live, the higher

the percentage chance that you are going to develop a cancer, that actually people are dying due to that.

- 2.65 But cancer is not just decreasing. Cancer death rates are not just decreasing for old people, they are decreasing across the board; they are decreasing even for children. Those are very tough cancers to be able to cure, but we are managing to do exactly that. That's what that particular slide shows you.
- 2.66 HIV-AIDS obviously used to be a death sentence until the mid-1990s. Today, because of new drugs and much cheaper drugs, new infections with HIV-AIDS are basically minimal, and people no longer die from HIV-AIDS, so people just keep on living with it, which is what that chart shows you. The share of deaths caused by HIV-AIDS peaked about 3% of global deaths around 2002, 2003. Now, it's basically two-thirds less of an issue than it was just 20 years ago.
- 2.67 Deaths caused by diphtheria, tetanus, measles, tuberculosis, yellow fever, hepatitis, meningitis, all down, not for cervical cancer. We don't live in an age—I mean, human progress means progress, doesn't mean miracles, so there are occasional stories where things don't work out.
- 2.68 Smallpox, one of the very few diseases eradicated, completely eradicated. It doesn't exist except for some labs. I hope they keep it safe. But in the wild, smallpox has basically disappeared. Along with malaria, it's probably the biggest killer of humanity in history, completely eradicated. Speaking of malaria, we see a happy trend where fewer and fewer people are dying due to malaria. Polio used to be a scourge in the world. Famously, FDR had polio. Polio has been eliminated from rich countries, and it is minimal in less developed nations as well.
- 2.69 Tuberculosis, deaths by regions, once again, on the way down. And I thought this was interesting that, you see, because drugs are getting so much better and so much cheaper over time—you know, patents expire, competitors come in, they start producing generics, et cetera—we are able to then share that knowledge and share those medicines with people who are very poor. When that happens, you can even have very poor countries, let's say Ghana or Sierra Leone or Ethiopia, and they have lower tuberculosis death rates than England was at that level of economic development, because maybe they didn't have those drugs or maybe those drugs were very expensive.
- 2.70 So the point of this chart is to show that death rates are lower in poorest countries today than they were in richest countries in the past, because those richest countries in the past didn't have those kinds of technologies and those kinds of drugs. Often, there is a very strong relationship between GDP per capita and good things in life. However, once you have enough technological and medical dispersion of knowledge, it's possible that, even in very poor countries, you can still get very good health care outcomes, simply because of that knowledge being dispersed.

- 2.71** Share of children vaccinated with vaccines from diphtheria, hepatitis, measles, polio, rubella, virus, etc, all increasing. This is a very difficult chart to read, but it basically shows you vaccine-preventable diseases in the United States—diphtheria: 100% reduction, measles: 99.9% reduction, mumps: 97% reduction. You've got pertussis, you've got rubella, you've got smallpox, again, 100% reduction, tetanus, hepatitis, varicella. These are the sorts of things that kids would be suffering from and dying from. Today, we are able, due to vaccination and better medical knowledge, to basically reduce them sometimes by 100%, thereby reducing a tremendous amount of suffering amongst the children.
- 2.72** Education and literacy—in the early 1800s, 10% to 15% of the population literate. Today, 80% of the population literate. Whereas before, 90% of people couldn't read, today, 9% of people cannot read. That was a global picture. This gives you a sense of literacy increases in global regions and individual countries. You can see that the slope of increasing literacy has become much steeper in more recent years.
- 2.73** For example, it took a country like, let's see, it took Spain, for example, it took Spain, what is it, 500 years to get from maybe 5% literacy to 100% literacy. But if you live in Papua New Guinea, for example, or Egypt, it may have taken you maybe 100 years to get from 5% literacy to get to 70% literacy or so. So, it used to be much more difficult to get people to be literate in the old days. People were poorer. People lived in rural areas. They weren't concentrated in the cities. They didn't have access to schools. Maybe they didn't have access to books, or the books were too expensive. But because of all of those things progressing, you can now be in a situation where even very poor countries are becoming literate at a very fast click.
- 2.74** Obviously, the goal here is to get men and women to be literate at the same rate, equality and all that. What's important is that we still have work to do, where there are certain countries and certain regions where many more men will be literate than women. Women will be kept from education. However, this is mostly now limited to countries in, say, for example, the Middle East, such as Afghanistan and Africa, but in other places in the world, North America, Europe, and so forth, women and men are literate at the same rate or a very similar rate.
- 2.75** The average years of schooling are increasing all over the world. It doesn't matter whether you're a poor country or a rich country. People are simply spending more and more time at school. Basic education, again, we see a flip from 20% with basic education to 80% with basic education and vice versa.
- 2.76** The share of people using the Internet—how often do I hear stories of somebody traveling through sub-Saharan Africa, maybe being on a safari in Kenya, and the local cattle herder will pick up a smartphone and maybe transfer money or look for news and whatever else. So your cell phone, your smartphone, contains the access to all human knowledge, the access to the Internet. And that's becoming dirt cheap. And

even in sub-Saharan Africa, now people have access to, maybe 30% or 40% of people have access to the Internet, which is very, very good.

- 2.77** Nor is it true that somehow we have stopped intellectual development, that because we are so rich, we become lazy. Maybe we read less, but it certainly doesn't mean that we are publishing fewer books. The number of books published in the West today is higher than it's ever been. Books are still being published. People still read books. Maybe that's a good sign. I certainly think that's part of human progress.
- 2.78** Electricity—over the last 25 years or so, what we see is that, back in 1998, maybe one and a half billion people didn't have access to electricity. This is down a few hundred million. So more and more people have electricity, almost universal access. That's what we see here. When it comes to the world, I believe it is about 80-90% range. Sub-Saharan Africa is still lagging behind, but here you see the regional differentiation. Again, the richer the country, the more access to electricity.
- 2.79** Going back to the slide with hidden hunger—because words matter, I was curious if that was a new term, or if that is something that you use at Cato Institute, or because the meaning behind it of malnutrition. Basically, hidden hunger describes it so much more deeply, as to what the issue is, and so I was curious about that moniker for it.
- 2.80** I think the answer to your question is, first of all, these are usually international agencies such as the World Bank and the Eurostat, which track down, maybe even OECD, that track down these kinds of numbers. And then different scholars can basically coalesce them into one index, which is called the hidden hunger index. But I guess hidden hunger is an interesting concept because it's not simply about people being hungry in the moment. It may also be, as indeed the definition says, it may be something like a vitamin A deficiency, which you may not see in the person.
- 2.81** Or maybe it is stunting. So you can come across a child, and you ask them, "Are you hungry? Have you had your meals today?" And the child will tell you, "Yes, I've had my meals. I'm not hungry." But then you compare his or her size to the peer group, maybe in other countries. And then you do a blood test and look for vitamin A deficiency, and suddenly you find that actually this child, in spite of having his stomach full, is actually not doing very well.
- 2.82** But this in itself is, of course, a sign of progress, is that we've stopped— we still worry about hunger, but we are now expanding the definition of hunger so that we don't miss out on some important components of hunger that we didn't care very much before, because back then we simply cared about: have people been fed? But now we also start caring about appropriate nutrition. Did they get enough vitamin A, etc.? This is what I would say is a typical process that we see within the human progress studies, is that we start redefining problems to become more nuanced, sometimes more encompassing. We'll talk about that in the subsequent lectures.

Lecture 3: Further Evidence of Progress

At a glance — Further evidence of progress, deepening the empirical case.

- 3.1** In the first lecture, we talked about what is human progress, and I've given you my definition, by no means uncontested, that human progress is a humanistic concept that is all about human flourishing. Then in the second lecture, we looked at some of the most important indicators of human well-being, including happiness, for example, but also child and infant mortality, maternal mortality, and so forth. We have seen how important income is, because it correlates very highly with those indicators of human flourishing.
- 3.2** What I also tried to do in the second lecture is to give you a long-term perspective, not just snapshots in time, showing you where, for example, incomes or poverty, were in 2024, 2025, but tried to show you a long-term development, what life was back until 200 or 300 years ago, and then how life has improved over the last two or three centuries.
- 3.3** In this lecture, what I want to do is to finish off with the evidence. In other words, before we can talk about causes of human progress, and after I've given you the definition of human progress, we need to be certain that we have evidence for human progress. We are all rational, empirically based human beings. We want to know that somebody is not bamboozling us. We want to know that, in reality, we can prove that human progress is happening. In this lecture—this will be the second and last lecture on the evidence for human progress—what I want to do is to start by looking at things like, for example, hunger.
- 3.4** Once again, you will see a long line going back to the 13th century. This is the United Kingdom. I will repeat it one more time—hopefully for the last time, and I'm sure that all of you understand me by now, but maybe people who'll be watching this course later— why do we do that? We do that because we always want to bring in, if we can, a country with the longest data as we possibly can. From that, we can deduce, because everybody was poor back then, that other people would have suffered from similar maladies or similar problems, that standards of living would be very similar to those countries, like, for example, the UK in the 13th century.
- 3.5** Now, when we talk about hunger, you will very often hear about 2,000 calories per person per day. So if you go to the USDA, the United States Department of Agriculture, they say an optimal diet should be about 2,000 calories per person per day. Now, that is true, but a little bit misleading. A child needs very different amount of calories per person per day than a sedentary woman or a heavily physically active man.
- 3.6** What 2,000 calories per day means is that, if a country can produce or import 2,000 calories' worth of food per person per day on average, and those calories are appropriately distributed, then everybody should have enough calories for the day. So

the 2,000 calories per person per day is a bit of a rule of thumb. But it is kind of important, because if you have a society, a country or a region which can only produce or buy many fewer calories than 2,000 per person per day on average, then you are probably going to have severe food shortages or even famines. And famines used to be widespread throughout the history of our species.

- 3.7** There were centuries, like in 17th-century France, where there were eight famines, where a region simply ran out of food, and millions of people had died. So what you want to do is to live in a country, basically, where the country produces or imports, is able to buy at least 2,000 calories per person per day. They don't have to produce it themselves. Hong Kong doesn't have vast agricultural enterprises, but it can import stuff. It produces other stuff that it can sell, maybe financial services, and earns enough money to be able to buy, I don't know, mutton from New Zealand and butter from Holland and whatever else.
- 3.8** Now, the reason why I talk about it that way is because there are a lot of countries which didn't have 2,000 calories per person per day for a very long time. France, for example, around 1700, is still able to either produce or buy roughly 1,500 calories per person per day; and the 17th century and 18th century are very hungry periods in the history of France. Indeed, the French Revolution is, at least in part, produced because there are food shortages, and bread suddenly becomes much more expensive. And it is only really in the early 19th century that France, for the first time, is able to produce enough food to be able to avoid famines.
- 3.9** UK, going back all the way to the 13th century, there you can see that UK, in some ways, had a better diet, or maybe I should say, access to more calories than France for a very long time. But even there, access to calories was moving between 2 and 2.5 [thousand] calories per person per day. Today, in Western developed countries, we have access to roughly 3,500 calories per person per day on average, which means that we are also suffering from the problem that was created by the elimination of a problem. We had a huge problem with hunger, then we had a problem with obesity, and now we have to tackle that problem.
- 3.10** But as you can see, over time, the situation has improved. Africa, which is the world's poorest continent, now has access to 2,500 calories per person per day. Who would have thought that? When I was growing up, in the 1980s, the TV was filled with images of African children starving, people dying by tens of thousands. And we thought this was the future, that this was what the situation was going to persist. Well, it didn't. So even in poor nations today, we have much more food than we used to, even in the rich countries.
- 3.11** Famines have disappeared from the world outside of war zones. Yes, when you have war zones, wars, man-created wars, you can still get famines. But the sorts of problems that humanity was used to for thousands of years have basically disappeared. This is the total daily supply of calories per person between 1961 and 2022. When I give you

these long-run statistics, sometimes you have to zero in and see what's been happening in different regions.

- 3.12** Here, over the past 60 years, you can see how the situation has improved in different parts of the world. North America in the lead—maybe that is why we have an obesity problem. Africa at the bottom, but even the situation in Africa is much better than it used to be. And I already talked about famines. They have pretty much disappeared. Look how many people— this is the death rate from famines per decade. So, death rate, how many people per 100,000 would die in famines—as you can see, 1870, bad, 1940, bad, 1960, bad. But by the time you get to modernity, famines are pretty much a problem taken care of.
- 3.13** These are regions, as you can see in the 1960s, for example, the biggest problem with famines was in Asia. 1960, that's basically communism in China. That shows you that problem. I couldn't help myself but show you famines by political regime, because you can see how extraordinarily devastating communism was, when it came to hunger and famine, basically, be it in the 1950s or in the 1960s, even in the 1930s and the 1920s. Communism, China, Russia, those were the great killers of humanity.
- 3.14** Share of people who are undernourished—so this is again working a little bit on that theme of hunger and hidden hunger—you can see how that's been declining. Now, after COVID, we had a spike in undernourishment, partly because a lot of people lost their jobs during the COVID pandemic. Many economies have suffered great shocks. They were shut down, they weren't growing. So a lot of people have fallen back into poverty.
- 3.15** And you can see how some of the progress which Africa has experienced in the early part of the millennium has been eroded by the COVID pandemic. But there is absolutely no reason to think that it will continue to rise. In fact, we would expect it to decrease, but 2023 is the best number I could get. It's about two years out of date. We'll get new numbers. I would bet that it's going to look much better.
- 3.16** Share of children who are stunted. There you can see, again, a decrease since the beginning of the millennium. Life expectancy. Here we have a statistic going back to the 16th century from the United Kingdom. Here you can see how life expectancy would fluctuate between 20 years and 40 years. That used to be the case even in the world's most developed countries until the late 1800s, early 1900s. In 1900, in the richest parts of the world, Europe, Western Europe, and the United States, life expectancy was about 50 years. Today, globally, it is 73 years.
- 3.17** And then, after about 1900, you see this massive takeoff in the Western nations, whereby in much of the West, you can expect to live to about 80 now. Life expectancy in the United States fell a little bit during COVID, but now it's about 79, 80 years, and we can see similar processes happening elsewhere. And the gap which was opened by industrialization, where the West took off, and the rest of the world stayed behind, that

is also shrinking slowly, with Africa and Asia seeing massive improvements over the last 100 years or so.

- 3.18** So, to put it slightly differently, it's not just that human progress has given us life: 50% of children under 15 are no longer dying, mothers are not dying at a very high level in maternity and when giving birth. Human progress has given us not just life; it has given us an extra life. We no longer live to 35. We live to 73 or 80.
- 3.19** Annual working hours per worker. In the second lecture, I gave you a snapshot in time. I showed you what the working hours are in the developed world. Here, I'm showing you what happened over long periods of time by region and by country. And what you can see is that, essentially, in a place like Germany, for example, you could expect to work to about— no, you could expect to work about 3,400 hours in 1870, and now they are down to about 1,500. That myth about hard-working German workers probably doesn't apply anymore as much as it used to in the past, but it's a relationship that we see with economic development.
- 3.20** Also, the number of holidays, vacations, days of work, paid leave, have also increased drastically. There's a famous clip in *Downton Abbey*, where the Duchess asks, "What's a weekend?" Because she doesn't know what a weekend is. And indeed, for most of our history, people maybe took the Sabbath off, you know, Sunday. But nowadays, not only do we have weekends off, but there are plenty of federal holidays and state holidays and religious holidays and whatever else, and that too, seems to me, shows human progress.
- 3.21** Data on child labor are difficult to come by. There is a myth that started in the 19th century, when Charles Dickens was talking about chimney sweeps and things like that, that children were mistreated because they had to work for a living. In reality, children always had to work for a living. Childhood, the idea of a childhood, is something that is very recent. In ancient Rome, the moment that you could contribute to the domestic economy by making fire or bringing water or whatever else, you were put to work. In the agricultural times, all the way up till the 18th or 19th centuries, children on farms would have to contribute something, be it going to look for wood or milking the cows or whatever else.
- 3.22** We were producing so few calories that every child would have to be—every pair of hands would have to contribute something. Children always worked. To give you a sense of how bad the situation was, you can look at Italy. If you look at Italy, what you see is that about 65% of Italian children could expect to work in the late 1800s. Our statistics stop in 1960, when it basically reaches almost zero. The situation in the United States and in England was much better because those were richer countries. And so, by the 19th century, fewer children had to work. But globally today, roughly 10% of children still work. 300 years ago, it would have been 100%. Unless you were a member of nobility, you were working.

- 3.23** Human Development Index basically takes into account life expectancy, education, and income per capita. We already discussed this in the second lecture. Here you can see the increase in Human Development Index as measured by the United Nations across all different regions, and it's measured from zero to one. Zero, you are miserable. One, you are excellent. Everywhere, we see upward trends.
- 3.24** Population. The extraordinary thing about the world is just how many people it has, and yet it can feed, and it can provide with more education, and it can provide with higher standards of living, and higher GDP per capita and fewer working hours and so forth. There's a view that some people have, we call them Malthusians, and we'll discuss them in subsequent lectures, that the more people you have, the more difficult life must get. Think about it like if you're buying a pizza, and it's just you and your boyfriend, and you know that each of you is going to get three slices. If suddenly another couple comes to dinner, suddenly it's many fewer slices per person. And this view, the Malthusian view, is, the more people you have, the worse life must become.
- 3.25** Well, look at this chart. Look how many people there are in the world today as opposed to in the past. There are 8 billion of us. There were maybe 300 million people in the world in the year one, the birth of Jesus or the emperorship of Augustus. 300 million people. Now we have 8 billion of us, and we are much more prosperous. So clearly, the Malthusian vision, which is that more people must mean more misery, doesn't work. There is something else going on. In the subsequent lectures, I'll be talking about humans as not just an empty stomach, but also a pair of hands and a brain that is capable of creating innovations which make life better for everybody else.
- 3.26** Urbanization. This is a very important thing that we are witnessing. Most people were always agriculturalists. They lived in rural areas. There were very few people in the cities. But already, over half of human population lives in the cities, and by 2100, it is expected that 80% of humanity will be living in the cities. Why is this important? Well, it's important for a number of reasons. One is that humans who live in cities use much less energy and many fewer resources than people who live on farms. And that means that we can be more environmentally conscious.
- 3.27** But more importantly, people who depart land, who no longer live off land, who no longer live on farms and in agricultural areas, are also not doing damage to those agricultural areas. When you leave the agricultural area and move to the city, nature can reconstitute itself. Flora and fauna can bounce back. And before you know it, try to leave your house for two or three years, come back, and you'll see what nature does to it, right? So nature will bounce back if we can just move to the cities; we can leave more and more land to nature.
- 3.28** Here's a share of population of people using safe drinking water. Once again, we see tremendous improvements around the world. Sanitation—this is basically not to be too crude, but you don't want to mix humans and excrement. It leads to all sorts of bad things. And so, having proper sewage systems, being able to have proper toilets and

plumbing, is very important for all sorts of reasons, including infections and so forth. So improved sanitation facilities are actually very important.

- 3.29 And here we are seeing tremendous progress, especially in lower-middle-income countries, countries which are very poor, but they are growing. Unfortunately, the situation in the low-income countries, the poorest of the poor, is not very good. So if you go to Botswana, which is a middle-income country, you can see basically Western-style sanitation facilities. But if you go to a place like Burundi, which is also in Africa, but it's much poorer, there, they haven't been able to generate enough capital to create this kind of sewage system and piping and so forth.
- 3.30 Countries that are democracies and countries that are autocracies—the picture there is mixed, but it's still, in general, positive. If you look in the late 18th century or even the middle of the 19th century, very, very few countries actually were democratic. Most countries were autocracies, essentially. Now, here we have a chart showing you autocracies, electoral autocracies, electoral democracies, and liberal democracies.
- 3.31 My preferred way of looking at human progress is who is living in a liberal democracy, a democracy which has certain, shall we say, protections for minority rights; the majority cannot simply decide one day to confiscate your house or commit you to an asylum or kill you or whatever else.
- 3.32 We see tremendous progress from 1850 to about 2010, 2013. Liberal democracy has been in decline. There is no question about it. But the situation is still much better than it has been, let's say, 20 or 30 years ago. So liberal democracies have been giving a way to electoral democracies, which means that there are fewer rights that are being observed, fewer rights for the minorities which are being observed by the majorities. So fundamentally, what this chart shows you is that roughly half of the countries in the world are either electoral or liberal democracies.
- 3.33 But of course, countries have different populations. So if you look at this slide, this slide shows you how many people live in electoral or liberal democracies. Here you can see that roughly 2 billion people live in electoral or liberal democracies, but 6 billion people live in autocracies of some kind. So, we have a lot of democratic countries still, but we also have a lot of people who are not living in democracies, because different countries have different populations, sizes.
- 3.34 Countries with universal voting rights—as you can see, universal voting rights for men and women is a very new thing. I believe that New Zealand was the first country where women could vote, and that starts in about 1897 or 1893, I can't quite remember. But since then, basically, women voting and male voting has been universalized, and men and women can vote together almost everywhere. It didn't used to be the case.
- 3.35 Human rights index, which is basically an expert estimate of the extent to which people are free from government torture, political killings, forced labor, have private property rights, and enjoy freedom of movement, religion, expression, and association, is

measured by a very good organization called the V-DEM. Here you can see that, even though we have seen slight decline in human rights index starting about 10 or 15 years ago, it is still pretty high compared to, let's say, 1789, where very few people in the world had those kinds of rights.

- 3.36** This slide shows you countries by share of women in a parliament. In the bottom right-hand corner, you can see countries where women have 50% of seats or more. But then, you can also see that you have many more countries where women have between 40% and 50% of the seats in parliament, 30% to 40%, and so on and so forth. The bottom line is that women now form an ever-larger share of parliaments around the world, and you can see it by the increasing slopes on that particular chart.
- 3.37** Number of countries with key economic and social rights for women, so whether it is paid leave for mothers, or employment discrimination prohibited, or domestic violence being prohibited, you can see that basically all over the world, women's rights are being equalized with men. Once again, I think this is a very good example of human progress. The number of countries giving or protecting, in some ways, rights of the LGBTQ, gays, lesbians, and bisexuals, is once again increasing. It used to be that same-sex acts were illegal in many countries. Now, more and more countries, at the very least, permit same-sex acts between adults. But countries with same-sex marriage, being able to adopt children, and so forth, are also increasing.
- 3.38** Homicides. I may have mentioned already, if I didn't, I apologize, that at the end of the 14th century, beginning of the 15th century, 75 Italians could expect to be murdered out of 100,000 in their lifetimes. 75 Italians out of 100,000 could expect to be murdered in their lifetimes. The last figure for Italy that I saw was 0.3. Now, in the United States, our murder rates in some regions are elevated. New England murder rates are very similar to Europe; they're higher in the South, and so forth. But they are still nothing—well, they're not nothing, but they are still much lower compared to homicide rates as we experienced them in history.
- 3.39** We are so parochial. We are so myopic in terms of thinking about where we are in 2025. We read the newspapers and we keep reading about how violence is getting worse and murder here and there, etc. And yes, it is true that violence has been on the uptick, homicide rates, but it's nothing compared to the past. And I think that having that historical perspective, understanding that when homicide rates in the United States are 5 per 100,000, it's still high, by European standards, it's still high, but it's nothing compared to 40 per 100,000 in Germany or 75 per Italy. So it's important to have this perspective.
- 3.40** This is global homicide rates per 100,000 people. There, the situation is also very positive. We keep reading about the world getting more and more violent. But in reality, as you can see from this chart, homicide rates around the world are falling, not increasing. People are becoming gentler toward each other. So right now, the world

average is roughly where I think the United States is, at 5 per 100,000. We want to get it as low as possible, but let's keep perspective.

- 3.41** Terror—there, we obviously had a massive increase of deaths from terrorist attacks during the War of Terror, which has been decreasing in recent years. So there, the situation is much more mixed. We are coming out of it, but it was pretty bad for some time. These are deaths from terrorism around the world. So this has nothing to do with the United States. These are people dying in India and in Afghanistan and in Pakistan, due to terrorist outrages. Here are global incidents of terrorist hijackings and deaths. There, you can see long-term trends downward, except obviously for 2001, when 9/11 happened.
- 3.42** Let's look at wars. Wars, certainly since the fall of the Berlin Wall, have not given us a clear trend. These are actually not wars, these are conflicts. These are interstate conflicts, meaning conflicts between countries. These are conflicts intrastate, within state. These are conflicts also that include nonstate actors, maybe terrorist groups. And these are conflicts which involve one group of people discriminating against another, say, maybe a pogrom of some kind. And so, we are not talking about wars, looking at this slide; we are looking at the number of conflicts. And the number of conflicts, there is no obvious trend. It stayed pretty much the same.
- 3.43** However, if you look at wars, putting conflict aside, but if you are just looking at wars, then after the Second World War, we have been living through a relatively peaceful period. We have had many fewer, especially interstate wars, wars between states, and you can see it on this slide. But I think another way of looking at conflict and looking at wars is how many people die in those conflicts, and the death rate per 100,000 people has actually been decreasing.
- 3.44** So, we have fewer wars. We have as many conflicts, but conflicts are getting shorter, and they are killing fewer people. As a result of which you can still live in a world where you have a lot of conflict, but fewer people die, because there are so many more of us, but conflicts are getting less deadly. And that is what we are seeing on this slide after the end of the Second World War. And the interstate wars, you can all think about interstate wars. In 2024, you can see Ukraine versus Russia there. You can see intrastate wars. You can see things like the Syrian war, etc. But nonetheless, I think it is a good sign that fewer people are dying, even if we cannot say that there is less conflict in the world.
- 3.45** This is a slide showing us a peaceful versus a hostile relationship between states. So, basically, this slide calculates how many wars you could be in, given how many foreigners you are exposed to. If you are a country with two neighbors, like the United States, we have Mexico and Canada, we could potentially be in a war with two countries. If you're a country with eight neighbors, you could be potentially in war with eight neighbors, etc. And this slide tries to basically calculate how bad the conflict is

between you and your neighbor, as opposed to how bad it could be if you were in conflict with all the neighbors that you have.

3.46 What you can see here is that, basically, something like 18% of the world is in a security community. So, the EU, for example, in the EU, everybody has a lot of neighbors, but they are actually in a security community. They're in NATO, but they are also in the European Union. You also have warm peace, which I don't quite remember the definition of, but it seems like a good thing, so up to 20% of countries are in either a security community or warm peace.

3.47 One-fifth of humanity, is that a lot? Is that too little? Well, look at 1900. Almost everybody was at negative peace. They were at peace, but they were resentful of each other. They didn't like each other. So I guess that's the difference between negative peace and warm peace. In negative peace, you dislike each other. You are preparing to invade each other. You are building up your militaries. Whereas in warm peace, you say, "I just don't care. Yeah, go ahead. Live your life and let me live my own." So again, very positive picture there. Could be better, yes, but compared to 120 years ago, relationships between countries are much better than they used to be.

3.48 This chart shows you the number of nuclear weapons over time, or maybe I should be more specific, and this is nuclear warheads, because, of course, a missile can carry more than one warhead. And what is remarkable about this chart shows you that, in the mid-1980s, the world had roughly 70,000 nuclear warheads, whereas now it's down to about 10,000 nuclear warheads. The United States reached its peak nuclear warheads in the early 1960s at 30,000, and then the Soviet Union, Russia, in the mid-1980s, when it had about 40,000 nuclear weapons. But both countries, because of a variety of treaties that they have signed for denuclearization, are now sitting on about 4,000 nuclear weapons or nuclear warheads each. So we have fewer nuclear warheads. Could they still destroy the world? Probably. But I would say that we have a better chance of survival if they all go off than we did in the 1980s.

3.49 Agriculture—I think this is a very important chart because what it shows you is that, for an average human being to survive in the world for the last 2,000 years or so, he or she would have to have access to roughly 1.5 hectares of land from which to derive the food on which to live. Part of the reason why we can discern what living standards were like 10,000 years ago, how many people could survive in the world 10,000 years ago, is because we have a good sense of how much land people needed to produce food on which to live.

3.50 Now, look what happens in the last, well, certainly in the last century or so; the amount of land per person collapses. Why is that? We have now reached peak land in the world, roughly 10 years ago or so. What that means is that, even though the global population is still increasing, the total amount of land used for pasture, for animals and for crops is beginning to fall. So we are able to feed more and more people on less and less land. How is that possible?

- 3.51** It is possible because we are becoming so much more productive in terms of our agriculture. Back in the day, maybe you could produce five bushels of wheat per acre of land. Now you can produce 10 or 15 and whatever else. I don't have those numbers in my head, but this chart shows you that, if you needed an acre of land to produce one unit of crops back in 1961, by 2022, you needed one-third as much land to produce the same quantity of food. We have become simply much more productive. We have better seeds, we have genetically modified crops, we have better fertilizers, and as a result of that, we are able to produce more and more food, on less and less land.
- 3.52** Here is a slide showing you arable land use per person in different parts of the world. This, I think, is a very telling chart because it shows you the increase in population, but also the increase in cereal yield and the land use at the same time. So that basically explains what I was talking about a minute ago, that you can have an increased population, but because you have much more yield, your agriculture is much more productive, you can keep the land use constant and perhaps even decrease it over time.
- 3.53** Let's look at air pollution. Here, you can once again see that when it comes to a number of pollutants produced from agriculture, be it sulphur dioxide, nitrogen oxide, carbon dioxide, black carbon, organic carbon, they're all post-peak, meaning they are now in decline. We are still producing more and more food, but our use of these pollutants is actually decreasing, with the exception of ammonia, which is a fertilizer.
- 3.54** The death rate from air pollution, from indoor and outdoor pollution, is also decreasing. This is for the world. In a previous lecture, I showed you air pollution in India and Britain as a snapshot in time relative to GDP per capita. Here I'm showing you, essentially, the 200 years or 300 years of the Industrial Revolution. Once Britain starts industrializing, there's a tremendous increase in air pollution, and then eventually it starts dropping. This is what's called an environmental Kuznets curve. The richer the people get, the more they are willing to take care of their environments. We would expect similar things to happen in Beijing, in Delhi, and in many other places around the world.
- 3.55** Animals—when it comes to African elephants, for example, we have seen a spike, then a slight decline, but still much better than what it was 30 years ago. So there is some positive news here on the elephants. Wildlife is coming back in Europe. Ursula von der Leyen, who is the president of the European Commission, recently had her horse eaten by a wolf. And that happens because there is just so much wildlife that is being rewilded in Europe. We can see massive increases in beavers, bison, seals, goats, wolves, jackals, lynxes, deer, boars, etc. All of them are increasing in Europe, as Europe has become richer and more environmentally conscious. They are rewilding, essentially. The Europeans are also getting away from land, so there's more land left for the animals.

- 3.56** Seafood production—note that, in 1960, virtually all the seafood that people ate was caught in the wild. If you wanted a fish on your plate, you would expect that there was a fisherman in the morning, who went out on the coast of Maine and caught a fish or a lobster, and then put it on your plate. Today, less than 50% of fish eaten in restaurants around the world was caught in the wild. The rest is bred and grown specifically for the purpose of being eaten—on farms, in ponds, in whatever else.
- 3.57** And that means that we have a reason to hope that wild fisheries can bounce back; that the overfishing, which was a massive problem around the world and is still a problem in certain parts of the world, will stop; that the oceans will be able to recover, because most of the seafood that we'll be eating will be grown for that specific purpose, but no longer caught in the wild. So I think that's a very positive statistic over there. This is just a different way of saying that aquaculture is now a much more important source of fish and crustaceans than capture. And East Asia and the Pacific lead the way in aquaculture production, as you can see here from the regional, this aggregation of who actually grows these fishes.
- 3.58** Not everything is going great. We still estimate that roughly 30% or so of the world's fish stocks are being overexploited. That is not a sign of human progress. That is actually not a sign of progress at all. But there are ways, such as aquaculture, that hopefully can fix that.
- 3.59** Number of whales killed globally per decade. When I say that fish stocks are overstretched and that it's not a sign of progress, but that I'm hopeful that the future of fisheries can be positive in the long run, this is the kind of statistics that I have in mind. When humans put their minds to it, when humans decide that they care about a specific type of species, like we started caring about the whales, we can see dramatic improvements. These are number of whales killed globally per decade, and look, in the last 40 years, whattremendous progress we have made. So there is hope.
- 3.60** Forests—as you can see, you have a long-term trend here, and then we have a short-term trend here. This goes back a thousand years ago. You can see how, for example, in China, roughly 30% of the Chinese land mass was filled by forests, and by the time that you get to the late 20th century, that fell to 10%. But China, once again, has decided that that's not the way they want it to go. They started reforestation, and now they're almost to the levels they were in 800 years ago. Forests are good. We shouldn't fetishize forestry in the following sense: you do need places for people to live. You need spaces for cattle to be raised. There's no reason why the world should be covered 100% by forests.
- 3.61** Nonetheless, during the agricultural period and then during industrialization, when wood was used for almost everything, we really did a terrible job on the forests. But now we are making amends. So there are some places in the world where forestry is growing, or forest coverage is growing, and some places where it is decreasing.

Nonetheless, it is true that, globally, we are still losing forest coverage. Every year, a certain percentage of forests gets cut down, and new forests are being planted.

- 3.62** So what we are concerned about is what's called net change in forest area. We want to know that, after you take account for the forest that was cut down and the forest that was planted, is the world still losing forest coverage or not? And the truth is that the world is still losing forest coverage, unfortunately. However, there is a silver lining, which is to say that, from the 1990s, every decade, the world has been losing less acreage or fewer hectares of forest. So we lost about 80 million hectares, but we cut that in half, almost in half, so that by 2010, we lost only 47 million hectares due to deforestation. So, not a pretty picture, but we are on the right trajectory if we can keep it going.
- 3.63** Here are number of parties, meaning number of countries, which are signed up to different multilateral environmental agreements. So whether you are looking at the protection of the ozone layer, or United Nations Convention to combat desertification, or the Kyoto Protocol, or whatever else, basically, a lot more countries are taking environmental stewardship of the planet seriously.
- 3.64** Oil spills—technological progress, better construction of ships, for example, means that we are now exposed to many fewer oil spills than we did back in the 1970s. Look at this extraordinary decline. I think that if somebody told me in the 1980s that we would be experiencing so few oil spills from tankers in 2024, I wouldn't believe them, because that seemed like the only thing that people were talking about.
- 3.65** The share of plastics emitted to oceans is a problem. Here, I would simply draw your attention to the relationship, once again, between wealth and technological sophistication and environmental pollution. Basically, the more sophisticated and the richer the society is, the more sophisticated the economy is and the richer it is, people are able to prevent plastic, basically, from harming the environment. If you look at how much plastic escapes to the ocean from Europe and from North America, and then you compare it to, for example, what is happening in Asia, it is staggeringly different.
- 3.66** So, it is certainly hoped that, with time, in Asia, the environmental Kuznets curve can once again kick in, and Asians, as they become richer, will start taking better care of their environment and will flush fewer plastic bags into the ocean through their rivers and so forth. These are rivers contributing to plastic ocean pollution. Philippines, Malaysia, India, Philippines, Philippines, Philippines, Philippines. Apparently, the Philippines are very keen on flushing all of their plastic into the ocean.
- 3.67** However, there is a silver lining here, which is that we are recycling much more of the plastic than we used to. If you compare recycling of plastic in India, for example, between 2000 and 2019, it has risen from 5% to 13%, in China, from 5% to 13% as well. Even in Asia, what we are seeing is greater concern for plastic pollution and attempts to

do something about it. But certainly, it ought to give us pause. It ought to give us concern.

- 3.68** Energy—every dollar produced in the economy requires certain energy to go into it. The positive—let's say that you're making a car. It will take a lot of energy to create a car. Are you building a house? You're making this clicker. Everything requires energy. So the key is to use as little energy as possible in order to create a dollar of output. And if you look at this chart, what you see is that, all over the world, the amount of energy used in order to produce a dollar of output is decreasing, so becoming much more efficient. That doesn't mean that the total amount of energy used is decreasing, because the economy is growing. The economy is now \$160 trillion. So obviously, we are using a lot of energy, but per dollar of output, it is getting smaller.
- 3.69** And we've come to the finale of this particular lecture. And I have only one slide to share with you, and that is space. What's happening in space? The extraordinary flourishing of our ability to fly ships into space, powered by the United States, specifically by Elon Musk's SpaceX, and perhaps even Blue Origin at some point, the price of getting a kilogram or a pound of stuff into space has basically collapsed. We are able to send ships up and bring them back down, reuse them, revamp them, send them back up again. We have visionaries like Elon Musk precisely making that happen.
- 3.70** Why is that important? Well, why is that impactful for human progress? For a variety of reasons. There are a lot of research projects, experiments, for example, which cannot be conducted on Earth. They have to be conducted in a weightless environment. So being able to get a lab up to space for cheap is actually beneficial to humans. It is beneficial to humans that we can put up satellites that allow us to communicate with each other within seconds on the other side of the world.
- 3.71** It's good for humans because we want to put up telescopes that can tell us if there's a huge rock that is flying toward Earth and that we need to somehow bump off its trajectory, so it doesn't kill us like it killed the dinosaurs hundreds of millions of years ago when it splashed into the Gulf of Mexico, or the Gulf of America, depending on where you stand. So there are a lot of benefits to space becoming more accessible and becoming cheaper.
- 3.72** With that, I hope that I gave you an overview of the evidence for human progress, how it impacts our lives, our societies, the environment, even space. I hope that I convinced you, with data which comes from third parties, data which I haven't created, but come from a gold standard of international statistics, which is the Our World in Data outfit at Oxford University, I hope that I convinced you that we are longer living, perhaps not wiser, but certainly better educated, that we are vastly richer, we are better fed, that fewer children and mothers are dying, and those are all good things.
- 3.73** But it's not just that. It's not just the material progress that has happened. I would argue that the evidence shows conclusively that we are also nicer to one another. We

are nicer to women. I didn't have a slide showing you, for example, attitudes to race in the United States, to interracial marriage, where people, 90% of Americans, didn't like it. Now, everybody does. I hope that I showed you that we are nicer to gay and lesbian people, that we are nicer to our environment, that we are better stewards of it.

3.74 And whilst there's still a lot of environmental damage going on—I acknowledge that, and it troubles me—there's also a lot of reason for hope. If we can protect the whales, if you cannot open a newspaper every week without learning about a new initiative being launched around the world about how we are going to save this or that animal that we care about, it shows that we have agency, it shows that we have resources, it shows that we have commitment.

3.75 And so, no, the world is not without problems. The solution of the big problems, hunger and so forth, has created other problems. But I'm confident, or at least I think there is a very high probability— because I don't want to be histrionic about this, I don't want to fall into the presumption of inevitability that Karl Popper warns us against— but I think there is a very high probability that we'll be able to tackle the problems that remain in our future. Thank you very much.

Lecture 4: Progressophobia

At a glance — Progressophobia — why people resist or deny clear evidence of progress.

- 4.1 Welcome to Lecture 4. In lecture one, I tried to define human progress. I told you what I think that it means. And in Lectures 2 and 3, we looked at the evidence for human progress, where I tried to use the best available data to show you in how many different ways human flourishing continues to progress, that we are actually better off. But not a lot of people feel like that. There is actually a lot of negativity out there in the world. If you ask an average person, "Is the world getting better or not?", almost invariably, you get an answer that the world is actually getting worse. So, in spite of there being a lot of good news, we don't feel particularly good about the world in which we live.
- 4.2 Today, what I propose to do is to talk about progressophobia, why some people actually don't like to think about the world getting better. I'm also going to be talking about negativity biases, why, in spite of all the good news, we continue to focus on the negative. Finally, I will discuss our ignorance of the facts about the world and why that ignorance matters for the future.
- 4.3 So, let's start with progressophobia. Progressophobia basically means hating the idea of progress. Steven Pinker, when he wrote his book, "Enlightenment Now," and then eagerly awaited the reviews, many of the reviews were very negative. And he said, "What really surprised me is how many people who call themselves progressives actually hate the idea of progress." Why should people who call themselves progressives, for example, hate the idea of progress?
- 4.4 I think that part of the reason is the tendency amongst many intellectuals and commentators to reject or sneer at the idea that the world has gotten better in some important way. Optimism sounds naive. It sounds shallow or even immoral. To a lot of people, it sounds very American. "Things are getting better." "Oh, you must be some kind of an American." It's very shallow, and that sort of thing. If you want to be really deep, you want to be pessimistic. Somebody once said that, if you come and warn somebody about the world ending, you come across as caring. But if you are coming across to somebody and saying the world is getting better, it feels like you are trying to sell them something.
- 4.5 So serious thinkers, progressophobes say, should be focusing on oppression and injustice and catastrophes, not on feel-good stories about success. People who talk about progress, about the world is getting better, they are dismissed as Pollyannas, people who feel that everything is going to work out, or technophiles, people who believe that, if only we can find a new gadget, we can solve any problem that is out there.

- 4.6 But my point is that the discussion here should not be about optimism versus pessimism. It should be about accuracy versus inaccuracy. People always say, "Oh, Marian, you're so optimistic." I don't think I'm particularly optimistic. I am just a guy who is interested in facts, and my point is that the real state of the world is much better than most people realize. So, am I an optimist? No, I'm a realist, and my point is that, if you look at the real data, the world is better than it seems. And denying progress when it has actually occurred is as misleading as denying that there still are problems out there to be tackled and resolved. So, understanding what has gone right is crucial if we want to repeat those successes and extend those successes.
- 4.7 Why are so many people pessimistic? Well, partly it's because there are good political uses, or pessimism is politically useful. On the left, you may encounter arguments like capitalism and globalization is making everything worse. There is more inequality, there is more exploitation, there is more environmental destruction. So capitalism is bad. It's making the world worse. And if you say that there has been progress, the leftists believe, then you are weakening the case for radical change. You cannot admit progress because then you are taking the wind out of the sails of the left-wing populists.
- 4.8 But on the right, you can see on this slide, this is the horseshoe, where the far left and the far right actually, over time, get closer together. On the right, we also hear a lot of denial of progress. There is moral decay. The world is getting worse. The young people no longer respect the old. This has been the right-wing commentary for 2,500 years, at least. Immigrants, cultural liberalism, they are destroying traditional values, many people on the right believe. We need to make America great again, right? So, progress is actually making America worse. We have to go back to some sort of a golden era, like Hesiod, from Lecture 1—that may have been quite attractive to him.
- 4.9 So my point would be that both sides, both the left and the right, cherry-pick facts and ignore large improvements that do not fit in their narrative. Global poverty is less of a problem than it used to be. So in that sense, the left wing is wrong. But the world is also less violent. We have fewer murders, and in that sense, the right wing is also wrong. And both sides, the left and the right, can still criticize genuine problems out there, while acknowledging that overall things have improved.
- 4.10 Another problem as to why people fear progress or why they deny progress is because they feel that, if you admit progress, people will feel complacent. People may feel that, if you admit that, for example, women have more rights than they did 100 or 200 years ago, then we will stop fighting for those problems where anti-female discrimination, for example, still persists. I don't think it follows, but it's certainly a worry that these people have. So instead, you emphasize how bad things are. "No, nothing has improved for women in the last 200 years," because that way, you can still keep the forward momentum, even though it is actually unconnected to reality, as we have seen from our evidence for human progress in Lecture 2 and Lecture 3.

- 4.11 And I think that's counterproductive because social movements very often succeed precisely because they can show that they have achieved certain outcomes. Why would you support an organization or a movement which hasn't accomplished anything? Actually, I think it would be a better idea for them to emphasize how much they have succeeded; and recognizing that certain things have improved, such as, for example, women's rights, can inspire. It can inspire people to do more work in that arena. So, despair and doom is actually counterproductive because it may demotivate people from pursuing their goals. It may feed fatalism.
- 4.12 Negativity bias. Part of the reason why so many people feel so negatively about the world is because we have evolved negativity biases, and I'm going to be talking a little bit about them. First of all, is there a negativity bias? Well, on this slide, what you can see is, between 1995 and 2015, citizens of the European Union have been asked: when it comes to the expectations of the years to come, will the economic situation in your country be better or worse?
- 4.13 Now, if you look at the red line, it shows you that, actually, it is moving a lot. Sometimes people feel good about it, sometimes people feel bad about it; sometimes they feel good about it, and then bad about it. But what is so interesting is that when you ask them, "Do you think that, in the year to come, your personal economic situation is going to be better?", the numbers of people who say it's going to get worse are very low, and people who say it's going to stay the same are elevated. So that is the situation in the European Union. People expect their situation to remain the same, but they very often expect the situation in their country to become worse.
- 4.14 What about the situation in the United States? It's very similar. If you ask Americans about trends in life satisfaction, in their personal lives, "How satisfied are you in your personal life?", 80% of Americans between 1978 and 2024 were very satisfied with their own lives. But if you look at how they felt about the United States, there were times when only 10% of people in the United States felt that the country was going in the right direction. Now, this is mathematically impossible. You cannot have 80% of your citizenry saying, "I'm satisfied with my life," but 90% of Americans saying that the country is going to hell.
- 4.15 Let's now look at the UK. There, you can see that people, even people who feel that things are generally becoming worse, tend to feel that things are generally becoming worse more about their country than about their local area. So you ask, "Is immigration a big problem?" For the country: 80%. In my local area: 50%. You ask them about unemployment. In my local area, it's a 70% problem. But in the UK as a whole, 90% of the problem. Crime, in my local area, 50% say it's a problem, but 80% for the whole country.
- 4.16 We have this massive gap about how people feel about the countries in which they live, and then about how they feel about their own lives. This is called the optimism gap: my life versus the world. Most people feel optimistic about their own future, but

pessimistic about the country and the world. It seems to be saying to us that the broader society is in crisis, but individuals are somehow getting by. Why would that be?

- 4.17 Well, perhaps one of the arguments is that there's a perception error. People can estimate how well they are doing in their personal lives much better than estimating how the country or the world is doing. You know about your life much more than you know about the country and about the world. When you look at your life, "it's OK," people are telling us time and time again. But they see the world through the prism of disasters and crises that they see on TV, and they feel much worse about it because they have much less of a direct and concrete understanding of the world. They understand themselves, but they only understand the world through the prism of the media.
- 4.18 That's the gap that I'm trying to breach with my work. In your personal lives, you're doing all right. At least that's what Americans are telling us. But the world is going to hell. And what was Lecture 2 and 3 all about? It was about showing you that the world is actually improving; it's not just the United States.
- 4.19 So, where does this negativity bias come from? Well, our species, *Homo sapiens*, is about 300,000 years old. And for most of that time, change, let alone improvement, remained very, very slow. The dramatic rise in our standard of living is only perhaps 200 years or so. So we've been around for 300,000 years, but our prosperity is maybe 100, maybe 200 years old. So that accounts for maybe 0.08% of our existence as a species.
- 4.20 My point is that we have evolved in an incredibly dangerous environment. We are primed to look for the negative news. Why should human beings appreciate high standards of living when we have only lived in that environment for 0.08% of our time? And for 99.92% of our time, we lived amongst incredible drudgery and violence and poverty and so forth.
- 4.21 And in fact, Daniel Kahneman, a famous psychologist, tells us in his book, "Thinking, Fast and Slow," that natural selection favored threat prioritization. So we overweight on bad news. "Organisms that treat threats as more urgent than opportunities have a better chance to survive and to reproduce." When, 300,000 years ago, everything could eat you or kill you, it made sense to prioritize bad news. But since that prioritization is now part of us, it's part of our psyche, it's part of our genes, we still react to bad news in the ways that our ancestors would have, rather than in the way that would reflect our improved fortunes.
- 4.22 Peter Diamandis, in his book "Abundance" in 2012, introduces a very interesting idea, the role of the amygdala, a part of our brain, in this threat assessment. So let me read you what he has to say. "Because our brains have limited computing power, they must separate that which is paramount, which is important (possibly a poisonous spider, for example), from that which is mundane (such as a clear sky). Because survival is more important than any other consideration"— your organism tells you you have to survive,

so survival is more important than any other consideration—" most information is first sifted in the brain through the amygdala, that part of our brain which is responsible for primal emotions like rage, hate, and fear."

- 4.23** Information relating to those primal emotions gets our attention first because the amygdala is always on the lookout for something to fear. Here's how it works. Maybe you're walking through a forest, and suddenly you see a branch on the ground. That information first goes to your visual thalamus and then straight to the amygdala. And if it looks like it might be a snake, you are going to jump. Why? Because the amygdala sits right on top of your nervous system. So your organism, your evolution, even your hardware, sight, visual cortex, visual thalamus, amygdala, nervous system, tells you "jump."
- 4.24** It is only a little later that that information about the branch goes into the visual cortex, where information about the world is stored, and the branch gets assigned to a branch. It is only a fraction of a second later that your visual cortex matches memories of other branches to that on the floor, on the ground, and you realize, "Oh, my goodness, it is not a snake, it is a branch." But by that time, you've already jumped. So that's the role of amygdala. It's always on the lookout for something to fear.
- 4.25** This is even true in laboratory settings. If, for example, you specifically ask students to come in to the lab to appreciate only the good news, if then on a split-screen computer, you show them good news and bad news, eyes gravitate toward the bad news, because the system, your psychology, your evolutionary makeup, forces you to look at the negative news, even though you are deliberately there in that space, in that laboratory, with the desire to look at the good news.
- 4.26** News amplifies pessimism. News is about what happens. You never hear any reporter standing in the middle of a city which hasn't been bombed, saying, "I'm now reporting from a boring city where there is no catastrophe." You never hear a reporter reporting from an airplane which landed safely at the airport, saying, "Thank goodness, we've just arrived fine." No. Those things are never reported on. News is about dramatic things that happen. If it bleeds, it leads. Especially if they're gory, if they are terrible, then they're going to lead the news cycle.
- 4.27** So all the positive stories, that the fact that 99.9999% of humanity will go to bed tonight safely, and a vast majority of mankind is going to be well-fed and healthy, that's never going to get reported. It's only the bad news which gets reported. And I would actually argue that social media also increases the immediacy and the intimacy of bad news.
- 4.28** So, for example, I remember smartphones had gotten so good by 2011 that we were able to watch the Japanese tsunami in real time, basically, happening on our phones. It makes you feel vulnerable. It makes you feel like you are participating in this great catastrophe. So, through social media and through smartphones, you may feel that

something that happens on the other side of the world may potentially threaten you. You certainly develop an empathy and a feeling of endangerment that previously you wouldn't know about. If something happened 200 years ago on the other side of the globe, you would probably have never learned about it. But right now, no matter what happens anywhere in the world, you feel like it's impacting you because it is immediate and it is intimate.

- 4.29** Perhaps the most interesting thing about negativity biases and how the media interacts with the news is the fact that, in spite of all the things that have improved in the world over the last 100 or 200 years, the media reporting has been getting consistently gloomier. This chart, for example, shows you the tone of New York Times reporting, going back to 1945. In 1945, the Second World War had just ended. And all the way to 2010 or so, the tone of the reporting gets constantly darker and darker and darker.
- 4.30** I'm choosing New York Times not because of any kind of bias, because this is the statistic I had, but I promise you that if you looked at any other newspaper or TV reporting, you would find that things have gotten darker and darker and darker, in spite of the fact that absolute poverty, for example, has fallen from 40% in 1980 to less than 10% today; in spite of the fact that more people have access to electricity, and fewer babies are dying, and fewer mothers are dying in childbirth, our reporting just gets worse and worse and worse and worse.
- 4.31** So, because we have evolved to look out for the negative, newspapers, TV, radio, millions of websites that report on the news, they have an incentive to produce news which is as dark as humanly possible. In fact, there is some evidence saying that, if you can add one negative word into the headline or the title of an article, 2.5% more people are likely to click on that article, precisely because it has that negative connotation. So we have hyperactive media, TV, radio, newspapers, websites.
- 4.32** We have this negativity bias in our brains, in our psychology. Is there any surprise, therefore, that the media coverage is becoming darker and darker and darker? If it bleeds, it leads. If you want to make money, if you want to keep those eyeballs glued to the screen, give them as bad news as humanly possible. So, we have these negativity biases. And the flow of apocalyptic warnings is continuous.
- 4.33** My friend Matt Ridley, who wrote an important book, "The Rational Optimist" in 2010, one of the passages I like to quote, he talks about how the different warnings about how the future was going to end badly for humanity, how it has changed over time. He writes, Matt Ridley writes, "In my own adult lifetime, I have listened to the implacable predictions of growing poverty, coming famines, expanding deserts, imminent plagues, impending water wars, inevitable oil exhaustion, mineral shortages, falling sperm counts, thinning ozone, acidifying rain, nuclear winters, mad-cow epidemics, Y2K computer bugs, killer bees, sex-change fish, global warming, ocean acidification, and even asteroid impacts that would presently bring this happy interlude to a terrible end. I cannot recall a time when one or other of these scares was not solemnly espoused by

sober, distinguished, and serious elites, and hysterically echoed by the media." All of these warnings are constant. They change all the time, but they are constantly reported on, because if it bleeds, it leads. If you can freak out people, you do it.

- 4.34 Another part of negativity bias is what is called an availability heuristic. Availability heuristic is just a fancy name for saying that vivid, traumatic events seem more frequent than they really are. We recall those planes flying into those New York skyscrapers to this day because it was terribly, terribly vivid and terribly traumatic for the American people and for much of the Western world.
- 4.35 But in the years since 9/11, Islamic terrorists have managed to kill about seven people a year in the United States, in contrast with lightning, which kills about 46 people a year, or accidents caused by deer, which is another 150, or drowning in bathtubs, which is 300. But none of us ever thinks that bathtubs are more dangerous than terrorism, because a terrorist attack is something that lives vividly in our memory. That's the availability heuristic: our mind pulls out of the memory file in our brain the stories which are most impactful, most traumatic, rather than those which are not.
- 4.36 Bad is stronger than good. Bad is stronger than good. Let's say that you have an interview with your boss, and he or she tells you, "You had a great year, and 99% of the comments on your work were absolutely positive." But he or she will include that one thing that you messed up on. I guarantee you that, when you walk out of that office, you will remember that mess-up, that one single thing where you screwed up, rather than the 99 things that you got right. Why? Because bad is always stronger than good.
- 4.37 We fear losses more than we delight in gains. So we continue to remember setbacks more than we relish successes. We resent criticism more than we are encouraged by praise. Positive changes are slow, while negative shocks happen quickly. There is a basic asymmetry in life between the positive, which is difficult and takes time, and the negative, which is much easier and takes less time.
- 4.38 Compare the amount of time you need to bring up and socialize an adult person from childhood to adulthood, compared to the amount of time needed in order to kill that adult in an accident, like a car crash, for example. Compare the amount of time it takes to build a home versus the time that it requires to destroy it. So positive changes are gradual, incremental, they take a long time, but negative things happen very quickly.
- 4.39 Another part of the negativity bias is what is called the prevalence-induced concept change in human judgment. This is just a very, very long and difficult phrase to say that, when we do a good job at resolving a problem, that becomes our baseline, and we judge new instances of that problem based on that new baseline. Daniel Gilbert, who came up with this concept, says that, "as we reduce the prevalence of a problem, such as discrimination, for example, we judge each new behavior in the improved context that we have created."

- 4.40 We don't judge discrimination in 2025 relative to 1968 or 1868. Our baseline is 2025. Another way to say this is that solving a problem causes us to expand a definition of a problem. Racism and racist discrimination was a serious problem, but now we don't complain about that; we complain about, because we've done a pretty good job at tackling, now we complain about microaggressions. So when problems become rare, we count more things as problems.
- 4.41 And Gilbert and his colleagues said, "Our studies suggest that when the world gets better, we become harsher critics of it. And this can cause us to mistakenly conclude that the world has actually not gotten better. Progress tends to mask itself." Whenever we succeed, that becomes the new baseline. We forget all about the much worse stuff that has happened and so forth.
- 4.42 Another thing that happens is called reminiscence bump. Adults recall disproportionate, vivid memories from adolescence and early adulthood. As we grow older, we start recalling more and more stuff from our childhood. And because these early years contain very emotionally significant firsts, first kiss, for example, etc., these are identity-forming milestones. They are encoded very strongly in our brains. So, as we go through life, this reminiscence bump is making our past seem happier, richer, more meaningful.
- 4.43 Finally, when it comes to negativity bias, we tend to underestimate human ability to adapt and to change the trajectory, the negative trajectory, on which we are. Human adaptation is very important. Morgan Housel likes to say that, if we extrapolate college tuition fees, they become prohibitively expensive in 10 years. If we extrapolate government deficits, we'll be bankrupt in 30 years. All of these problems may be causes of pessimism if you assume no future change or adaptation, which is crazy, given our long history of changing and adapting.
- 4.44 So, very often you find people who will say things like, "Oh, yes, in the past, we were able to solve problems, but not now. Somehow this generation is special. We are at a turning point in history. We were able to do all those things in the past, but moving forward, we won't be able to address the problems that remain." Which brings me to Thomas Babington Macaulay, an English historian who wrote in 1830, "We cannot absolutely prove that those are in error who say society has reached a turning point, that we have seen our best days. But so said all those who came before us, and with just as much apparent reason. On what principle is it that, with nothing but improvement behind us, we are to expect nothing but deterioration before us?"
- 4.45 So, the main takeaway that I would like us to take away from our discussion of negativity bias is that negativity bias is deeply ingrained in our brains. It cannot be wished away. We have evolved to be that way. But what we can do is to realize that we are suffering from negativity bias, because once you know that you're doing that, and once you've been exposed to all the real, truthful information about the state of the world that I tried to present in Lectures 2 and 3, then you can interact with the world

on a much more realistic basis. We cannot wish away our negativity bias, but we can understand that we are suffering from it.

- 4.46** Finally, in this lecture, I want to talk about ignorance and about the consequences of ignorance about the state of the world. Gapminder is a wonderful organization that spreads the word about the improving state of the world. A few years ago, they gave people in a number of countries 18 questions about whether the world is getting better or the world is getting worse, 18 questions, whether the world is getting better or getting worse. On average, human beings got only three questions right. Monkeys got six questions right. By pushing on random buttons, monkeys got six questions right and human beings only three. What we are witnessing here is not just a lack of information. What we are witnessing here is disinformation.
- 4.47** Let's look at some of those questions that Gapminder asked. "What share of population in high-income countries, like Germany and the United States, live in extreme poverty, on less than \$2 per person per day?" Only 23% of respondents in Denmark, Finland, Norway, Sweden, UK got the answer right, which is that less than 1% of people in rich countries lives on less than \$2 a day. On average, 77% of respondents thought that anywhere between 11% and 21% of people in rich countries lived on less than \$2 per person per day.
- 4.48** "What share of the world's population don't have enough food to meet their daily needs?" The correct answer is 11%, but only 21% of respondents got it right. 79% of respondents thought that around 23% or even 37% of the world's people did not have enough food to meet their daily needs.
- 4.49** "What happened to the global suicide rate in the last 20 years?" asked Gapminder. Only 6% of respondents knew that suicide rates in the world decreased by 25%. 94% of respondents thought that suicides either remained the same or increased by about 25%.
- 4.50** Another question: "In low-income countries across the world, what share of girls went to school until at least the age of 11?" This is before the pandemic. The correct answer is 60%, but only 14% of people answered correctly. 86% thought that only 40% or even 20% of girls in low-income countries went to school.
- 4.51** "How many companies in the world have a woman as a top manager or CEO?" The correct answer is 18%, but only 11% of respondents knew the correct answer. 89% of respondents thought that only 10%, or even 2%, of companies had a female top manager or CEO.
- 4.52** "How many people in the world have access to safe drinking water in their home or close by?" The correct answer is 70%, but only 18% of respondents knew the correct answer. 82% thought that it was only 50%, or even 30%, of people who had access to clean drinking water.

- 4.53** Another question by Gapminder: "In 1990, 58% of the world's population lived in low-income countries. What is that share today?" The correct answer is 9%, but only 4% knew the answer. 96% of people thought that the share of the world's population living in low-income countries was either 37% or even 61%.
- 4.54** "What share of countries in the world have laws against sexual harassment at work?" The answer is 75% of countries, but only 8% of respondents knew the answer. 92% thought that the number of countries that have laws against sexual harassment were much lower than that.
- 4.55** Here's another one. "Biologists have evaluated the status of 120,000 plants and animals. How many are endangered or threatened?" Well, according to this study, about 30%. But only 38% of the respondents knew the right answer. Some 13% on average thought that up to 90% of plants and animals were endangered or threatened. And on average, 62% of respondents got the answer wrong.
- 4.56** What share of all plastic waste in the world ends up in the ocean? The answer is 6%, but only 14% knew the right answer. 86% thought that between 36% and 66% of plastic waste goes to the oceans. So, there is a widespread ignorance about the real state of the world. Unfortunately, the bad news is that some of the most ignorant people about the real state of the world live in rich countries.
- 4.57** Max Roser, from Our World in Data, who presents the next slide I'm going to show you, writes, "The countries I marked with a star are those that were a low-income or lower-middle-income countries a generation ago (which is to say 1990). In these poorer countries, more people understand how global poverty has changed. People in rich countries, on the other hand, in which the majority of population escaped extreme poverty some generations ago, have a very wrong perception about what is happening to global poverty."
- 4.58** So, public perception of change in extreme poverty. Survey question. "In the last 20 years, the proportion of the world population living in extreme poverty has decreased, increased, or remained the same?" In China, 49% understood that it has decreased, which is the correct answer. But only 9% in Japan knew that global poverty has decreased. In Kenya, 37% of respondents understood that absolute poverty has decreased. But only 9% in Italy understood that absolute poverty has decreased. In India, which saw great economic growth since 1990 or so, 35% of the respondents understood that global poverty has declined, but only 9% of Frenchmen knew that global poverty has declined.
- 4.59** Here's another question. "In the last 20 years, has the child mortality in developing countries increased, decreased, or stayed about the same?" In Senegal, which most people will find difficult to find on the map, 77% of people understood that child mortality has decreased, but only 20% of Japanese knew that child mortality has

decreased. In Kenya, 69% of people knew that child mortality has decreased, but only 26% of Americans understood that.

- 4.60 Why does all this ignorance matter? Max Roser, once again, reminds us that pessimism about what is possible for the world matters politically. "Those who don't expect that things get better in the first place will be less likely to demand actions that can bring positive developments about." So expectations about the future matter.
- 4.61 If you ask a question, "Over the next 15 years, do you think living conditions for people around the world will get better or worse?", in Kenya, a very poor country, 68% of people expect that things will get better, but only 10% in Japan. In Nigeria, 67% of people expect that things will get better, but only 13% in France. In India, 65% of people are optimistic about the future, but only 18% of Italians. Even Americans, only 27% of Americans expect things to get better. And that, I think, is a problem.
- 4.62 Why is it a problem? Why is this negativity a problem? Why is it a problem that we continue to think that the world is in much worse shape than it really is? Why is it a problem that, in the United States, we feel that the United States is in much worse shape than it really is? Remember that chart I showed you, where 80% of Americans are satisfied with their lives, but only 10% of Americans believe that the country is going in the right direction? Why does it all matter?
- 4.63 I think that the answer goes something like this, and this is just my personal perspective, so take it with a grain of salt. There are certain institutional, economic, cultural, and ethical underpinnings of modernity. The institutional underpinnings of modernity may be things like liberal democracy or the rule of law. Economic underpinnings of modernity may be things like competition or free trade. Cultural underpinnings of modernity can be things like reason or toleration. Ethical underpinnings of modernity may be things like individualism, equal dignity, equality before the law.
- 4.64 I believe that it is these underpinnings of modernity that have made the present so much better than the past. However, if people do not understand how much better the world is today than it used to be, why should you support those institutions of liberal democracy, the rule of law, economic competition, free trade, reason, toleration, individualism, equal dignity, equality before the law? If everything sucks in the world, if America was never in worse shape, why should you support all those things?
- 4.65 Is there any reason to believe that, without those underpinnings of modernity, the future will be better? What makes us think that by getting rid of all those institutional, economic, moral, and ethical presets that have gotten us to this place, what makes us think that without them, the future will be better? What makes us think that, out of the ashes of modernity, something better will emerge? That is a question for next lectures to come.

- 4.66 You showed that suicide rates are dropping, and wealth is correlated with happiness globally yesterday. However, in rich countries where basic needs are met, what about things like depression and addiction? Once basic needs are met, meaning is still needed to avoid despair, and I also think there's a trend towards having cultural guilt in some of the Western, richer countries. I'm just wondering if you've investigated the linkage between abundance and despair.
- 4.67 Thank you. This is a difficult subject that we are only now beginning to study. First, I would argue that the most obvious indicator of despair, lack of meaning, is suicide, when you decide that life really is so horrible that you cannot take it anymore, and you take your own life. As a matter of fact, in the OECD countries, the richest countries in the world, suicide rates have been dropping. Not in the United States, they are slightly elevated, but in the rest of the rich countries, they are dropping. That makes me think that there's something else going on, certainly with regard to the United States, that we cannot blame economic success, for example, or that we cannot correlate economic success with a lack of meaning.
- 4.68 The other point that you raise is about anxiety, and it is true that people in rich countries tend to report more anxiety than people in poor countries. However, we don't know why. There are alternative answers to that question. One could be that somehow modernity, human progress, wealth, makes people more anxious.
- 4.69 The other alternative is that, as society becomes more tolerant, more accepting, more in tune with its feelings, with the feelings of the people, that maybe more people feel comfortable saying that they are anxious or are suffering from trauma, than in societies which are, let's just say, more traditional, where you are supposed to basically suck it up.
- 4.70 Maybe we become softer as a result of human progress. We welcome, in fact, we encourage people to say that they are anxious, because if they come to the fore and they say, "Oh, I'm anxious," we are going to make special provisions for them. So not only is society more open to anxiety, to expressions of anxiety, but it may encourage them by providing people with, for example, more time to answer questions during tests or giving them less work in the office and so forth.
- 4.71 It may also happen that our definition of anxiety or trauma has changed. In the old days, having a traumatic experience meant being shelled by the Germans in the trenches of France during the First World War. Today, people say, "Oh, I had a traumatic experience because I broke up with my boyfriend or girlfriend." There are a lot of things that may be going on. I am personally encouraged by the fact that suicide rates are, on average, dropping in OECD countries. To me, suicide is the ultimate expression that life has stopped being meaningful. Thank you very much.

Lecture 5: Theories of Progress

At a glance — Theories of progress — what drives improvement in the human condition.

- 5.1** The first half of this course was intended to give you a sense of what human progress is and isn't. It was intended to provide you with evidence of human progress, and then it asked, why, in spite of all the improvements, are we so negative about the world around us? The second half of this course will focus mostly on history, and we'll try to ask a question, where does human progress come from? In this lecture, we'll be talking about theories of progress, which is to say, we'll be talking about men and women who proposed an explanation for why the world has improved.
- 5.2** Our species is 300,000 years old. For most of that time, we were very poor. We lived in a world that was incredibly violent, and progress could really not be measured all that much. But in the last few hundred years, we have seen that, if you recall from Lecture 2 and from Lecture 3, we have seen that extraordinary upward slope of human progress, the hockey stick of great enrichment. We are poor, poor, poor for a very long time, and then suddenly this upward blade of the hockey stick of human progress. Why did it happen?
- 5.3** In this lecture, I'll be talking about institutional, economic, scientific and technological, cultural, human capital, and resources and geography explanations for human progress. Let us start with those thinkers who have reasoned that human progress depends mostly, by and large, in the main, on institutions, explanations having to do with institutional emphasis of human progress.
- 5.4** One of the most important currents in this way of thinking is one that talks about inclusive institutions, which is to say, an increase in the fair rules and strong protections for property, contracts, which then helps people to invest, invent, build businesses, look toward the future.
- 5.5** People like Nobel Prize-winning economist Douglass North, Nobel Prize-winning economist Daron Acemoglu, and James Robinson, and Barry Weingast, what they're saying is that, over time, the risk of expropriation, for example, somebody taking your stuff from you, has declined; that new fair rules and courts were established, which encouraged saving, lending, and innovation. So as institutions improve, as human beings have more stable understanding of what is their own and what they can do with their stuff—I can start my own business, I can do with my property what I like, I don't have the king taking it all away from me—this is what's called an inclusive institution. So that's one explanation.
- 5.6** But of course, a common pushback against those kinds of explanations is that it is hard to know if it is the good rules, the good institutions, which cause economic growth, or if it is growth that leads to better rules. Some places, some countries in the past, did not have inclusive institutions, but yet, they grew economically, and their people became

more prosperous. South Korea becomes prosperous during dictatorship. Taiwan, same. Chile, under General Pinochet. Francoist Spain, or post-1978 China. China has grown tremendously, we all know that, but it's still a totalitarian communist dictatorship.

- 5.7 Another way, which I call a subsidiary of inclusive institutions, is an explanation promoted by Dr. Ogilvie from the University of Oxford, who talks about the ends of guilds and serfdom. Guilds were institutions in medieval Europe, into early modern Europe, whereby, for example, if you wanted to make shoes, you needed to be a member of a guild. The guild ensured quality, or so they claimed; they would apprentice you, and after many years of working for a master, you would then eventually have the opportunity to make shoes by yourself.
- 5.8 So there was a guild of shoemakers. There was a guild of dyers of clothes. There was a guild for just about everything. The problem with guilds was that they restricted access to the making of whatever product. If you were just an average Joe who, in our society, has every right to make shoes if you can, you were prohibited from doing so. And so, it is the decline of the guilds which happens in the UK in the second half of the second millennium and in Europe after Napoleon's conquest, it is the decline of the guilds, the freeing of the individual to do with his labor whatever he wishes, that this particular theory believes is at the root of progress.
- 5.9 Another important aspect is the end of serfdom, the fact that, in Europe, slavery and serfdom ends a thousand years ago, and all those people who are formerly obliged to work for their lord and who cannot move from one land, from one piece of land to the next, and do with their body and their labor whatever they want to, that these people are free to do whatever they please. So, according to this theory, it is the freer labor and the freer product markets—you are free to do with your body anything you want, and you can do with your body and you can make whatever you want—that is at the root of progress.
- 5.10 But there's a common counterargument to that, which is that guilds were also teaching skills and, in short, quality, and sometimes they were important for the checking of noble power. So maybe your local lord wanted to confiscate property or he wanted the city-state to move in a certain direction. And it was the guilds, like in Italian city states, like in Florence, who actually were the checks on the power of the lord. So guilds may actually have been an important part of human progress, rather than a detractor from human progress—at least that's what the critics say.
- 5.11 Another theory behind what caused progress is an increase in state capacity. This is a theory that stronger and better-run states—with some checks and accountability—built better tax systems, had better laws, better infrastructure, and that those things supported markets, expansion of the economy, and enrichment. The way it works is that regular taxation and credible public debt, backed by competent administration, could fund a lot of public goods, what we call public goods, such as roads, canals, courts, and national defense—goods which some economists believe cannot be

provided privately. I happen to think that many of them can be provided privately, but nonetheless, that is just my personal opinion.

- 5.12 Common pushback against this theory is that, yes, strong states can build things, but wars between very competent, strong states can destroy and wreck economies. And high taxes, whilst they may be required for the building of public goods, may also hurt economic growth, especially if the leaders are unaccountable.
- 5.13 A subsidiary of this theory is the financial revolution and deep capital markets theory. The way it works is that, in places like the Dutch Republic and England, trustworthy public debt and banks mobilized savings for businesses and for war, at scale. So, when you have competent states, when those states can be trusted, what tends to happen is that you can borrow at a lower rate, which is good for your economy because then you can invest without having to pay too much. So lower interest rates can channel money to more productive uses. If you have to build a canal or a road at 20% interest rate, that's much more expensive, and maybe you decide that you don't want to do it. But if it's 10%, maybe you will do it. That's the theory behind it.
- 5.14 But of course, deep finance can also cause a lot of bubbles and crises. If the Dutch Republic and England are the examples of financial revolution, deep capital markets, they are also the places that have been privy to some of the biggest bubbles and biggest financial crises of all time, such as the tulip mania in Holland and the South Sea Bubble in England. And deep markets that we are talking about tend to work best when they are underpinned by institutions, such as inclusive institutions that Douglass North talked about.
- 5.15 Patents, prizes, and intellectual property. The idea here is that, when an innovator can be certain that his invention or innovation is going to be protected by a patent, that he has intellectual property in that invention or innovation, more inventors and innovators are likely to produce inventions and innovations. So patents give temporary rights, temporary property, intellectual rights, to some inventions and innovations, which spurs the effort of people to produce more inventions and innovations, which can then contribute to greater economic growth.
- 5.16 The criticism of that position is that patents can also slow the spread of ideas. If, for example, I develop an airplane, the first airplane, I have a 20-year patent to it, as in fact happened, and I decide not to do anything with that patent, if I don't continue to innovate, then maybe innovation has been postponed by 20 years until my patent expires. And once it has expired, then people can take my innovations and improve on it. So, do patents actually encourage more inventions, innovation, and growth? Or do they actually discourage innovation and growth, precisely because other people cannot use that intellectual property to do with it what they please?
- 5.17 We are moving now from institutional emphasis onto economic emphasis. One theory is that the Atlantic trade had given rise to the merchant class. The theory goes

something like this. Ocean trade in the Atlantic after 1500 opened big markets and strengthened merchants who pushed for pro-market reforms in places, once again, like England and the Dutch Republic. So the bigger markets and profits raised the power of groups that supported secure property and lowered trade barriers such as tariffs.

- 5.18** The idea here is—we're coming back to it. Whether you believe that any of these particular theories are right or wrong, there is something that unites them, which is the idea that private property rights, secure private property rights, are important; that people who believe that they cannot be expropriated, some general or a ruler cannot come and take away their stuff, are much more likely to use their wealth for the benefit of themselves, but also of society.
- 5.19** If you are able to develop a new machine, a new invention, with which you help yourself and your family, you're also improving the society, because eventually those machines will be used elsewhere. The idea here is that there is a certain class of people in Europe at that time, the merchants, who, by receiving the profits from transatlantic trade, can exercise greater power. They can protect themselves from the predation, from exploitation, from expropriation by the rulers, and that this is somehow important.
- 5.20** The counterargument here is that trade alone may not do much unless domestic policies have also improved; in other words, that unless you also have a ruler who likes trade, it may come to nothing. An example here would be somebody like the Hongwu, the Ming Emperor in China, who decides one day to burn the Chinese fleet, and the Chinese can no longer trade in the Indian Ocean. So they could have benefited from trade; trade would have enriched the domestic merchants, but the Emperor says no. So, is it trade which is important, or is it the institutions, meaning the ruler and the rules by which he rules the country that is important? There's also the argument that richer and better-run places may have traded more to begin with. So that's another criticism of this particular theory.
- 5.21** Columbian exchange and population increase. So after Columbus sails to the Americas, there's a lot of stuff that comes from the Americas to Europe and from Europe to the Americas. And one of the things that happens is that Europeans import from the New World the potato. And as a result of the potato, there is an increase in calories and population, which leads to economic expansion, or what economists call extensive growth—not to be confused with intensive growth, about which we'll talk in a moment.
- 5.22** Extensive growth basically means that you are able to support an increasing number of people at the current standard of living. You are not increasing—intensive growth, meaning increasing standards of living of every individual, that's not what's happening here. It just means that there are more potatoes, more calories, and therefore more people. They are still incredibly poor. There is just more of them. But because you are able to now support more people, the total aggregate amount of stuff produced by the

economy, economic growth, actually does happen. Economies in Europe grow. GDP per capita doesn't grow, but the overall GDP grows.

- 5.23** The pushback against this theory is that New World crops, such as the potato, for example, but others as well, have also moved to China by the Europeans and then to Africa. So it wasn't just Europe which received these New World crops, but China and Africa. And yet the great enrichment doesn't happen in China and Africa, it only happens to Europe. So how could it be all that important? That's the question that the critics are asking.
- 5.24** Now, we come to intensive growth. Intensive growth is not only the entire economy becomes bigger because you have more people, but they have the same standard of living; now you have a situation where you have more people, but everybody's standard of living is higher. So GDP per capita increases. This is called intensive growth. And the idea is something like this. The way this works is that ideas are nonrival. Just because I have an idea and share it with you, it doesn't mean that I have lost that idea. I still have that idea, but now you have it, too.
- 5.25** So, when you have bigger populations— the potato maybe increased the population of Europe somewhat, so your population is bigger— and you have also a better connected population, so it's not just Europeans communicating with each other, but now you have them communicating with the population in North America and South America and Asia and China and so forth, that when you have a bigger population of the world, maybe because new crops, better food, and because they are connected together, maybe by trade routes, they are able to mesh those ideas together. They're better able to come up with new ideas, which then fosters economic growth.
- 5.26** So, according to this theory, it is easier to create more economic growth when you have very large populations that communicate with one another, and swap ideas and learn from each other than when it's just a small group of people. Because in a small group of people, you are going to have that amount of ideas, but in a big group of people, you're going to have a bigger amount of ideas. In other words, more people means more potential inventors and tinkerers, and each new idea can be reused and recombined with other ideas. Then, when you have these dense networks of people working and thinking together, that can lead to higher economic growth.
- 5.27** The criticism of this idea is that, just because you have more people, that doesn't mean that you are going to have more ideas that can be implemented. After all, China and India have been incredibly populous countries for many millennia, but they remained very poor. So what you need is to have actually an institutional setting, which allows people's ideas to be, first of all, expressed and then transformed into practice, into something useful.
- 5.28** Let me put it slightly differently. Yes, having more people and more ideas is a good thing. But if they are not free to put those ideas into practice, if they are not free to

build on those ideas, if they are not able to do something essentially with their ideas, because the ruler says that you cannot, then a big population is not that helpful. In other words, the critics are saying, it's not really the population that matters, or maybe it matters only a little; what's important is the institutions.

- 5.29** China is the most populous country for 2,000 years, but it remains poor because it's a totalitarian system where people cannot express themselves freely. They cannot take their idea and then transform it into a product that everybody wants to buy because it's a totalitarian communist system. It's only when the Communist Party, after 1978, allows the Chinese people a little bit of freedom, where if you have an idea, then you can transform it into useful products that other people want to buy, that China actually becomes wealthier. So it's not the number of people, it's the institutions of freedom that matter, according to the critics of this theory.
- 5.30** Another theory, which is related, by my colleague, Johan Norberg, at the Cato Institute, and by my friend Matt Ridley, is that it is really openness, it is really openness. Open societies that welcome trade, migration, and free speech create big connected markets where ideas mix, driving innovation and growth. So it's not just the number of people. The people have to exist within an open system, a system that is open to exchange of goods, ideas, and so forth.
- 5.31** Trade and migration, according to these two scholars, enlarge the pool of skills; dense networks of firms, ports, and cities recombine knowledge; free speech and press spread new ideas and let people challenge bad policies. As Matt Ridley calls it, "Ideas are having sex." It's the freedom, it's the openness, that really matters.
- 5.32** A common pushback against this theory is that openness often comes bundled with other reforms, better institutions. So is it institutions, really, or is it the openness? The causality of openness, of freedom, is hard to isolate, these people say. Trade and migration create winners and losers. Therefore, they need good domestic rulers and good institutions in order to survive. So is it really the openness or is it good institutions? That's the question.
- 5.33** Here's another theory of human progress. Western takeoff, human progress in the West, doesn't happen because of anything the Europeans have done. It happens because they colonized and because they enslaved other people. It was the profits from colonies, slavery, silver from the South American mines, that have helped to finance European states and trade. But there are criticisms of this position. Every society was a slave society. Ancient Romans were slavers, Egyptians, Greeks, the Ottomans, the Chinese; everybody had slaves. If slavery is so important, why is it that economic takeoff really only takes place in Europe? Native Indians had slaves. African Americans enslaved each other. Why didn't economic revolution—why didn't the great enrichment happen in Africa?

- 5.34 What about exploitation of natural resources and colonization? Well, the world's two biggest and most exploitative colonies, colonial empires, were Portugal and Spain. They are also Europe's poorest countries. If it matters so much that the Spanish have gotten silver from Latin America, why did it go bankrupt eight times in the 17th century, 16th and 17th centuries? If colonialism is so important, why is Portugal one of the poorer nations in the OECD, whereas countries that never had colonies or slaves, such as Switzerland, Sweden, are much richer?
- 5.35 In fact, the critics argue, colonialism may have been bad for a country like Spain. Why? Well, for at least two reasons. One is that it may have retarded, it may have slowed down institutional development. In England, if the king wanted money to buy new clothes or build an army or whatever else, he had to go to parliament and ask it for money. The king of Spain could simply get money from the New World. So the parliament never acquired the kind of power in Spain that it did in England.
- 5.36 The other problem is the so-called Dutch disease. We don't have to go into the details, the mechanics of it, but basically, all that money that poured from Latin America to Spain made everything in Spain incredibly expensive. There was hyperinflation. So what happens then is that, actually, nobody has any incentive to do anything at home, to build anything at home. The Spaniards were buying stuff from abroad, and their domestic economy actually collapsed. I'm unconvinced by this theory.
- 5.37 Let's move on to scientific and technological explanations of human progress. One of them has to do with the printing press and the contributions of the printing press to literacy and the information revolution. Cheap books— after the printing press takes off due to Johannes Gutenberg's improvement of a Chinese model in the 1450s—helps to spread ideas and literacy and technical knowledge. Printers and readers formed networks that sped up learning and debate. That's the theory.
- 5.38 But there's a criticism of this theory. China and Korea also had early printing presses long before Europe, and they didn't have great enrichment. Sophisticated printing presses also spread in Italy, but Italy was a very late industrializer, so printing alone cannot possibly be enough.
- 5.39 Another theory, open science norms and priority races. Open science is the exact opposite of patents. You know how I told you previously that some people believe that only when you have patents can you incentivize the inventors and innovators to actually invent something, because you give them a temporary right to those innovations?
- 5.40 Well, these people, like Joel Mokyr, and especially my friend and colleague, Terence Kealey, believe the exact opposite. They believe in open source. They believe that, when you are sharing methods, when you are given credit for first discoveries—in other words, your benefit is that you are considered the creator; you don't have necessarily any kind of exclusionary rights, like a patent; you just bask in the glory of being the

inventor or innovator, but you are sharing your knowledge with everybody else in an open source—that this is what drives progress forward.

- 5.41** But the obvious pushback against this theory is that, if you don't give people patents, if you don't give them temporary right, intellectual right, to their invention or innovation, they may be disincentivized to innovate. My colleague and friend Terence Kealey would say that it doesn't matter. What matters is that the inventor and the innovator gets the recognition and feels good about himself. But because there is no property right, everybody can immediately take that idea, that invention or innovation, and can start building upon it and improving it, making it better. And that's what drives the world forward.
- 5.42** Yet another theory, which is promoted and defended by Joel Mokyr, who just received the Nobel Prize in economics, relates to the scientific revolution and the creation of useful knowledge. Maybe I should have included it in the cultural part of this presentation, but I decided to include it in the economic side. Joel says that open inquiry and experimentation produced useful knowledge. Scientists created new propositional knowledge about nature. Engineers and artisans turned it into prescriptive knowledge. And then they applied it, thus improving the human lot. So there's a lot in it. So I'm going to try to, I'm going to try to disaggregate it a little bit.
- 5.43** What is prescriptive knowledge? Prescriptive knowledge could be, for example, that Edward Jenner, an English scientist, realizes that milkmaids don't get smallpox. And he reasons from it that milkmaids don't get smallpox because they are exposed to a milder version of smallpox, which is cowpox. And so what he does is he takes some of the fluid from cowpox, and he injects it into human beings, and they become immune from smallpox. To this day, vaccination is derived from "vaca," which is cow.
- 5.44** So the prescriptive knowledge here—you take the fluid from the cowpox, you give it to a human, they become immune. But then we don't have any vaccines for the next 150 years. Why not? Because Jenner knows that taking the fluid from the cowpox and putting it into a human will prevent infection from smallpox, but he doesn't know why. He knows that it works, but he doesn't know why it works. And it's only later in the 1860s, when Louis Pasteur and others discover the germ theory of disease, that we have an understanding of this microbial world of pathogens.
- 5.45** That's propositional knowledge. The propositional knowledge is that there is a world of microorganisms, pathogens, the germ theory of disease. There's a prescriptive knowledge, which says that you can inoculate people or you can vaccinate people against pathogens by giving them vaccines. And then there is the application of useful knowledge, where you start developing more vaccines and giving it to all of humanity. So this is Joel's idea of scientific revolution and useful knowledge.
- 5.46** Republic of Letters is this amorphous institution in Europe where people swap letters, usually written in Latin, scientists all over Europe. There is no building. It's just that

people get used to sharing ideas, swapping ideas, explaining what works and what doesn't. It's called the Republic of Letters. It's a Europe-wide network of scholars who say, "Oh, that guy, he knows how to vaccinate people. Oh, that guy has discovered a microscope." And maybe somewhere out there, somebody is going to start looking at a microscope for the microbial world. So you've got this Republic of Letters. You've got improvements in propositional and prescriptive knowledge and an application of useful knowledge in the real world. That's Joel's theory.

5.47 But then the critics say, "Well, science and the Republic of Letters were pan-European. All people in Western Europe had access to this knowledge. But industrial takeoff, the great enrichment, only really happens or starts happening in Britain. Why didn't it happen somewhere else?" So maybe useful knowledge cannot alone explain where and why modern growth began. Maybe there was something more about England, maybe institutions, maybe something else.

5.48 Now we move on to cultural emphasis. One of the main proponents of cultural emphasis of human progress is Steven Pinker, with whose work we began this course. So Pinker believes that it is reason, science, and humanism, the hallmarks of the Enlightenment, that helped societies to solve problems: improved health, wealth, safety, and freedom. The way it works is that, by using evidence and experimentation, institutions adopt better policies and technologies. Rights and pluralism reduce violence and oppression. That's Steven's idea.

5.49 The criticism is that rising literacy, urbanization, and commercial prosperity can themselves generate more secular, individualist, and tolerant attitudes. In other words, it is because you have already become richer that you can now indulge in reason, science, and humanism. So Enlightenment values, according to the critics, are the symptom of deep economic and institutional shifts, not the driver of those shifts.

5.50 Max Weber, in his famous theory of the Protestant work ethic, Weber believed that Protestant traditions fostered work ethic of discipline and thrift and commitment to work as a calling by God, which supported the earliest capitalist development. So it was the Protestant doctrines of predestination—you showed that you were predestined to go to heaven by becoming successful in life—predestination that encouraged hard work, punctuality, savings. They were all visible signs of grace from God. And all of these forms, or rather, and all of these norms put together raised the labor effort. They raised productivity, investment, and reliability of contracts and credit.

5.51 Critics say that independent work ethic effects are difficult to separate from schooling, institutions, and income. The critics say that Weber is wrong. Work norms may have changed because economies grew, rather than causing economic growth in the first place. So it was because people already saw that, by working hard, you can make more money and you can become better off, that they have adopted all of these better work ethics, better discipline, more thrift. It had nothing to do with Protestantism, so goes the argument.

- 5.52** Joe Henrich from Harvard has a theory about the role of the Catholic Church in promoting human progress. According to Joe, the medieval Western Church basically destroyed European clans and tribes; that what you needed for capitalism, what you needed for great enrichment, was individualism, and you couldn't have that in a highly tribal society. What the Catholic Church does is it bans cousin marriages all the way to the seventh degree, which means that if you've grown up in a little hamlet or a little village or a little town, there's absolutely nobody you can marry, because you are related to all of them. You have to leave. You have to break the bonds with your family, with your tribe. You go and search for a spouse dozens of miles, maybe hundreds of miles away.
- 5.53** It is this creation of a nuclear family of individuals that is related to an increased trust in strangers. After all, you are now moving only among strangers. You are no longer moving among family. You had to abandon them, so you have to increase your trust in strangers, and you have to participate in impersonal markets. You have to maybe do business with somebody in London that you just met five minutes ago, rather than with Uncle Joe, whom you've known your entire life. So it is this destruction of tribalism and clannish nature of European society, which leads to more nuclear families, to more individualism, to more trust in strangers, and to more impersonal markets.
- 5.54** The criticism of this argument is that the enforcement of the cousin ban, or rather the cousin marriage ban, was very patchy. A lot of people ignored them, above all, the European royals. Local practice often ignored whatever the Catholic Church was demanding. The marriage ban arrived many centuries before the events which they are meant to explain. So that's the argument against that particular theory.
- 5.55** Another theory is that, in pre-industrial England, because that's where the great enrichment begins, richer families had more surviving children. And over time, traits like patience, literacy, work discipline, higher cognitive ability, which we call IQ, spread, which made the societies more ready for industrialization. So differential fertility shifts trade distribution across generations. The criticism of this theory is that, where is all that human capital now? If England had all of these thrifty and hard-working high-IQ people, where did it go? Why isn't England now the primary driver of human progress and great enrichment?
- 5.56** Another theory, which is pushed by the late David Landes from Harvard University, has to do with timekeeping and work norms. So it's a little bit related to what Max Weber was talking about, but to David Landes, timekeeping is fundamental to his theory, I think. And what he says is that, when you have the spread of clocks and watches in Europe, when they become cheap enough to put on every tower in every small town and into the pocket of every man and woman, values like time discipline, which is very important if you are going to be running a successful business, and trust, and respect for enterprise, basically, all of these are connected to causing the great enrichment in Western Europe.

- 5.57** So, was timekeeping really that important? Not everybody agrees. The criticism of Landes' theory is that culture, such as moving from having no conception of time and being four hours late for a meeting, versus being right on time and thereby showing that you are a serious person. The criticism is that culture is very hard to measure, and it can become a "just-so" story. In other words, you can invent a story in the past to fit whatever argument that you want to prove today. And who knows? Maybe timekeeping increased in Europe precisely because Europe was already richer, rather than vice versa.
- 5.58** My colleague at the Cato Institute, Deirdre McCloskey, has a theory pertaining to bourgeois dignity and pro-innovation culture. According to this theory, rising respect for commerce and invention encouraged creativity and enterprise. Social approval made entrepreneurship and tinkering more attractive. What does she mean by that?
- 5.59** Well, for the longest time, to innovate was very risky, and it was frowned upon. There is a famous story in ancient Roman literature about a man who comes to see Emperor Tiberius and offers him a bowl made from unbreakable glass, expecting a reward. And the Emperor has him executed because he worries that, by making this invention, he will put so many goldsmiths into unemployment.
- 5.60** So in the past, the argument goes, invention and innovation were frowned upon. But for some reason, in 17th- and 18th-century Europe, suddenly, inventors and innovators were appreciated. They were painted in paintings. They were celebrated in a similar way that Elon Musk, for example, is celebrated for many of his inventions and innovations today, and that that is what prompted the great enrichment in Europe.
- 5.61** But, the critics say, what caused that shift in the appreciation for inventors and innovators? Maybe you're right. Maybe the key to human progress is that we now appreciate inventors and innovators. But why was that that change? Why did it happen?
- 5.62** Another set of theories pertains to human capital and health. Here we have somebody like Chicago economist Gary Becker, who has an idea about religious competition contributing to human progress. Competition amongst denominations, he believed—he's dead now—race, literacy, and skills in some regions, improving human capital and governance. According to Gary Becker, Protestant insistence that believers read scripture pushed schooling, rival churches and princes invested in parish schools, local self-government blossomed. There was a rise in basic education.
- 5.63** Remember that this is different from Protestantism as a religious idea. Protestantism, as a religious idea, says that you work hard because God wants you to work hard, so we see a Protestant religious explanation for human progress in the previous section. This is a Protestant secular explanation for human progress. Gary Becker doesn't say that a Protestant doesn't contribute to human progress because God wanted you to be prosperous. He says that it was the competition between different denominations,

different churches, that encouraged Bible reading and literacy, and that is what prompted human progress forward. Nothing to do with religion, or rather, tangentially so.

- 5.64** Common pushback is that some Catholic regions also grew rapidly; and that when you do your studies, and once you control for education and institutions, religion as an independent factor in driving human progress forward actually diminishes in importance.
- 5.65** Demographic transition—the argument here is that because women have fewer children, every child becomes all that much more important. So you have rising return to skills. So when you know that you only have one baby or two babies, you really need to invest in their education, in their schooling, in their literacy, in whom they marry, because they are so much more important than if you had 10 or 20 children. You had to invest much more in them.
- 5.66** The criticism of this argument is that some economies took off before schooling became all that important. In other words, these countries were rich a long time before schooling was widespread, so that's the criticism of this particular theory. Robert Fogel, a Nobel Prize-winning economist, sadly dead now, his argument is that rising nutrition and health raised workers' strengths and skills, boosting output long before modern medicine. So this would be probably a little bit related to the idea that, with the Columbian exchange and better food, we have better diets in Europe, we have improved bodies and brains, and therefore, an increase in lifetime productivity.
- 5.67** The critics of Fogel say that a higher income may raise food intake and health inputs, so that the observed height-wage link may actually be income-height link. To repeat, it is precisely because Europeans were already rich, the argument goes, that they were able to afford better food, leading to better nutrition, to increased height, rather than better nutrition and increased height leading to more economic growth and greater prosperity. So disentangling which way the causal reality goes, as you can see from this lecture so far, is very important. We all have stories, ideas, theories about how the world works. But being able to say whether it's going this way or that way is actually kind of important and very difficult.
- 5.68** Robert Allen, a famous British economist, says that great enrichment happened because, in Britain, wages were already so high that you wanted machines to do that work. In other words, instead of employing workers who were very expensive, it made sense to invent new machines to do those jobs. Obviously, the machines that powered the Industrial Revolution in England were steam engines powered by coal. So England got lucky. It had an expensive labor force. It had a lot of coal. And if you wanted to increase your productivity, if you wanted to increase your economic output, it was better to develop machines which can use that coal, rather than pay for workers, which are very expensive. That's Allen's theory.

- 5.69** The counterargument is that coal cannot explain every industry that powered the Industrial Revolution. The cotton mills, for example, which became so important during the industrial revolution, were powered by water, not by coal. And high wages may reflect prior productivity, institutions, and market size. So using them to explain new innovations, the onset of industrial revolution, may be reversing causality. That's the argument against Robert Allen.
- 5.70** Mark Overton and Edward Glaeser, they have a theory which talks about a Second Agriculture Revolution and urbanization. The First Agricultural Revolution happens 12,000 years ago, when people start farming. The Second Agriculture Revolution happens 200 years ago or 300 years ago, when you have better crops, rotation between crops, better land rules in modern Europe, which raise farm output, feed workers from the countryside, and help fuel growing cities where people learn from each other and found better matches for jobs and ideas.
- 5.71** The idea here is that, during the Second Agricultural Revolution, let's say, 1700-1800s Europe, you have better technologies, better crops, more scientific understanding on how to grow more crops on a specific piece of land. You produce more food, and that means that people can start moving off land and into the cities, because obviously people in the cities, they are not farmers. They are consumers of food. They are doing other things. They are making shoes. They are making carriages. They are developing microscopes and telescopes.
- 5.72** But to be fed, you have to have farmers who are productive enough to create a farming surplus, which can then be sold to the city where people are going to be doing other stuff. So the idea here is that, because agriculture becomes more productive, a lot more people move to the cities, they are fed from the countryside, but they do other stuff. They live in close relation to each other. They share ideas, they learn from one another, and they create more ideas, more inventions and innovations. So that's the theory.
- 5.73** The criticism of this theory is that agricultural gains depend on institutions such as secure private property rights. If you didn't have good institutions, if you didn't have good private property rights, then it would make no sense for you to try to make your land more productive. And big cities, are they really that important, according to this theory, or the critics of this theory, when so many people were dying in the cities because the health in the cities was so much worse than it was in the countryside? So the critics say, neither increased agricultural output nor cities can actually be all that important to human progress. Far too many people died in the cities for cities to be important. They lived together. Maybe the sanitation wasn't too good. It just doesn't make sense to the critics.
- 5.74** Geography and diseases—Jared Diamond, whom we met in Lecture 1, says that it was the importance of climate, waterways, and resources, as well as diseases, which shaped chances for early growth. Some countries, he says, had the right animals. They may

have been exposed to the right diseases. Other countries didn't, and therefore, some countries were able to prosper, and some didn't.

- 5.75** A perfect example would be when the Europeans arrive in North America. They bring with them pathogens, which wipe out 90% of the original population. It was unintentional, but it did happen because native inhabitants of the Americas were never exposed to pathogens which Europeans were, often from Asia. It was from Asia that the Europeans got the bubonic plague, from which about a third of Europe died in the 14th century. But because, on this Eurasian landmass, all of these pathogens were going east and west and so forth all the time, people eventually, everybody got exposed to these new pathogens and developed eventually resistance toward them, people who didn't die, whereas people in the Americas didn't have that.
- 5.76** So he argues that, because Eurasians were more resilient to the pathogens, and also because they had certain animals which the Native Americans didn't, such as horses, for example, and because they had access to the sea— many of the main rivers in Europe flow into the Atlantic, so from there, you can take a ship all the way to the Americas—that, Jared Diamond believes, was at the root of European takeoff.
- 5.77** The criticism of this argument is that China was exposed to the same pathogens. China had horses. China had rivers. China could have done the same job as Europeans have, but they didn't. So how important really is geography and pathogens and animals, if China, which is on the same landmass as Europe, couldn't do what the Europeans had done? And I also think—and this is my personal view—that Jared Diamond downplays the Latin American geographical advantages. There's plenty of land for agriculture in Latin America. There are plenty of rivers that are navigable on which you can take your crops either upstream or downstream. Those will be the critics of geography and diseases, theories of human progress.
- 5.78** So, in conclusion, what can we say? In Lecture 1, I read you a statement made by Tyler Cowen, a prominent American economist, and Patrick Collison, who said, "By 'progress,' we mean the combination of economic, technological, scientific, cultural, and organizational advancements that had transformed our lives and raised standards of living over the past couple of centuries."
- 5.79** My personal takeaway from this lecture, as I was preparing it, is that I would be comfortable, and I would suggest to you that we should focus on four distinct but interrelated drivers of human progress. The first is institutions, better institutions, institutions which restrained predation by the elites. They restrain predation, exploitation by the elites. They protect property. They support impersonal markets, markets between me and you, whom we just met, rather than just members of my family, and institutions that create the rule of law. That's number one.
- 5.80** Number two, expanding domestic and global market competition and trade. Domestic and global market competition and trade. It is very important to have competition and

trade domestically, so that you are not just relying on buying cars from the Ford company or GMC, but you can choose from Tesla and Toyota and others. There has to be competition domestically, and there has to be competition globally between the car manufacturer in the United States and the car manufacturer in China or Europe, like Volkswagen or BMW. It is the competition between companies in different parts of the world that drive progress forward. So, it is competition and trade that incentivize specialization and innovation.

5.81 Three, third driver of human progress, an epistemic or knowledge system that promotes scientific and technological discoveries, thus making useful knowledge cumulative. We'll talk much more about this in the subsequent lectures. But basically, what you want to do is to live in a country and work in a country where scientists and innovators are allowed to function, where you don't have one authority telling you, "You can research on this, but you cannot research on that," because it goes against the interests of the state or interests of the clergy or interests of the main group of rent-seeking businessmen, etc. You want to live in a society where anybody can innovate new scientific and technological discoveries.

5.82 And that, I guess, is point number four, the fourth driver of progress, is a culture of openness, optimism, and skepticism, one where you can question the ruling classes, one where you can question the received wisdom, one that is open to new ideas, new innovations, one that doesn't punish innovators, and one that is optimistic about the future, a sense that if there is a problem, that you can hopefully fix it, because you live in an open, competitive society that rewards success. Thank you very much.

Lecture 6: Progress Through Innovation

At a glance — Progress through innovation and its central role.

- 6.1** In this lecture, we will be talking about the role of innovations in human progress. The point that I will be making, the most important point that I'll be making, is that human progress comes from innovations, the tested and adopted use of ideas and inventions in real practice. We will start by discussing the difference between ideas, inventions, and innovations. We will then look at different kinds of innovations, and we will also talk about the history and speed of innovations. But as we go through this lesson, simply remember that my main claim, my main claim is that human progress comes from innovations.
- 6.2** So let's look at the difference between ideas, inventions, and innovations. People have trillions of ideas every day. Most of them are very bad. Ideas ask what might be possible. Inventions show how it can be done, and innovations prove that it actually works for many people in the real world. That's the difference. Ideas are about what may be possible. Inventions are showing how it can be done, and innovations are applied inventions that show how it can be done around the world.
- 6.3** Let's just think of an idea. Idea is a thought, a concept, or proposal about how something might work or be improved. It exists mainly in somebody's mind or on paper. A perfect idea would be the idea of human flight. The myth of Icarus, the ancient Greek myth of a boy and his father who are trying to escape prison by building wings from feathers of birds. And then Icarus flies too close to the sun, and the wax which holds the feathers together melts, and he falls to the ground and dies. Or think about the sketches of a helicopter, which Leonardo da Vinci did in the 15th century, so it's an idea in the mind.
- 6.4** Invention is a concrete or original design or method that embodies an idea in a working prototype. A perfect example of that would be the Wright brothers' first engine-powered flight back in 1903. And finally, innovation is the widespread adoption, scaling, and refinement of inventions, so they can change everyday life, institutions, or markets. A perfect example of that would be that you move from an idea of flight, in the mind of Michelangelo, sorry, of Leonardo da Vinci, then to the first flight of the Wright brothers, and you end up in a world filled with commercial aviation networks, budget airlines, air cargo logistics—basically, the way that we move around the world today.
- 6.5** But it doesn't just have to be a technological invention; it can be actually an idea itself, which then leads to invention and innovation. An idea could be something like the moral claim that all humans have the same basic worth. That's just an idea. You turn it into an invention by creating concrete rules or institutions that treat people equally under the law, such as, for example, the 1776 Declaration of Independence or the 1789 Declaration, French Declaration of the Rights of Man.

- 6.6** We have an idea: all people deserve equal dignity. We then have an invention, such as the Declaration of Independence or the Rights of Man. And then we have an innovation, which is the wide practical adoption of those rules, so that everyday life actually reflects this equality of dignity. You could see that, for example, during the US civil rights movement or the end of the Jim Crow and apartheid, the fall of feudal and caste privileges, and the post-1945 human rights regime. So, those are the differences between ideas, inventions, and innovations.
- 6.7** Let's look, first of all, at the most important political, institutional, and legal innovations that happen in Europe in the last 1,000 years or so. Why Europe? Why do we talk about Europe so much in this course? Well, it is because human progress, and especially the Great Enrichment, starts in Europe. It is Europe that leads the rest of the world in terms of ideas, such as equality under the law, then perfected in North America, in the United States. But the Industrial Revolution, which then creates all of this economic wealth that allows us better sanitation, better hospitals, universities, and so forth, depends on having money.
- 6.8** We talked about the close relationship between GDP per capita and all the other good things in life. We want to know why and where this all started. It starts in northwestern Europe, sometime during the second millennium. People differ in terms of when exactly is the beginning of human progress, but in this course, I'll be taking the 1,000-year look. Let's look at the political, institutional, and legal innovation in Europe.
- 6.9** One of the first ones that you see starts right as the second millennium begins in northern Italian city states. This is an emergence of chartered corporations or self-governing towns, when kings and lords are forced to grant charters to towns, guilds, and universities, treating them as legal persons so they can hold property, make rules, sue, and be sued. This emergence of towns that have autonomy or semi-independence stabilizes rights. It allows durable organizations for trade, education, and mutual aid. It creates a template out of which many other institutions will grow.
- 6.10** So what are the institutions that we are talking about here? We are talking about the emergence of the north Italian city states, such as Milan, Florence, Genova, and Pisa, who formed the Lombard League, and gradually became self-governing communities who elect their own officials. As I will be explaining in subsequent lectures, what's important is really the competition between different territories in Europe.
- 6.11** As I will explain, if you have just one ruler, very often that ruler can make bad decisions, decide that certain innovations are inappropriate, just like Tiberius did with our unbreakable glass that we spoke about in the last lecture. But if you have dozens, maybe hundreds of different territories that compete with one another, it is much more difficult to restrain innovation from taking place. And so we start in northern Italy in the first century of the second millennium, precisely because it is in this part of the world that we see the emergence of self-governing and autonomous communities who can then compete with one another.

- 6.12** The second step here is the Peace of Constance, where the Holy Roman Emperor, Frederick I, is forced to recognize the autonomy of these city-states. They still have to acknowledge his ultimate sovereignty, but he acknowledges that they are autonomous, so they can compete with one another. Subsequent innovations are, for example, the Magdeburg Law, which becomes a chartered city. And then, based on this model, hundreds of other cities in central and Eastern Europe also become semi-independent.
- 6.13** In 1215, the city of London, liberties and its charter is confirmed by King John, which is then echoed in Magna Carta. So once again, here we have the city of London getting more independence. In 1226, Lübeck is recognized as a free imperial city within the Holy Roman Empire. And then later, Lübeck is at the beginning of the mutual aid treaty, otherwise known as the Hanseatic League. All of these cities become important centers for scientific and institutional and cultural innovation. So we want to start there because, as I said, it is in these autonomous cities that the human spirit is allowed to take flight, to invent and innovate and move humanity forward.
- 6.14** Another crucial step in political, institutional, and legal innovation are professional courts and the general legal system. So, in the second millennium, we have the revival of the Roman law, which was lost for many centuries after the collapse of the Western Empire, and Roman law spreads through the continent. We also have the growth of the common law in England, which produces trained judges, written procedures, general rules. Over time, there's codification of law, such as the Napoleonic code, in 19th-century France. And this creates clear, predictable civil law, which reduces disputes, lowers transaction costs, and supports long-distance commerce.
- 6.15** So once again, it is when you have improved and professionalized legal systems, what the economists call transaction costs can be lowered. Say, for example, that in the olden days, if somebody stole something from you, you would get a bunch of your kin and you would go and wipe out that particular village in order to punish them—a very high transaction cost, a lot of people die. But if you now have a professionalized court system that you can trust because it is impartial, nobody has to die. You just throw one guy into the prison, and everybody can move on.
- 6.16** What are the innovations that happen in Europe? Beginning in the 12th century, we have the Assize of Clarendon under Henry II in England, who strengthened the royal courts and itinerant judges. Back then, judges didn't just sit in one place; they would go around England, dispensing justice. In 1178, we have the origins of Common Pleas; but also Magna Carta in 1215 now requires that judges sit in one place, so instead of judges going to some place to deliver justice, people now come to judges who sit in one court, and this is how it happens to this day, 800 years later.
- 6.17** In 1234, we have Liber Extra, the official papal compilation of canon law that standardizes church courts across Catholic Europe and trains a pan-European class of professional jurists. This becomes very important in the next lecture, because you now have two competing legal systems emerging in Europe. On the one hand, the secular

legal system of, say, Henry II in England, but also the papal and the Catholic legal system that we call the canon law. And so, it's not just a competition between different rulers and different city-states. You now have competition between the secular and religious authority. And it is between these competing authorities that you get a lot of innovation squeaking through the middle. We'll be talking about that.

- 6.18 In 1495, you have the creation of the Reichskammergericht, the Imperial Court Chamber, in the Holy Roman Empire—again, a big move toward professionalization of the court system. Habeas Corpus gets passed in 1679 in Britain, which secures the right to challenge unlawful detention in court, and it is a major constraint on arbitrary imprisonment. And ultimately, you have the Civil Code, or the Code Napoleon, in France at the beginning of the 19th century.
- 6.19 We have the rise of representative assemblies and the consent to taxation. Very important in terms of institutional development, that the king must now come to the people and ask them for money, rather than the king simply being able to get money on his own. If a king is able to get money on his own, then there is no need for representative assemblies. So, kings are forced to summon representatives of towns, nobles, and clergy to approve taxes and sometimes even laws.
- 6.20 The norm, no taxation without representation, gave economic actors some control over fiscal policy. If you are a merchant, you're no longer worried that the king was going to take all of your stuff away because now you have a say in parliament. You can say no. And over time, obviously, these parliaments get more and more power. The franchise is enlarged, and it makes governments much more accountable.
- 6.21 The origins of this can go back 1,000 years. The Cortes of León in Iberia in 1188 includes town representatives and asserts that new taxes require their consent. In 1215, Magna Carta says that new taxation must be approved by the "common counsel of the realm." In 1265, Simon de Montfort, the rebellious Englishman, enlarges the types of people who can sit in parliament. The États Generaux in France takes place for the first time in 1302 under Philip IV.
- 6.22 In the 16th century, we have the Dutch Estates General and the Act of Abjuration, where the Dutch basically say, "We want to be independent. We no longer want to pay Spanish taxes and live under a Spanish king." And that Estates General, it's still called Estates General to this day, becomes the representative body of the Dutch Republic, and later, the Dutch Kingdom. And of course, in 1776, we have the American revolt against taxes being imposed on them by Parliament in Westminster. And finally, we have the democratization of the franchise with the British Reform Acts in 1832, '67, and '84.
- 6.23 Even in the United States, where the franchise is not, as you well know, universal in 1776 or 1787, a greater share of Americans have a right to vote than anywhere else in the world. We all know in which ways the American Constitution and the American

Declaration of Independence fell short. What we don't appreciate is that, even though the franchise was limited in the 13 colonies, it remained the most democratic system until the 1840s. A smaller share of people in other parts of the world had a vote than even the restricted American franchise.

- 6.24 We have the increase in constitutionalism and higher law limiting rulers. So it's not just that people have the ability to impact law and the creation of new laws. But there is a sense that starts creeping in, in the 13th century, that certain laws are more important than others. In the United States, for example, we have a sense that freedom of speech is somehow more important than that people can drink at the age of 21. That's constitutionalism, right?
- 6.25 So documents like the Magna Carta, and later the Bill of Rights, and a written Constitution, treat some rules as superior to others. There's the idea that there's a higher law enforceable in courts and legislatures, restricting power of the—arbitrary power of the government, protects private property and speech, and creates a general template for limited government in many countries. Magna Carta is the obvious example in England in 1215. But we also have the Golden Bull in the Roman Empire in 1356.
- 6.26 Then we have the Glorious Revolution, truly a fundamental step in the history of mankind, when the British rulers, William III and Queen Mary, basically agree that they will become kings under the law, that they cannot do anything they want. There is an unwritten constitution that the Parliament is sovereign. In 1776, we, of course, have the American Revolution and the Declaration of Independence. The constitutional Bill of Rights is adopted eventually in 1791, I believe, and the French Declaration of the Rights of Man and Citizen in 1789.
- 6.27 We have the creation of the modern bureaucratic nation-state. I'm iffy on this issue because very often the modern bureaucratic nation-state can actually be the enemy of innovation. Nonetheless, a lot of scholars, perhaps even a majority of scholars, believe that having bureaucratic nation-states was an important step in terms of human progress. States, which could build standing administration, which could conduct censuses, that could impose regular tax systems instead of relying on ad hoc levies and feudal dues, performed better.
- 6.28 Predictable revenues and professional administration allowed governments to build public goods, such as creation of defense or building of roads and ports and basic education. Again, I'm a bit iffy on this issue, partly because many of these public goods, such as education, was provided privately before the state stepped in. Nonetheless, when it comes to this particular driver of human progress, surely the most important date there is 1648, the Peace of Westphalia, which ended the 30-Year War and codified the nation-state system in which we live today.

- 6.29** Then you have the French Jean-Baptiste Colbert, who starts reorganizing France in the 17th century. Prussia follows in the 18th century. Even Spain, after it changes monarchs from the Habsburgs to Bourbons, starts reforming. These are the main innovations when it comes to modern bureaucratic nation-states.
- 6.30** In conclusion for this section, we have the creation of chartered towns and guilds, which create pockets of self-government, with more predictable, negotiable rules and a lot of competition. We have professional courts and general legal systems, which protect property and contracts beyond local custom. We have representative assemblies that tied taxation and lawmaking to some form of consent, forcing the rulers to bargain with the populace.
- 6.31** We have constitutional settlements and bills, which placed speech, religion, and enterprise under a higher law. We have modern bureaucratic states and regular taxes, which replaced ad hoc exactions, and more reliable administration of public goods. And together, these changes gradually produced a more open environment in which initiative, cooperation, and long-term planning could take place.
- 6.32** We move on to scientific, technological, and epistemic innovation. One of the most important innovations there is the creation of the medieval and enlightened university system, which again begins in the first century of the second millennium with the creation of the University of Bologna, which emerges as a law-focused "studium" with corporate privileges and creates a template for other universities. So universities create permanent communities of scholars with common rules, degrees, and curricula.
- 6.33** And what began, as the University of Bologna did, as a school of law and theology, became then a multi-faculty institution. And by the time that we arrive in the Enlightenment, in the 18th century, some universities are not just teaching, but they are explicitly devoted to research, creating new knowledge, not just passing on old knowledge, but creating new knowledge. And it is this creation of universities that then provides Europe with a never-ending stream of clergymen and lawyers and doctors and officials and scientists.
- 6.34** So, the University of Bologna is the beginning, but the University of Paris follows. You have the University of Oxford and Cambridge, then the University of Prague in 14th century, which is the first university in central Europe. The University of Heidelberg is the first one in the German-speaking lands, in the 14th century. Göttingen is created especially as an Enlightenment university, back in the 18th century; and in the 19th century, 1810, we have the University of Berlin, which becomes the first research university. And basically, modern universities, whether we are talking about Harvard or Stanford, they are basically based on the 1810 German model of not just a teaching, but research university.
- 6.35** Next, we move on to Renaissance humanism and critical reading of texts. After the Roman Empire falls, many scholars, I would include myself among them, talk about the

Dark Ages in Europe, when knowledge disappears. That knowledge then gets rediscovered during the Renaissance and during the period of Renaissance humanism. And that's a good thing, because people are now at least able to internalize information that the ancients have. The ancients were here. During the Dark Ages, knowledge went down here. And now we are getting back up to the place where the ancients were.

- 6.36 That's a critical step in terms of human progress. But as we will learn later, the other critical step is to overcome the ancients and become better than the ancients. So what we want to do is to go from here to here to here, and then to there. So eventually, we want to transcend ancient knowledge. But in this period, which is also important for human progress, we are simply rediscovering ancient knowledge which was lost.
- 6.37 So humanists rediscover classical authors. They promote written, elegant Latin. They develop textual criticism, which we will discuss in a moment. They compare manuscripts, they correct errors, they ask who wrote which text and why. And they start questioning old authorities. Maybe some things that the ancients thought they knew are no longer true—very important in terms of human progress. It is good that you have rediscovered the knowledge that you previously had, but maybe some of that knowledge is wrong, so you have to criticize it, you have to be skeptical about it, and you have to build on top of it.
- 6.38 Some of the things that humanists talk about is human dignity, personal agency, equality under the law. Civic virtue—that a ruler should not just rule for himself, but he should rule for his community—becomes a central thesis of humanism. And there's a rising status of learning, public debate, and participation in urban life.
- 6.39 One of the earliest humanists, perhaps the earliest one, was Petrarch, who rediscovers Cicero and other human authors and launches "studia humanitatis," the study of grammar, rhetoric, history, moral philosophy, and poetry in Europe, which lost many of these things during the Dark Ages. Leonardo Bruni, in the 15th century, links classical learning to active citizenship. In other words, he says, learn from the Romans. Look how they believed in law and liberty. Learn from them and repeat that for the modern era.
- 6.40 Lorenzo Valla is a very important textual critic who exposes the Donation of Constantine as a forgery. The Donation of Constantine, basically, says— well, it was a forgery, which basically said that Constantine the Great granted Western Europe, the Western Empire, to the Pope. And Lorenzo Valla, by undertaking textual analysis, basically is able to prove that the Donation of Constantine, this text, is a forgery. It is a much later creation because many of the words which are used in the Donation of Constantine didn't exist in classical Rome.
- 6.41 Erasmus follows in Valla's footsteps. He produces the first printed edition of the Greek New Testament, with a new Latin translation. And he notes that there are errors and variants in the Vulgate. Vulgate is the old, uncorrected biblical text, the Latin text. And

basically, by comparing the Vulgate with the new translation of the Greek New Testament, he's able to point out that there are many inconsistencies and many problems in the way that the Bible is being read in Europe at that time.

- 6.42** We return back to printing, Protestantism, and literacy. Innovations there include the Gutenberg printing press and Bible in the mid-15th century. We have the spread of the printing press to England in 1476. We have Aldus Manutius, who produces inexpensive italic type and octavo editions in Venice. So the original books that are produced by Gutenberg are very expensive to produce. They're called incunabula. But this particular Venetian is able to create cheaper types; he's able to decrease fundamentally the price of books, which means that more people, even ordinary people, can now buy them.
- 6.43** So now we have cheaper books, we have printing presses, and the next step, of course, is Luther, and him pinning his theses to a church door, which are then spread through Europe in huge numbers. It is also Luther who completes the first German Bible in the vernacular, which is to say that people are not only able to read the Bible by themselves, as opposed to go through the medium of the priest, but they are able to read it in their own language. They're able to engage with the Bible in their own way.
- 6.44** So you've got printing, you've got cheap books, you've got Protestantism, and you've got a spread of new ideas. The central authority of the Catholic Church, as a moral authority, collapses. Now, some people think that that's not a good idea. Other people will be very happy about it. But the bottom line is that, when you have a central moral authority collapsing and a new one emerging, you suddenly have a competition of ideas. Christians actually come up with better arguments against one another.
- 6.45** So, if you are an atheist, you may say, well, it doesn't really matter what arguments you have in favor of defending Protestantism or Catholicism. But it does matter to Christians, whether you're listening to better ideas in favor of Catholicism or better ideas in favor of Protestantism. And because you now have collapsed central moral authority, you have different moral authorities—you have Luther, you have Calvin, you have the Pope—they all now have to compete with one another with better arguments, in order to get people to stick with their particular denomination.
- 6.46** We have the emergence of the scientific method. So from the 13th century onward, European scholars began to treat knowledge of nature as something that had to be understood through experimentation and observation, rather than commentary on ancient authorities. Forget about what Aristotle said about flora and fauna; some of it may be true, but we want to move beyond that. We want to collect more knowledge. We want to improve on Aristotle.
- 6.47** And building on earlier works, even on Islamic scholars, for example, European thinkers argue that controlled trials, measurement, and careful records were the best way to understand the world. So by the 17th century, this experimental and inductive approach has been organized into shared rules and institutions for doing science.

Inductive as opposed to deductive. Deduction is where you start with a theory, and then you work your way down. Inductive is where you start by collecting data and experimenting with them and building a theory back up.

- 6.48 The beginner in this story, at least in my view, is Roger Bacon, completely unrelated to Francis Bacon, about whom we'll be talking about, who insists on experiments as foundations of natural knowledge in 1260s. Leonardo da Vinci, of course, uses systematic observation, dissection of the human body, and practical experiments in the 1500s. Francis Bacon, in 1620, criticizes deduction, theory working downwards. He sets out a program of inductive reasoning, working from the bottom up, and calls for an organized collaborative research with detailed records.
- 6.49 And in the 1660s, you've got the creation of the Royal Society of London and Philosophical Transactions, which formalize—Philosophical Transactions is a publication—they formalize this experimentation. They formalize reporting, replication, peer scrutiny, thereby turning the scientific method into collective practice.
- 6.50 Another very important move in terms of human progress is precision science. Between 1500 and 1800, the spread and routine use of new instruments extended the senses and allowed nature to be measured with unprecedented accuracy. These tools made it possible to test theories, to compare results across places, and to build cumulative sciences such as astronomy, physics, and physiology.
- 6.51 What are the main innovations here? Well, telescopic astronomy and navigation, in the 1600s. Telescopes enter observatories and navigation. Remember pirates with those long telescopes? We have sharpening of maps. We are able to see the surface of the moon for the first time. We have improved long-distance sea travel. By the mid-1600s, we have microscopy. Microscopes become standard lab tools. We begin to understand the microbial world, although the germ theory of disease has to wait another 200 years.
- 6.52 In the 1650s, we have experiments with vacuums. Air pumps, for example, enable us to test gases and sound and combustion. This is a huge step. Aristotle says that a vacuum is impossible, but here we are discovering that it is possible to do a vacuum. Sometimes very cruel methods. You put a bird there, you vacuum out all the air, and the bird dies. Well, we know there is a vacuum.
- 6.53 1656 onwards, we have the spread of clock-disciplined science, accurate pendulum clocks, astronomical observation and experiments. In 1700s, we have shared temperature scales, standardized thermometers, Fahrenheit and Celsius, which give science and medicine common numerical measures of heat. In 1730s and 1760s, we began to have reliable longitude at sea. Sextants and marine chronometers are adopted in navigation, making ocean routes safer and enabling global data collection.
- 6.54 We have the emergence of new cosmology, possibly my favorite step in terms of human progress. In the Aristotelian and the Ptolemaic cosmos, Earth stands in the

middle of the universe. It stands still in the center, below the celestial spheres. So this Earth is in the center. It is being circumvented by stars, by planets, by the Sun.

- 6.55** And this view of Earth being in the center is then being matched in the European thought by what is called "the great chain of being," with church and state ranks being treated as God-given and immovable. So just as the Earth sits in the middle, still surrounded by planets and stars and the Sun, so the Pope sits in the middle, surrounded by cardinals, bishops, and priests. The king sits in the middle, immovable in the center, surrounded by princes and dukes and viscounts and barons and people, ordinary merchants then peasants, and ultimately, slaves or serfs.
- 6.56** Copernicus comes along, and in 1543, proposes that this is not so, that Earth is one planet amongst many. So the universe no longer looks like a tidy static hierarchy, and the argument that social order must mirror the fixed cosmic order is destroyed. You can no longer argue that my place in society as the king is right, immovable in the center, because the Earth-centric concept of the cosmos has been undermined.
- 6.57** Galileo, Kepler, and Newton then show that heavenly and earthly motions obey the same mechanical laws. For the longest time, Europeans believed that there are celestial spheres about the Earth which are perfect; that's where God lives. But Earth itself is corrupted. But Galileo and Kepler and Newton say that both the cosmos and the Earth behave according to the same laws of nature, such as gravity, for example. And so you can no longer talk about the uncorrupted versus corrupted. We are part of the same whole.
- 6.58** And finally, the new cosmology changes how Europeans understand their relationship with God. God before Kepler, before Newton, before gravity, is a harmonizer. He is omnipresent in your daily life. He gives you pleasure, he gives you pain, he gives you love, he gives you treachery. Everything that happens in your life is because God interferes with you every single moment, according to his will.
- 6.59** But after the new cosmology is introduced, after the Aristotelian and Ptolemaic models are removed, God becomes the great architect. He becomes the great mechanic. He's the one who creates the rules of the game, but then steps back, allowing you to live your life, free will, making mistakes, but also achieving great accomplishments. God is no longer a micromanager. He is not a harmonizer, pulling on different levers according to what suits him. He is a law-giving architect.
- 6.60** So nature now is governed by impersonal rules. Yes, they are God-given, but they are impersonal; and priestly, monarchical, and feudal hierarchies no longer appear sacred. There is more space for criticism and for reform. So here the innovations are, as I said, Copernicus' heliocentric model, which challenges geocentrism in the 16th century. Tycho Brahe in the 16th century also makes observations of comets and undermines the idea of celestial spheres.

- 6.61** Galileo, and his telescope discoveries, discovers lunar mountains. He discovers the phases of Venus. He discovers moons around Jupiter. It's not just the Earth that has a moon. Other planets have moons. We are not that special, he says. Kepler, in the 17th century, introduced the three laws of planetary motion. And finally, in 1687, gravity. Universal gravitation and laws of motions unify terrestrial and celestial mechanics in one mathematical framework. God is a mechanic, not a micromanager.
- 6.62** We also have the emergence of new engines, machines, and power which move human progress forward. In 1589 and after, we have mechanized knitting. Coke-based iron production arrives in 1709. Steam engines begin to drain mines from too much water in 1712. Factory cotton spinning in the mid-1700s. Steam power becomes general industrialized power in the 18th century. And by the late 18th century, we have fully mechanized textile mills, which are able to produce plentiful clothing for human beings for the first time in history.
- 6.63** We already talked about the importance of scientific societies which organize knowledge, journals which allow for peer review, and we talked about the Republic of Letters. These institutions, journals, and so on set up regular meetings, peer scrutiny, and priority rules for discoveries. This open competitive system rewarded useful findings, discouraged secrecy, and coordinated research efforts across countries. It linked propositional knowledge, theory, with prescriptive knowledge, techniques, and sped up innovation, such as vaccination.
- 6.64** We already talked about it. We talked about the Republic of Letters, between the 16th and 18th centuries; Royal Society of London, founded in 1660; Philosophical Transaction as a publication in 1665; the Académie Royale of Sciences in Paris; the Prussian Academy of Sciences in 1700; and the Royal Society of Edinburgh in 1783.
- 6.65** Another step in human progress is standardized measurement and the metric system, which replaces thousands of local units. Standard measures improve science, engineering, and trade, since experiments and contracts could be replicated and compared across regions without disputes over conversion. France, in 1795, becomes the first country to adopt the metric system, followed by the Netherlands in 1816. British Weights and Measures Act standardizes imperial units, which are still shared by the United States today, in 1835, and even Spain finally gets there in the middle of the 19th century.
- 6.66** So, in conclusion, we have the Renaissance humanism and critical reading, which raise the status of learning and train people to question inherited texts and authorities. We have the medieval and Enlightenment universities, which gave learning durable homes with corporate privileges, degrees, and regular curricula. We have printing with movable type, which spreads texts cheaply, so scholarship, religious debate, and practical manuals reach a far wider public.

- 6.67** We have experimental scientific method, which reframes knowledge as testable and cumulative, rewarding observation and replication rather than commentary. New instruments such as telescopes, microscopes, clocks, and barometers extend the senses and turn qualitative impressions into precise measurements. The Copernican-Newtonian cosmology shows us that the same simple laws govern heaven and Earth, strengthening confidence in a universal, intelligible order.
- 6.68** We have scientific societies, journals, and the Republic of Letters, which link investigators into a competitive, cooperative network that validated and diffused discoveries. We have engine-powered production systems and standardized measures, which translate growing knowledge into reliable power, productivity, and engineering in everyday life. And taken together, these changes turn knowledge into a cumulative, shared, and applicable enterprise, creating a culture in which questioning, testing, and inventing became normal ways of understanding the world and improving everyday life.
- 6.69** We turn to economic innovation, oceanic voyages, which opened new sea routes that linked Europe, Asia, Africa, and the Americas into a single trading world. Chartered trading companies and colonial empires then organized regular flows of silver, spices, and textiles, and other goods across oceans. In the 19th century, governments began dismantling mercantilist barriers, lowered tariffs to trade, and shipping restrictions, which deepened global markets and raised productivity. It starts in some ways with Columbus in 1492, when he sailed for the Americas, and the Columbian exchange takes place, where crops from Europe go to the Americas, and then American crops come to Europe.
- 6.70** Vasco da Gama reaches India in 1498 by sailing around the Cape of Good Hope in South Africa, thus creating a direct route to Europe via the Cape, because Europeans can no longer go to India by land, because Constantinople falls to the Ottomans. So they have to find a new route, and they find it around the Cape of Good Hope. Magellan, between 1519 and 1522, is able to circumnavigate the globe, thereby proving that the world is actually round.
- 6.71** Between 1600 and 1602, the English East India Company and the Dutch VOC are founded as charter companies to organize long-distance oceanic trade. In 1846, we have the repeal of corn laws in Britain. By corn the Brits meant crops such as wheat, not just corn. And the market in Britain was heavily protected because the aristocrats owned land and they didn't want people to buy crops from abroad, because that would lower the price of grain and thereby reduce their profits. But in 1846, the Brits opt for free trade, and suddenly the price of their food decreases.
- 6.72** In 1849, we have the repeal of the Navigation Act, under which if you wanted to bring stuff into Britain, it had to be done on a British ship. Now anybody can do that. We have progress in finance. Merchants and bankers create new ways to record transactions, pull capital, and move money that make commerce more reliable and

scalable. Here we have such things as partnership contracts in Italian city states, all the way to the 11th century.

- 6.73** Very often, in the old days, if you undertook a venture and it went south, if the ship sank, for example, not only did you lose all of your investment, but you may lose your house, you may be put into debt slavery, you may literally be put in prison because you cannot pay the debt. But now what starts happening is that we have pooled the risk. A lot of people are coming together in order to reduce the risk of partaking in commerce.
- 6.74** In the 13th century, Italian merchants who sell their products in France, they no longer carry with them actual coins because it's very dangerous to travel in Europe. Instead, they carry promissory notes. And those promissory notes eventually give rise to paper money that we have on us today. In the 16th century, we have the first bourses, which start in Antwerp in 1531. So the stock exchange that we associate with Wall Street, that really starts in Antwerp in the 16th century.
- 6.75** In 1555, in England, we have the emergence of joint stock trading companies, which means that if, as a businessman, you lose your money on a venture which doesn't work out, you only lose the amount of money that you put into the venture. But everything else—your house, your immovables— you retain. So that makes business-making easier. And in 1587, we have the establishment of the first public bank, which is used for deposits and transfers under state supervision, which again makes financing of state debt and therefore paying for public goods easier.
- 6.76** We already talked about patents. The first patents appear to have emerged in Venice in the 15th century, but then they spread to England in the 17th century. The first US patent is extended in the United States in 1790; and in 1791, you have the Patent Office established in France. So in conclusion, when it comes to economic innovations, we are talking here about oceanic voyages and new sea routes which tie Europe, Asia, Africa, and the Americas into a single trading world. We have charter companies and regular shipping, which turn scattered exchanges into a stable global flow of silver, spices, textiles, and other goods.
- 6.77** We have partnership contracts, double-entry bookkeeping, the public banks, and we can now move funds without hauling coins. We have bourses and joint stock companies, which create deeper markets for shares and public goods, spreading risk in financing distant enterprises. We have patent statutes and modern patent offices, which grant time-limited rights in return for disclosure, rewarding invention while building public store of technical knowledge. Together, global trade, sophisticated finance, and patent systems mobilize resources and ideas into an unprecedented scale, supporting sustained innovation and growth.
- 6.78** Let's now move on to moral and ethical innovation. Let us first talk about the Enlightenment and secular morality, which emerges in the second half of the second millennium. Thomas Aquinas, in the 13th century, argued that basic moral norms flow

from human nature and reason, though he ultimately traces that nature to God. So, this natural law framework, as he called it, treated all humans as sharing a rational nature and therefore common duties and rights. So you have natural law, which is ultimately based on religion, on God.

- 6.79** Enlightenment thinkers push this line of thinking a little bit further. And they increasingly justify morality, law, and political authority in terms that did not depend on the church and scripture. They basically say that you can come up with morality based on secular logic, rather than the necessity of revelation. A lot of people disagree with that, but nonetheless, that is what the Enlightenment thinkers believed.
- 6.80** Hugo Grotius, in the 17th century, says, "What I have said would be relevant even if we were to suppose that there is no God, or that human affairs are of no concern to him." He gives us an outline of a moral view that he says works, even if there is no religious scripture to support it. John Locke, in 1689, recast natural rights to life, liberty, and property, and broad toleration, in largely rational terms. Diderot and d'Alembert, in France, diffuse secular critical knowledge and portray morality as improvable by reason and education. American Declaration of Independence in 1776 states that governments derive just powers from the consent of rational equals with rights.
- 6.81** And in 1785, Immanuel Kant develops a purely rational, universalist morality, the categorical imperative, independent of theology. He says that you should act in such a way that your actions should be universally applicable; that is the categorical imperative. But it is a version of the golden rule, which says that "do unto others as you would have them do unto you." So the Enlightenment thinkers believe that you don't actually need, like Aquinas thinks he does, religious affirmation of morality, that basically by me being me and you being you, us being human beings with equal dignity, the same rights that I claim for myself must also be applicable to you. There is reason and reciprocity, which is supposed to produce secular morality.
- 6.82** We have the increase in the freedom of conscience and religious toleration. After Protestants split from the Catholic Church, years of conflict follow. The 30-Year War between 1618 and 1648 is one of the bloodiest conflicts in European history, in which one-third of central Europe perish. After long religious wars, thinkers and even entire countries accept that individuals should not be forced into a specific belief and worship. As a general rule, this new toleration, after the conflicts such as the 30-Year War between 1618 and 1648, in which one-third of central Europe dies, after these conflicts, a new spirit of toleration rises in Europe.
- 6.83** There is reduced persecution of minorities, especially religious minorities. Religious minorities are allowed to participate in public life, and it creates a more open intellectual environment for science and philosophy. Some of the innovations here are the Edict of Torda in Transylvania, which grants freedom to several Christian denominations in 1568; the Union of Utrecht in 1579, the Edict of Nantes in 1598, in which Henry IV extends religious toleration to the Huguenots. We have the Edict of

Potsdam in 1685 in Prussia, and ultimately, we have the Patent of Toleration by Joseph II in the Habsburg lands in 1781. In all of these cases, we have more religious toleration and more participation of religious minorities, such as Protestants and Jews even, in public life.

- 6.84** Another aspect of human progress, another innovation in human progress, I believe, is equality before the law and abolition of legal privilege. Revolutionary and reform movements attacked noble estates, guild monopolies, feudal exemptions to the equal rules of the game, arguing that everybody should play according to the same rules. That didn't matter whether you were a commoner, a peasant, a nobleman, or a clergyman. Everybody needs to be equal under the law.
- 6.85** And this emphasis on equality before the law weakened traditional hierarchies in Europe. And it opened careers to talent. You no longer had to be a nobleman, for example, to buy a commission in the French army or a British army. What mattered was a talent, and one of the first people to understand that is Napoleon. And he starts promoting people on purely talent and manages to conquer all of Europe, before other European countries catch up onto the secret sauce that Napoleon has.
- 6.86** So some of the main innovations here are the August Decrees in France, which abolish feudal privileges in 1789; the US Constitution in 1787, which abolishes aristocracy, or rather, it says that the United States shall have no aristocracy; the Rights of Man in 1789 in France, we already discussed, as well as Stein-Hardenberg edicts in Prussia, which begin to abolish serfdom and noble privileges in that country at the beginning of the 19th century.
- 6.87** We also return to Deirdre McCloskey's idea of bourgeois dignity, the cultural change in how societies in Europe viewed merchants, entrepreneurs, and inventors. So between 1600 and 1800, as I mentioned in the last lecture, there's a shift which Deirdre calls bourgeois revaluation. Rhetoric, or what people say about trade, profit, innovation, becomes much more respectful and even admiring. Commercial life is legitimized, encourages ordinary people to see improvements as honorable. If you come up with an innovation, it doesn't mean that you've done something wrong. Quite the contrary—it is something that is good.
- 6.88** And that helps to unleash waves of experimentation, enterprise, technological adoption, that moves society forward. And some of the main innovations here are the celebration of merchants in the Dutch Republic in the 17th century, the English and Scottish Enlightenment that defends commercial society as a source of liberty and mutual benefit. They say that, if I trade with you, even though I don't know you, I may not even like you, but we have to get along because that's the only way the two of us make profit. We also return to the idea of weakening of guilds and corporate privileges, especially after the Napoleonic conquest of Europe and the victories for free trade in the 1840s.

- 6.89** Another massive move in terms of human freedom is, of course, the abolishment of slavery and serfdom. Religious and secular reformers reframe slavery, bondage, as violations of basic human rights. Their argument, their campaigns, led European states to abolish slave trade, then slavery and serfdom. And then they supplied a general moral precedent for later struggles against other forms of coercion. So as I noted, slavery was a universal and eternal phenomenon in the world, but Britain, which abolishes domestic slavery hundreds of years ago, decides in the late 18th century to abolish slavery not only in England, but also throughout its empire.
- 6.90** Britain, in the late 18th century, beginning of the 19th century, becomes the first empire to abolish slavery. There have been many empires before; it didn't occur to anybody else to do that. The Brits were the first. Among the innovations there are the famous *Somerset v. Stewart* case in 1772, when I believe it is an African slave is freed because he was brought into England by his slaveholder. But in this particular judgment, the judge says, "There is no slavery in England. The man is free." In 1807, the British abolish slave trade, so slavery is still permitted, but trading of slaves is forbidden.
- 6.91** In 1814, at the end of the Napoleonic Wars, at the Treaty of Paris, European powers are all cajoled by the British to start working toward ending of the slave trade; and in 1833, slave trade is actually abolished throughout the British Empire. In 1863, you have the Emancipation Proclamation in the United States, where slaves are freed by President Lincoln in the rebelling states. And by 1861, you have the abolishment of serfdom in Russia. Serfdom is not exactly slavery, but it is certainly an infringement upon human freedom. And Russia becomes, I believe, the last European country to do so in 1861.
- 6.92** So in conclusion, natural law theories grounded in morality, in shared human nature, and in reason, supplied a language of universal rights and duties. Freedom of conscience and religious toleration reduced persecution and widened the space for open debate, science, and dissent. Equality before the law and the abolition of feudal privilege weakened hereditary estates and guild monopolies, opening careers to talent.
- 6.93** Bourgeois dignity raised the status of merchants, entrepreneurs, and inventors, making profits and innovation socially respectable. Abolitionist and humanitarian movements condemned slavery, serfdom, and violations of basic rights. And together, these changes shrank the moral realm of coercion and privilege and enlarged the legitimate sphere of voluntary cooperation, enterprise, and individual initiative.
- 6.94** Abolishing slavery seems like the obvious example of moral progress. In previous days and sessions, we've talked about things like economic progress and accumulating wealth and meeting basic needs. I'm just wondering, how do you think about the potential for moral progress in the future in the West?
- 6.95** Well, just as our ancestors are condemned by us for doing things which seem barbaric and cruel to us, I suspect that there will be things that will strike future generations

that we do today as barbaric and cruel. The obvious candidate for such a condemnation would be probably eating of animals. You can imagine a future where you take the best imaginable cow, the Wagyu A5 cow in Japan, you just prick her with a needle, take a few cells, and then you basically use those cells in order to build the best possible steaks for all of humanity forever.

6.96 Instead of murdering animals, what you do, you simply use their DNA in order to create food in labs, which may be cheaper and which may be safer because you don't have to inject it with antibiotics or hormones because it is created in a lab and whatever else. And so, you can imagine a future human who is enjoying a steak from a Wagyu A5 Japanese cow that is cheap, that is safe, that is not filled with hormones or antibiotics, and wondering, "How crazy were those people back in 2025, who enjoyed a ribeye from an actual animal who suffered a cruel fate, and then food which was not as perfect as the one that I'm enjoying today?"

6.97 It was a tough one. I hope it was OK. Thank you so much.

Lecture 7: Competition & Modern Prosperity

At a glance — Competition and the roots of modern prosperity.

- 7.1** Humans have always innovated. If we didn't innovate, we would probably perish. So the question is: can we show in an empirical way that, in the last 1000 years, especially in the European context, innovations really took off, that it was a special period in human history, where we have an unprecedented amount of innovations that have then contributed to human progress, first in Europe and then the rest of the world?
- 7.2** What I did was to compile 100 most important innovations in the economic sphere. I then compiled 100 most important innovations in the moral sphere, 100 most important innovations in the scientific and technological sphere, and finally, 100 most important innovations in the institutional sphere. My criterion was innovations that have contributed most to human flourishing, such as higher life expectancy, lower child mortality, and so forth.
- 7.3** Now, these are my top 100 innovations, and no doubt there will be people out there who will quibble on the margins. But I would venture to say that people who will review this course and who will look at it critically will accept at least 90% of them. We may quibble with some of them, but we all agree that, for example, the discovery of penicillin or antibiotics was an important move in terms of human progress, or for that matter, discovery of stone tools, or bows and arrows, or the steam engine.
- 7.4** So this is a list of the top 100 legal, institutional, and political innovations. And they are separated into yellow, which represents Africa; blue, which represents Asia; green, which represents the Middle East; red, which represents the West and Western offshoots, so Europe, the United States, Canada, Australia, New Zealand, and so forth; then we have orange, which represents Central and South America. And this is a look at these institutional, legal, and political innovations beginning roughly 3000 BC.
- 7.5** And as you can see, most of these legal, institutional, and political innovations actually start in the Middle East. That shouldn't be surprising because, as we all know from grade school, civilization, cities, writing, starts in the Fertile Crescent, in the Levant, in Mesopotamia. And so, here we see, at the beginning, state-backed writing systems for recording taxes and contracts. In fact, there is a view that writing starts as a mechanism for accounting. The taxman needs to know how many bushels of wheat you have delivered, and so you need to keep these records.
- 7.6** At any rate, as we can see here, looking at these 100 most important legal, institutional, and political innovations, they start in the Middle East, which is depicted by green. Then, if you look at the insert, the insert is the last 1,000 years. What you can see there is the massive prevalence of institutional, political, and legal changes in Europe. Truly, the great takeoff in terms of these kinds of innovations happens over the last 1,000 years, and primarily in the Western world.

- 7.7** Here we have a visual confirmation of the speed of these innovations speeding up, especially over the last 1,000 years, and then especially after 1800, which is the period of the Great Enrichment. Now, let's look at the top 100 scientific, technological, and epistemic innovations—epistemic innovations is in terms of how we understand knowledge. When it comes to scientific, technological, and epistemic innovations, we actually start in Africa.
- 7.8** That, once again, is commonsensical because our species, *Homo sapiens*, comes out of Africa. So it is in Africa that we would expect the first technological innovations to take place, such as stone tools, roughly two and a half million years ago. Then we would expect these technological and scientific innovations to move to the Middle East because that's, again, the start of civilization in Mesopotamia and the Fertile Crescent, and then moving on to the Western world. That's indeed what we are seeing.
- 7.9** On the X axis, which is to say the horizontal axis, you have Africa in orange. Those are the stone tools, 2.6 million years ago. Conquest of fire, maybe bows and arrows, etc. But it takes a very long period of time between those points of discovery. People are innovating, but they are innovating at an extraordinarily slow rate. The previous slide showed you a period of 3,000 years. This one shows you a period of two and a half million years. And look how sparse the innovation is for the first 2 million years or so.
- 7.10** From Africa, we move, as I said, into the Middle East. Then, here you have an insert, which is the last 1,000 years. Here, once again, you can see a lot of red dots. Those are scientific and technological and epistemic innovations in the Western world, beginning after 1500 or so. Once again, you can see the very strong, very rapid, very steep slope upwards.
- 7.11** Here we have the 100 most important moral and ethical innovations. Here, once again, we start in the Middle East, which is the origin of civilization. This particular chart doesn't have an insert because it's short enough; it's only 2,000 years. Once again, you can see how the geography of innovation moves from the Middle East, then to Asia, and eventually into the Western world. Especially after 1500, we have this massively steepening upward slope in terms of innovations in the West and the Western world.
- 7.12** Finally, we will look at the most important 100 economic innovations, or rather innovations that contribute to economic growth. This slide goes once again back to the Stone Age, back to the discovery of the stone tools, because stone tools are both. They are a technological innovation, but they are also an important economic innovation. Economic growth is basically your ability to do more per hour of labor. Before stone tools, the Australopithecines, or whatever the species were called before we came along, could produce only so much. With discovery of the stone tools, they could produce this much. Then with discovery of the steam engine, we can do this much per hour of labor.

- 7.13** So economic growth is really, how much can you accomplish for an hour of labor? So it's both a technology, but it's also an economic innovation, and that is why I included that. And that is why we have this slide going back to two and a half million years ago. And once again, you can see the start in Africa, then moving over to the Middle East, then Asia, and eventually to the Western world. And the insert, once again, shows us the last 1,000 years. And you can see how that is dominated by the Western world and by the steepness of the curve.
- 7.14** I have decided to separate the ages of innovations into four. Prehistory, again, going back all the way to the stone tools, contains roughly 2.5 million years. The ancient and classical world, roughly 3,500 years. Medieval to early modern period, roughly 1,200 years, and modernity, roughly from the birth of the Industrial Revolution, roughly 300 years. And what you can see here is that, in every period, the number of the most important innovations has increased.
- 7.15** On this chart, you can see the 400 innovations together. As you can see in the prehistory, we only have 24 innovations, be that technical, economic, legal, or scientific, only 24 in that massive span of time of 2 million years. In the ancient and classical world, we have 45 innovations, but spanning only 3,000 years. Between medieval to early modern period, roughly 1,200 years, we have 74 innovations. And then, in the last 250 or 270 years, we have an incredible 256 innovations. You can see how the speed of innovations is increasing.
- 7.16** This is the cumulative growth of innovations over time. We can see each innovation building on top of each other. The compound effect gives us this upward-sloping curve. Again, here we are looking at the entirety of the innovations, the 400 most important innovations, 100 per science, 100 per economics, 100 per legal, 100 per moral. On this slide, you can see the exponential acceleration in innovation rates.
- 7.17** So, innovation rates haven't just increased; they have exploded. The modern world, the last 300 years or so, innovates at about 95 innovations per century, compared to just 1.3 innovations in ancient times. Innovation today, in what I call modernity, the last 270, 250, 300 years, we are innovating at a 74-times higher rate than the people in the ancient world, and I'm not even talking about prehistory. If I took it all the way to the stone tools, 2.5 million years ago, we'd probably be innovating at 70,000 times more. But the bottom line is, we are innovating at roughly 75 times the rate of people in the ancient world.
- 7.18** Here are the innovations, types over time. So in prehistory, which is to say that very long period of time of 2.5 million years, technology dominated. 58% of all technologies were things like survival tools, conquest of fire, or control of fire, or bows and arrows, or stone tools. Then, in the ancient and classical world, we have morality and ethics. We have a lot of emphasis on that particular aspect of innovation. We have civilizations rising in the Middle East, in the Levant, in ancient Greece. People are now living together in cities. How do we get along? By what sort of rules are we going to live?

How are we going to treat each other? So a lot of emphasis is now put on morality and ethics.

- 7.19 Major religions, such as Christianity, for example, but also Buddhism and so forth, emerge in this period. Once we get to the medieval and early modern period, between 500 and roughly 1700, institutions are in the lead. We already talked about, for example, the Italian city-states. We talked about the Glorious Revolution. So this is a period when we have a disproportionate number of innovations happening in the institutional sector.
- 7.20 And by the time that we get to modernity, what is extraordinary is that we are innovating in all four categories at roughly the same rate. So there's a near-perfect balance between 24 and 27%. Innovations are proceeding at roughly the same pace in economic, institutional, moral, and technological spheres. It's an almost-perfect balance. We'll need to figure out why that's happening, but we'll leave that for the next lecture.
- 7.21 Here is the migration of innovation leadership. In prehistory, we have a lot of innovations happening in Africa, in Asia, and also in the Middle East. In the ancient and classical world, there is stuff happening in the West, no doubt about it, but the Middle East and Asia are in the lead. By the time we get to the medieval and early modern period, the Western world is definitely in the lead, and in modernity, the West, which is denoted in blue, is definitely dominant.
- 7.22 In the modern period, the last 270-300 years or so, 86% of all innovations come from the West and its offshoots. So if you are an innovator today, you want to be in Western Europe, you want to be in northern Europe, you want to be in the United States, you want to be in Canada, you want to be in Israel, you want to be in Australia, Japan, China, South Korea; you want to be either in Asia or in Europe.
- 7.23 So, finally, let me conclude by proposing a possible chronology of human progress. This is my personal view. But if I were asked to give a series of lectures on human progress, as indeed I was, I would want to make some chronological sense out of it. And so, I would probably pick the last 1,000 years. I would start with the medieval institutional and commercial revolution, which happens between roughly 1000 and 1500. I would then move on to the Renaissance humanism and printing between 1350 and 1550.
- 7.24 The voyages of discovery, Columbus sailing for America, Vasco da Gama sailing around the Cape of Good Hope, and the creation of global trade would feature very prominently between the 15th and 18th centuries. And those voyages then give rise to the Scientific Revolution from the 16th to 17th centuries, followed by Reformation, greater toleration and religious pluralism in, again, 16th-17th centuries.
- 7.25 This greater toleration then leads us to the Age of Reason, or the Enlightenment, in the 17th and 18th centuries, turning into classical liberalism and political revolutions of the late 18th and 19th centuries. Classical liberalism, by that, I don't mean liberalism as it is

understood in the United States today; by classical liberalism, I mean an emphasis on equality before the law, limited government, equality of dignity, respect for the individual, greater freedom.

- 7.26** Eventually, we have, of course, the rise of the Industrial Revolution and the Great Enrichment, also in the 18th and 19th centuries, after which the world is completely unrecognizable. The world today would seem like magic to people who lived 500 or 1000 years ago.
- 7.27** In this lecture, I hope to convince you that human progress is intimately linked to innovations, that people have always been innovating, but in the last 1000 years, and especially in the last 500 years, there has been a massive spike in terms of how much innovation goes on. People have always innovated, but the speed of innovation was incredibly slow. The question before us is: why has the speed of innovation increased? And in this part of the lecture, I'll be talking about the importance of competition between states in terms of driving innovations and ultimately human progress forward.
- 7.28** I will start by talking about what economic historians refer to as "efflorescences." Efflorescences is just a fancy word for saying that there had been periods in human history before, when you suddenly had a flourishing of human progress for a short while before it was snuffed out. We'll talk a little bit about efflorescences, and I will argue that innovation is always disturbing, distasteful to the powers that represent the status quo.
- 7.29** If you have an interest in the status quo, be it a religious leader or a financial plutocrat or a king, you don't like innovations. You are already sitting on top of the pile. You are sitting on top of the pyramid. You don't want anybody to upset the innovation. So, what tends to happen is that we have efflorescences of culture and innovation and wealth, but they get snuffed out by people who are threatened, by status quo powers, who are threatened by those innovations.
- 7.30** I will explain why, in Europe, innovation is allowed to continue, by making a claim that Europe was particularly competitive. The countries in Europe were particularly competitive, whereas that was not the case in other parts of the world, and I will explain what I mean by that. And so, precisely because European states were at each other's throats, precisely because they feared one another so much, they allowed innovation to happen, not because they liked innovation, but because it was necessary for them to survive. That will be essentially the point that I'm making.
- 7.31** I will conclude by asking whether modernity, the period of history that we live in today, is something special; whether, in fact, we are not just another efflorescence, where we have, for a couple of hundred years, an increase in standard of living, culture, innovation, and then it collapses like all the previous efflorescences. Let's start with efflorescences.

- 7.32 Jack Goldstone, an economic historian, says that an efflorescence is a temporary burst of exceptional growth and creativity, often one to three centuries, when a society experiences rapid population and urban growth, rising per-capita incomes well above the long-run premodern baseline. Let me stop here. The long-run premodern baseline may be something like 0.01% a year. At that rate, GDP per capita doubles every 6,000 years, essentially. It's pretty much stagnant.
- 7.33 Essentially, you as a person will never experience a higher standard of living than your parents and your grandparents, and your children and grandchildren will never see higher standards of living than yours. Because right now in the United States, we are used to 2% economic growth rate a year, at least, and if it's much less than that, we are very upset. Well, people in the past, their economies grew at 0.01% a year. Incredibly slow rate.
- 7.34 What Jack Goldstone is saying here is that the long-run premodern baseline was very low. It doesn't mean that the economies, the efflorescent economies, were growing at our rate; it's just that they were growing at a higher rate than what they were used to. Goldstone says when a society experiences rapid population and urban growth, rising per-capita incomes, intense commercializations and trade, accelerated innovation and cultural and scientific output.
- 7.35 Perfect example of efflorescence: classical Greece. You have city-states, you have philosophy, math, drama, intense trade, high urbanization. This is the Greece of Socrates and Athens and Plato and Aristotle. Then we go to the Hellenistic Mediterranean. This is Alexander the Great leading his army, spreading Greek wisdom all over the known world. And after he dies, the Greeks still dominate much of Europe and much of the Middle East. So we have flourishing of science and engineering in Alexandria, in Egypt, for example.
- 7.36 Pax Romana, ancient Rome, from the end of the civil wars, the victory of Augustus, all the way to the end of the second century AD. We have integrated markets. We have the law being applied everywhere. We have the road system. Many of those roads are actually still used today. We have large cities and high living standards for antiquity—not from our perspective, but in terms of what people were used to, those higher standards of living were certainly there.
- 7.37 But efflorescences were not just in Europe; they were in other parts of the world. Gupta, India, between 4th and 6th centuries AD, flourishing of mathematics, literature, long-distance trade, huge urban and agrarian gains. The Abbasid Caliphate, the so-called Islamic golden age from about 8th century to 11th century AD. Baghdad was the center of science and of commerce. If you were an intellectual looking to learn, if you are looking for a liberal, small-L, place where you can express your opinions and where you are being celebrated as a scholar, that was the place to be.

- 7.38** Mughal India, from the 16th to the 18th centuries: textile manufacturing export powerhouse, big cities. Song China, between the end of the first millennium to the 13th century AD, incredibly productive. You've got printing; the original printing presses were in Song China. You've got navigations. The ships that the Chinese built are many times larger than what the Europeans can come up with. You've got a higher per capita income.
- 7.39** So you have these periods in human history, these efflorescences, where, for a short period of time, say between one and 300 years, you have a sudden spike in innovation. But all of these efflorescences have always petered out. They've ended. Why? The Nobel Prize-winning economist, Joel Mokyr, tells us, "Every innovation is an act of rebellion. It upsets political, economic, religious elites that have a vested interest in the status quo."
- 7.40** Let's say that you are the king with absolute power. You don't want people going around with crazy ideas such as equality before the law and no feudal privileges. Let's say that you are the Pope, and the Catholic Church has the monopoly on wisdom throughout the Western world and the colonies. You don't want Lutherans and Protestants running around, telling you, "We have a better idea." Innovations are always an act of rebellion. People who have a vested interest in the status quo don't like that sort of thing.
- 7.41** And so, efflorescences very often come to an end. In the Ming dynasty China, officials fear that Chinese people sailing abroad will pick up bad ideas and bring them back to China and destabilize the empire. An edict goes out that no more international travel will be permitted. The court officials cancel the fleet. They burn it down. They shut down major shipyards. There's a prohibition on naval trade. And the goal here is to preserve the elite control. So the Chinese shut off their global trade, just at the time when the Europeans are beginning to set out for the voyages of discovery, the worst possible time.
- 7.42** In the Ottoman Empire, the religious establishment blocks printing of books in Arabic, so there's a very slow diffusion of scientific and technical texts. So just as in Europe, you have the much-improved printing press by Johannes Gutenberg, in the Islamic world, you've got the religious establishment, which basically prohibits books from being printed in the language of the Ottoman Empire. In the Qing dynasty, the Qing understood that the world was industrializing, but the court officials were incredibly conservative, and so they repeatedly vetoed the building of railways and telegraphs. So ultimately, China itself is conquered in large parts by the European colonial powers.
- 7.43** So the question is: why has innovation been allowed to take off in Europe? Why has it persisted? As I showed you in Europe, we have a much higher speed of innovation in the last 1000 years, especially in the last 500. Why was an innovation throttled, strangled, destroyed, by the powers of the status quo? And so, here I want to talk about the competition between European states.

- 7.44 Competition was a feature of European institutional, legal, political, and even geopolitical makeup for at least a thousand years. It really starts in the 11th century, so roughly a thousand years ago, with a so-called investiture conflict. When Pope Gregory VII, Hildebrand, declares that only the Pope has the ability to appoint bishops in individual European countries, that the word of the Pope is superior to those of secular rulers. Well, the one person who doesn't like that idea is the Holy Roman Emperor, and a massive conflict begins between the Holy Roman Emperors and the popes.
- 7.45 But neither one wins. Neither one emerges victorious from this conflict between papal authority and between secular German authorities. And so, what happens is that you have the emergence across Western Europe of two authorities. Maybe in Ming China, you only have the Emperor who can decide for the entire empire. But in Europe, for at least a thousand years, you have at least two competing authorities: the church and the secular authorities. Each have their own separate bureaucracies, their own laws, their own courts, and both claim legitimacy.
- 7.46 And precisely because you have these two overlapping jurisdictions, they create a space for innovators to squeak through in between. Princes, cities, universities, merchants, and dissenters can appeal to rival authorities. They can bargain, right? "Oh, German prince, the Pope is going after me. Please give me protection, and in exchange, I'm going to give you some innovations." OK. "Oh, Holy Father, the German prince is going after me. He's going to kill me. Please give me refuge, and I'm going to give you my innovations."
- 7.47 So you have these two competing authorities, and innovative people can squeeze through the middle in between them, playing them off each other. Obviously, when you have competing authorities, it makes it more difficult to impose a single continent-wide policy that can permanently destroy innovation or experimentation with new ideas. Now, in China and in many Islamic empires, rulers are much more fully in control of religious institutions. You don't have this separation of church and state. Very often, it is the ruler who is presiding over the secular authorities, but also over the religious authorities. When that happens, you have less institutional rivalry and fewer protected niches for autonomy.
- 7.48 So, competition is there from the start. But then other things happen as well. Well, first and foremost, we have the emergence of the Little Ice Age of the 1300s and the subsequent decline in agricultural production. So the European population is already weak and starving in the 14th century when comes the terrible bubonic plague from China and wipes out anywhere between one-third and one-half of people in the West. The usual, or the old feudal way of doing things, the norm, the expectations of feudalism, the relationship between peasants and lords starts to break down. The old arrangements basically become unstable.
- 7.49 Labor becomes scarcer—so many people have died—and becomes much more valuable, especially in Western Europe. Because in Western Europe, which was much

more urbanized, people were living closer together, and many more of them died. And so, every human life becomes much more valuable, and that strengthens ordinary people vis-a-vis the elites. "If you want me to work for you, lord, my feudal lord—not the other lord, my feudal lord—you have to treat me better. You have to give me more days off work. You have to give me more money." In Western Europe, landlords, the feudal lords, have to start bargaining with peasants. They have to increase wages.

- 7.50** In many Asian empires, on the other hand, the rulers respond to these shocks because the Little Ice Age is a problem for the entire northern hemisphere, and the bubonic plague obviously goes through the entire Eurasian continent. But in other places, rulers respond to these shocks, to demographic collapse and the cooling of the planet, by clamping down, by reasserting control, and by reinforcing old, rigid social orders, rather than bargaining.
- 7.51** With the fall of population, you also have the shrinking of the tax base. When everybody was a peasant, the only way that you could raise taxes was by taxing the peasantry, but peasants are now dead. And so, the rulers try to compensate for the falling population by increasing taxes on people who are still alive and by waging wars in order to conquer more territory on whom the peasants, who previously belonged to a different lord, live.
- 7.52** We have uprisings of the peasantry throughout Europe, China, and even in India. Dynasties fall, empires crumble. In Europe, for example, we have the French Jacquerie and the English Peasants' Revolt, both in the 14th century. Life is very difficult.
- 7.53** So, there is a broad pattern of crisis which is the same across the Eurasian continent, but the political responses to these crises differ. In Europe, the breakdown of order produces renegotiation, bargaining, and fragmentation, rather than the recreation of large empires. In Europe, the mix of cities, nobles, and assemblies make it harder to rebuild a single empire. So the continent remains politically plural.
- 7.54** But outside of Europe, these same pressures create a different response, one that is based on centralization of power and on coercion, preservation of hierarchy, rather than bargaining. So as the old world dies, out of that world, a patchwork of rival states emerges in Europe. You no longer have a dominant power like ancient Rome or, later, the empire of Charlemagne. You now have a patchwork of rival states. Kingdoms, principalities, city-states, bishoprics coexist. England, France, Castile, Venice, Florence, the Low Countries.
- 7.55** This fragmentation creates many competing sovereigns. If one ruler overtaxes the population or suppresses innovation, people and capital can move to a different ruler. That is not an option in China. Merchant cities and smaller states can specialize and innovate without waiting for permission from the center. You have political exit options. So, if the ruler grows particularly oppressive, you as an innovator can opt for exit. You go to a ruler who is more tolerant of commerce, skill, and experimentation.

- 7.56 What is fundamentally at stake here is this. If you're an empire—Ottoman, Chinese, Russian— you are very, very large, and you are probably quite confident that you're not going to be attacked and conquered by somebody else. You are much more concerned about domestic tranquility and domestic stability. Your fear doesn't come from outside, it comes from within, that if there is too much innovation, too many new ideas at the center in Beijing or in Moscow or in Constantinople, they are going to lose control, and the empire will disintegrate. So your incentive, as a ruler of an empire, is to preserve domestic stability by not having new disruptive ideas take hold.
- 7.57 In Europe, the ruling elite's incentives are the exact opposite. These are small countries, usually at war with one another, and they constantly fear an enemy. They constantly fear that if you, France, develop new military techniques, I, Holland, may suffer. I may be conquered by you. And so I'm going to make a different bargain. In order to remain in power, I'm going to give up power. I'm going to encourage innovation, because innovation is going to create more economic growth, more money, more tax revenue, out of which I can pay for larger militaries. And I'm also going to allow for greater technological and scientific innovations, so that you can produce better weapons for me, with which I can defend myself against foreign enemies.
- 7.58 So, precisely because Europe is not a large empire. Yes, Europeans have empires overseas. The Spanish have their empires in South America. The Portuguese have an empire in South America. The British have an empire in North America. But Europe itself is not an empire. It is a patchwork of constantly competing states. And so, you're constantly worried, unlike an empire, like China, you're constantly worried about being conquered from the outside. And so, you have to start to generate more innovation, more revenue internally, in order to protect yourself from foreign enemies and to stay independent and sovereign.
- 7.59 So you can see how the incentive structure in Europe is completely different than what it would have been in China. In order to remain in power, I have to give up power. I have to start sharing it with parliaments and start sharing it with the merchant classes, and I have to protect their rights, and I have to let them speak, because otherwise they will leave and they will make somebody else rich.
- 7.60 Now, that is not to say that, in Europe, things could not have worked out differently. Europe came very close to becoming one empire under the Habsburgs. The Holy Roman Empire, Charles V, is also the king of Spain. He is in charge of the largest empire then known, the Spanish Empire in Latin America. His son will inherit Portugal. Charles V himself inherits the Low Countries, and on top of that, the Habsburg lands in Central Europe. And so, he's this close to uniting Europe under the Habsburg rule. And once you have the Habsburg rule, you have one emperor, just like the one sitting in Beijing.
- 7.61 But it never happens. The Habsburgs are opposed by the French, by the house of Valois. They are opposed by the German princes, because right at that time, you have Luther and the rise of Protestantism. They are opposed by the Ottomans, who attack

the Austrian lands of the Habsburgs from the south, and you have the Dutch Revolt. So the Habsburgs are tied down. And eventually, they participate in a catastrophic Thirty Years' War, which results in the Peace of Westphalia in 1648, where the Habsburgs, once and for all, give up any attempts to unify Europe under a single rule, and you have the creation of the modern nation-state.

- 7.62** After 1648, the interstate rivalry continues. Other men attempt to unite Europe under their rule. Louis XIV. Napoleon actually succeeds for a few years. Kaiser Wilhelm. Hitler. But at the end of the day, Europe always returns to its very competitive international system. So, Europe avoids a single authority capable of freezing rules, ideas, and technology across the continent. Westphalia stabilizes a multi-state system where competition is permanent, so that states that fail to improve in their administration, in their credit, access to credit, and in the military science, they fall behind.
- 7.63** So that states that fail to improve in their administration, access to financial credit, and military science, they fall behind and risk being swallowed by others. New knowledge is encouraged, not suppressed. Now, this is all relative, obviously, but compared to other parts of the world, new knowledge is encouraged, not suppressed. New knowledge becomes a policy. Rulers often help to fund, test, copy, and scale practical improvements, in order so they can get more security, more finances. They can defend themselves better.
- 7.64** But in other empires, China, Russia, Mughal India, Ottoman Empire, Safavid Iran, there is a reestablishment of order and rigidity. Courts, guilds, religious authorities gain protection from change, and military administrative power is used more often to freeze the existing social order, rather than to encourage disruptive innovation. Why? Because they fear innovation, disruptive innovation, at the center. They fear disruptive innovation domestically.
- 7.65** Can I prove what I just told you about the importance of competition and innovation? I think I can. I have three quotations for you. The first one comes from Frederick the Great and his book, "The Memoirs of the House of Brandenburg." In the first quote, Frederick the Great, the King of Prussia, reflects on Louis XIV, the King of France's revocation of the Edict of Nantes. As I explained to you in the last lecture, the Edict of Nantes was one where the French King, Henry IV, granted Huguenots much more religious toleration. His grandson, Louis XIV, who was more of a Catholic, revokes the Edict of Nantes.
- 7.66** This is what Frederick the Great says about that decision. Frederick writes, "The revocation of the Edict of Nantes, by which he," Louis XIV, the grandson of Henry IV, "by which he, Louis XIV, drove out 500,000 of his subjects, was one of those horrid political blunders. It ruined his commerce and depopulated the finest provinces in the kingdom (France). He would have done much better to employ those brave and laborious citizens against his enemies." So, here we can go back to the epistemics that Francis Bacon has talked about.

- 7.67 Frederick the Great is making an observation. From that, he's going to derive a theory. Observation, theory. And the theory of governance, Frederick the Great expounds upon in his essay on the forms of government and the duty of sovereigns. Frederick writes, "Would not one have to be demented to suppose that men said to one of their number, 'We are raising you above us because we like being slaves, and so we are giving you the power to direct our thoughts as you like'? On the contrary, what they said was, 'We need you to maintain laws which we wish to obey, to govern us wisely, to defend us. For the rest, we require that you only respect our liberty.'" So Frederick observes what happens when you are intolerant. From that, he derives a theory, saying that toleration and good government is the way to go.
- 7.68 Final quote comes from the biographer of Frederick the Great. "Prussian consulates became, in effect, recruiting offices, publicizing Frederick's enticing manifestos, advancing money for the journey and organizing itineraries. The prospect of land, settlement grants, free seed, livestock, and equipment, personal freedom, and exemption from military service was sufficiently alluring for about a quarter of a million immigrants who entered Prussia during the course of Frederick's reign."
- 7.69 Observation, theory, consequence, or practical outcomes. What could be more enlightened? So what caused human progress? This is a personal perspective, but based on my research, based on everything I told you, my personal view is that human progress is the result of gradual liberalization, small-L, nothing to do with American liberalism, political liberalism. Human progress is the result of gradual liberalization of the human body and mind from institutional, economic, technological, scientific, epistemic, and cultural constraints within a competitive international system.
- 7.70 So historically, liberty, which is so important to human progress, did not come about in one full sweep, but through the accumulation, or cumulative adoption, of liberties. Small steps. Instead of being burned at the stake for suggesting that the Earth is round, you are now praised. Instead of being banned as a woman from standing for office, you are now allowed to do so. Instead of being persecuted for creating a new innovation, you are now being praised. So historically, liberty did not come in one full sweep. It wasn't that one day Europe became free, but through cumulative adoption of liberties, over 1000 years, maybe even longer, small steps of freeing the human mind and body.
- 7.71 Small steps to a better world were taken, but steps that are positively reinforced. Maybe a better agricultural technology would allow you to produce more food, and more food will then lead to higher population. And then higher population will produce more ideas, and more ideas will lead to more innovations, which will then lead to higher productivity of agriculture. More people, more ideas, more innovations. "Small steps to a better world were taken, but steps that were positively reinforced, leading to the dramatic improvement in human well-being, once a critical threshold was met."
- 7.72 So, to use the words of Peter Boettke and Rosolino Candela from their book, "The Historical Path to Liberty and Human Progress," you've got accumulation of liberties,

slow accumulation of small liberties until you have institutionalization of liberty as a concept. Liberty under the law, which then leads to increasing returns, higher productivity, more people, more ideas, and so on, which then leads to human progress. That is essentially how I personally see human progress happening, where it came from—tiny little steps toward freedom, until one day you can say that you live in a free society, which then produces better and better outcomes, until you can say that human progress has taken place.

- 7.73** Finally, let us ask a question. Today, are we living through something special or just another efflorescence? This idea of modernity, the last 250 or 300 years, is the permanent state? Have we discovered some secret sauce that will allow us to progress indefinitely? Or are we just another efflorescence which is waiting to be snuffed out?
- 7.74** Well, in some ways, we are special. Incomes per capita in Western Europe over the last 200 years have risen 25 times, 25-fold. In the United States, 30-fold. During the past efflorescences—ancient Greece, Pax Romana— maybe you saw a GDP per capita growth of 1.5 or 2 or 3 times. During ancient Greece or Pax Romana, maybe you saw GDP growth per capita that was twofold or threefold. But in the Western world, over the last 200 years, we have seen GDP per capita rise by 25-fold or even 30-fold. So that's what makes it special.
- 7.75** But growth rates have also been slowing. Let's look at cumulative knowledge. Science today is organized in journals and peer-reviewed papers. There's a lot of funding. There's a lot of funding for science. We know that, in order to progress, each generation needs to build on top of the last in terms of new knowledge. So science is organized, it is well-funded. But what about the quality of science? How many papers fail to be replicated, because academics may have invented their data, or maybe they were not careful?
- 7.76** What if wokeism prohibits us from exploring certain ideas, and certain parts of the scientific endeavor get closed off? "You cannot research that because it is politically inconvenient." Maybe human progress also depends on new sources of energy—more fission reactors and fusion reactors. But are we going to be able to continue to build them? What if our extreme environmentalist friends say no? "Human progress may need a lot of energy, but we don't want you to build it because it is dangerous."
- 7.77** What about demographic transition? Very few children are being born into the world today. How are we going to sustain human progress if human progress depends on new ideas, new inventions, and new innovations, but we don't have more people to produce those new ideas, inventions, innovations? What if we don't have enough babies?
- 7.78** What about our institutions, the rule of law, security of private property? Can we be so sure that the rule of law is not deteriorating? Looking at the United States today, are we sure that the rule of law is not deteriorating?

- 7.79** What about innovation? We have a lot of money going to R&D for innovations. We have a lot of universities, a lot of researchers working to produce more innovations. But are we maybe too reliant on the United States? What if something goes wrong in the United States? Is there the sort of microcosm that we have in Silicon Valley which can be replicated in other parts of the world? Are we too reliant on the US?
- 7.80** Global diffusion and adoption of technologies depends on free markets. It depends on international trade. But how are we going to diffuse new knowledge and new technologies if trade slows because of protectionism?
- 7.81** Human progress relies on measurement and feedback, statistics, standards that we can all understand, that we all accept as true. But what if statistics are corrupted? What if we cannot rely on government agencies to provide statistics that we can believe in?
- 7.82** Human progress very often depends on shock absorbers, say, the ability to overcome a pandemic by being able to go into debt, raise money in the international markets. But our debt levels are already so high. Did we come to the end of the road? If there's a new pandemic, if there's a war, who is going to be funding our endeavors? Maybe we have reached the bottom of the barrel, and if Western nations cannot borrow anymore, will that lead to some public disruption, public dissent, revolutions, because the shock absorbers are at the end of their rope?
- 7.83** Human progress depends on innovations and on seeing innovators as laudable people, people whom we should cherish. Is that still true? Very often it seems that cultural figures like Beyoncé and George Clooney are much more greatly regarded than people who are coming up with technological and scientific breakthroughs.
- 7.84** I don't have an answer to that question. I don't have a crystal ball. Sometimes I feel like modernity truly is very special. At other times, I feel like so many of the preconditions for modernity to continue are being slowly eroded. Perhaps you have a better sense of whether we are just the latest efflorescence or whether we are something more permanent. I guess the future will show. Thank you very much.

Lecture 8: Critiques & Future of Progress

At a glance — Critiques and the future of progress, closing a personal journey.

- 8.1** This course on Human Progress was as much of a personal journey for me as it has been for you, in a sense that preparing for this course forced me to try to cover a truly vast area of research, history, philosophy, economics, technology. I tried to present you with an objective picture of what human progress is, where it came from, how it came about. But of course, ultimately, at the end of the day, it can still only be a personal perspective.
- 8.2** In this lecture, we will cover the critiques of human progress. We will ask ourselves, are there limits to human progress? Then I'll talk a little bit about why I think there may not be any hard limits to human progress because we are problem-solving animals. I call this approach solutionism. We will talk about the promise of a new age. We had the Agricultural Age, we had the Industrial Age, now we may be entering the age of intelligence. I will end with remarks about what out there still needs to be discovered. What else, what other problems need to be conquered? I will talk about protopia or the mastery over nature.
- 8.3** This last lecture is built in part based on the work of Steven Pinker and his book, "Enlightenment Now", as well as Jason Crawford's "Techno-Humanist Manifesto." So let's talk about the critiques of human progress. One of the most obvious critiques is the religious criticism of secular progress. The argument goes something like this. A religious person might argue, truth and morality come from revelation, scripture, church authority, not human reason.
- 8.4** Trying to build a secular scientific worldview, such as the one presented in previous lectures by the proponents of the Enlightenment, risks moral collapse and nihilism. If you want to have a proper understanding of morality, it needs to be based on scripture. It cannot be derived from reason and reciprocity, as the proponents of the Enlightenment thought. Modern societies, some religious thinkers think, are decadent because they turned away from God.
- 8.5** But there are counterarguments to that criticism. Secular institutions and science have sharply reduced disease, poverty, violence, and cruelty compared to many other, very often religiously dominated eras. Who could say, in all seriousness, that Europe during the Thirty Years' War was a better place to live in than today? I don't think many people could make that argument. And even religious people today benefit from vaccines, sanitation, markets, and fundamental rights, which have arrived to us or which have arisen from secular reasoning.
- 8.6** Human-centered morality, one that is not derived from scripture, can still value compassion. It can value pursuit of happiness, life satisfaction, and the pursuit of

meaning. And as I said in the first lecture, I think that pursuit of meaning is always easier on a full rather than empty stomach.

- 8.7** Another criticism of human progress comes from collectivists, who see individuals as expendable. Collectivists put emphasis on class, clan, tribe, sometimes church, sometimes race; or they declare that nation is really what matters. Individuals are just expendable cells in a larger body. Their lives may be sacrificed for the greater whole. Universal human rights are rights for the weak. It is strength and loyalty and self-sacrifice that collectivists believe ought to be praised; and repression can be justified to defend the group against decadent individualists.
- 8.8** Liberals with a small-L—again, I want to distinguish between classical liberalism of the 19th century from liberalism as it is understood in the United States today. But there is a counterargument to this collectivist tendency. Treating people as expendable has repeatedly justified slavery, totalitarianism, mass repression, gulags, the Holocaust, genocides. Modern gains in wealth, health, and fundamental rights came from treating individuals as bearers of rights, not as a fuel for some international ethnic conflict.
- 8.9** Societies that constrain the power of the government, of the ruler, societies that protect individual rights tend to be richer, safer, more peaceful. And universal dignity, international rules, cooperation, trade have helped to reduce large-scale conflicts. Even though the world is still a violent place, it is less violent than it used to be. Homicides used to be much higher. In 15th- or 16th-century Europe, major powers were at war with one another 100% of the time. So we may bemoan the imperfections of modernity, but we partake in the benefits that it had brought about.
- 8.10** There is the romantic or heroic criticism of human progress. Reason and science, these critics believe, are cold and dehumanizing. True greatness lies in passion, in will, in authentic emotion, in heroic struggle, not in comfort and security. Civilized life is seen as soft. Violence and war are celebrated as sources of meaning and renewal. "Dulce et decorum est pro patria mori." It is sweet and beautiful to die for one's country. That is the romantic, the heroic approach and criticism of human progress.
- 8.11** But of course, it was precisely this kind of romanticism that has fed nationalism and fascism in the 19th and 20th centuries. The idea that suffering and destruction are noble has produced true horrors, not spiritual uplift. And ordinary, peaceful improvements in medicine, in education, in the enlargement of fundamental rights to include women, to include ethnic minorities, gay people, have done more for human dignity than any romantic hero worship.
- 8.12** Another criticism of human progress comes from the greens, very often from the green anti-human environmentalists. Humanity, they believe, is a plague upon the planet. Technology, economic growth, industry are inherently destructive. The highest moral value to these people is an untouched ecosystem, even if it means lower human flourishing.

- 8.13** But as I will argue in a moment, people are not just destroyers of the world. We are builders. We are protectors. We solve problems. Remember that slide of the declining oil spills? Remember that slide of fewer and fewer whales being killed? Humans did that. We solve problems. Many pollution indicators are already improving in rich countries, especially as technology advances. Using fracking to get at gas to replace coal may seem insufficient to the greens because gas, after all, is still a fossil fuel. But burning of gas produces half as much CO₂ as burning of coal. So it's a step in the right direction. And nuclear fission reactors don't produce any CO₂ at all. That's how humans solve problems.
- 8.14** Another criticism of human progress comes from declinists, people who believe that everything is getting worse. The sorts of people we encountered in Lecture 1, Hesiod, in "Works and Days," the age of gold has given way to the age of silver, the age of bronze, and eventually the age of iron. Continuous decline. These same people today say Western civilization is in decline, is on the verge of collapse. Modernity, they say, brings with it alienation, consumerism, meaninglessness, environmental disaster. Golden Ages, Garden of Eden, lie in the past. Our era is corrupt and doomed.
- 8.15** But the data, two hours' worth of data that I have given you, show the exact opposite. Data on wealth, health, longevity, poverty, education, fundamental rights, all show long-term improvements, not collapse. We are not living amidst ruins of a previously wonderful, golden, perfect civilization. No, we have started from very rude beginnings—people who were cruel to one another, who treated each other horribly. We experienced short lives, mothers and children dying in childbirth. And it was out of that low beginning that we reached the heights of modernity, living in the West today.
- 8.16** It is not corrupt. It is not on the verge of collapse. Statistics clearly show that life in the West and also in the rest continues to improve. So this nostalgia for the past is highly selective. Past societies were definitely worse off. They had more disease, more poverty, more brutality, more ignorance—things that today we would never tolerate. And recognizing problems does not deny remaining problems, but the past shows us that problems that still remain can be solved.
- 8.17** We have the anti-scientific and postmodern skepticism about truth, yet another criticism of modernity and of human progress. These postmodernists tell us that science is just another narrative, a discourse of power, not a route to objective truth. Postmodernists tell us that claims to universal truth or progress are simply masks for Western, male, or colonial domination. They present skepticism to reason and evidence as sophisticated and deep.
- 8.18** Yet if all the claims, as postmodernists say, are just power plays, then the postmodern critique is also just a power play. In other words, the postmodern critique refutes itself. Science, which they bemoan and criticize, has given us vaccines, clean water, electricity, countless life-saving technologies. They work across cultures, not just in the West. Western medicine is not Western medicine; if it works, it's global medicine. You can

criticize the misuse of science, like the sort of misuse that we suffered from during the COVID pandemic, without denying that evidence and logic can, in fact, reveal real truths.

- 8.19** Part of this declinist narrative is based on the idea that progress at some point must hit a hard limit. It must hit a wall. The old fear, which I will discuss in a moment, said that we were running out of resources— land, coal, oil, metals, whatever. But there's a new fear that we'll run out of ideas, that we've picked all the low-hanging fruit in terms of innovation, and future innovation must, by necessity, become more difficult. But as I will argue, both fears misunderstand how progress works, because they ignore who humans are. We are problem-solving animals. We are innovative animals. We create new resources by discovering new ideas, new inventions, and new innovations.
- 8.20** What were the old fears? Well, we had the overpopulation fear and overconsumption. Population, Thomas Malthus told us, grows exponentially like that. But resources, especially food, only linearly, and the gap between them, exponential growth and linear growth, must lead to starvation. In reality, fertility rates fell, and agricultural innovation has produced more food than ever before. In Lecture 2 and Lecture 3, you could see how much more innovative, how much more productive, agriculture has become.
- 8.21** So yes, the world has many more people in it, but those people are better fed than ever before. So the claim that more people must lead to exhaustion of resources and starvation is false. We have more people, therefore more ideas, more inventions, more innovations, and ultimately, more food and more resources. Jevons, in the 19th century, worried about running out of coal at the time when Britain ran on coal because it was in the midst of the Industrial Revolution. But we have discovered new fuels, oil, natural gas. We have discovered new fields of oil and natural gas. We have discovered new coal deposits. We have new technologies, such as fission reactors. And so we did not run out of coal.
- 8.22** People worried about peak oil, that it would eventually be exhausted. In reality, once again, we have discovered new deposits of oil. We have discovered new technologies, such as fracking, which can get us oil that we previously couldn't get at. We have discovered new technologies which enable us to turn coal into oil. Germans did it during the Second World War. South Africa did it during the oil embargo in the 1970s and the 1980s.
- 8.23** People worried that we would run out of metal, such as copper, for example. But in reality, the prices of copper, adjusted for inflation, remained flat. Easy deposits were used up, that is true, but people became more skilled and more efficient. Our metal-using civilization has expanded. It did not shrink. There are at least nine different ways in which you can increase human resource base. First, you can increase it by increasing supply. So, if the price of oil increases, you have an incentive to go and search for new oil deposits.

- 8.24** You can also have technological breakthroughs. So maybe oil, you can no longer get it in this particular well with traditional drilling methods. But now we have fracking, which allows us to get at that oil. So you can increase deposits by looking for more oil fields, or you can use the old oil fields but get to oil and gas that you previously couldn't get to because of technological change, such as fracking.
- 8.25** We can increase efficiency in terms of using resources. A can of Coke used to weigh three ounces when it first came on the market in the 1950s, three ounces of aluminum or sometimes tin. Today, it weighs half an ounce. So you are increasing your relative efficiency. You can increase absolute efficiency as well. Some researchers show that, out of the top 100 commodities used by the American economy, 41 are post-peak. Let me rephrase that a little bit.
- 8.26** Normally, what you would expect is that, as your economy grows, the use of resources increases in tandem. But in 41 out of 100 commodities used by the American economy, we are post-peak. We are using absolutely less water, less copper, less iron. The total tonnage of copper and iron, in other words, is falling even though the economy continues to grow.
- 8.27** We can increase the value from the same commodity. Sand, when it was first melted, was used for glass beads to beautify people. Later, we discovered that we could use sand in order to create jars and cups. Later, we turned sand into windows. Today, we use sand in fiber-optic cables and in microchips.
- 8.28** We can turn things which are useless into things which are useful. A hundred years ago, when people were drilling for oil, natural gas would come up, and we didn't know what to do with it. In fact, it was very dangerous because if you lit a cigarette on an oil rig, you would kill yourself as well as everybody who was around you. And so we started flaring it. We started burning it off.
- 8.29** And today, if you go to developing countries, you still see that every oil well has a little flame on top of it. That's the gas being burned off. But we no longer do that in the United States. In the United States, we capture that natural gas and we use it to heat our homes and we use it in industrial production and so forth. So natural gas used to be useless, even dangerous. Now it's super useful.
- 8.30** We can substitute things for one another. We cut down our forests because we need it for military equipment, for furnishing, for heating. For thousands of years, we would cut down our forests in order to get at the wood until the time when forests were pretty much gone from much of the United States, as well as England. Today, forests are bouncing back because, instead of wood, we may be using coal or natural gas or nuclear fission.
- 8.31** We no longer kill whales in order to scoop up their heads, the fat in their heads called spermaceti, to make into candles. Now we just switch on the light. We don't actually care where the light comes from. It can come from a candle, from a killed whale, or it

can come from an LED. We care about the light. And so it happens that we have solved that problem. We no longer are killing whales. We are getting it maybe from natural gas, or we are getting it from nuclear fission.

8.32 During the second agricultural revolution, when our yields started to go up, part of the reason for agricultural yields going up was because we chanced upon guano, which is basically bird excrement that has accumulated in the islands of South America for millions of years, rich in phosphorus. We would mine it, and then we would sprinkle it on the fields of the Western world. We no longer do that. Today, half of humanity is fed by synthetic fertilizer, which is made from natural gas. It's called ammonia. So we have substituted one thing for another.

8.33 We can dematerialize. Your iPhone is not just a phone. It is also a camera. It is a compass. It is an alarm clock. It is a TV, very often. Imagine all the atoms, all those things that you would have to have, that now you have just in a small device, held in your pocket. Another typical example of dematerialization is that, if 15 years ago, if you walked into an American hotel room and you wanted to get internet, you would have to plug your computer into the wall with a blue copper cable. Those copper cables are all gone. That copper is being used somewhere else. Now you get your Wi-Fi through the air. We have dematerialized access to the internet.

8.34 We can reuse our resources. My favorite example is when Americans were building the atomic bomb during the Manhattan Project. The only way that they could get enriched uranium was through magnets, which required silver. So 14,000 tons of silver was lent by the American government to the Manhattan Project to create these magnets, to be able to enrich uranium. And after the Manhattan Project was closed down, all that silver was returned back to the American government, so you can reuse things.

8.35 Transmutation, a very important way of enlarging our resource base. For hundreds of years, crazy scientists would travel around Europe claiming to have discovered a way of turning base metals into expensive metals, turning lead, for example, into gold. They were all charlatans; there was no sorcerer's stone. But we already know how to turn lead into gold. We have done so before. Through neutron bombardment, 40 years ago, we were able to convert atoms of lead into atoms of gold. It's far too expensive. We cannot do it right now, with the technology we have. But there is no laws of physics preventing us from doing that.

8.36 What about minerals? The most expensive, the most desirable mineral out there is the diamond. The prices of natural diamonds are crashing to the ground. Why? Because our technology became so sophisticated and so good that now we can use synthetic diamonds to do the job of natural ones. We take a tiny bit of diamond, we heat methane until it becomes carbon vapor, and then carbon vapor deposits on top of that little bit of diamond, on top of one another, layer by layer, until you get a synthetic diamond, which is physically and chemically indistinguishable from a natural diamond.

Something like one-third of all the engagement rings in the United States are now synthetic diamonds, and the price is dropping.

- 8.37** Finally, just about every element that we have on Earth was created somewhere else. It wasn't created on Earth. It was created within stars under tremendous pressures and temperatures, and when those stars exploded, they seeded the universe with the elements that we can find on Earth. What is necessary in order to do that? Well, you need two things. One, you need hydrogen, because the Big Bang only created hydrogen and little bits of helium. Hydrogen is the most plentiful element in the universe, and our oceans are full of it. And you need fusion because, after all, stars are just fusion reactors, and we already know how to do fusion. So, theoretically speaking, between fusion reaction and hydrogen, you should be able to turn any element into anything else.
- 8.38** Should we ever need more resources, there's always space mining. Three different technologies are coming together to allow us to space mine. We have AI, we have solar power, and we have robots. Solar-powered robots syncing with AI. We could be sending them on Elon Musk's Starship to the asteroid belt to mine for water and whatever else is out there.
- 8.39** So no, we are not going to run out of resources. In fact, strictly speaking, there are no natural resources. The phrase "natural resource" hides the central role of knowledge. Raw materials on their own are useless without know-how, without knowing what to do with it. Iron ore is just a rock until you know how to smelt it and forge it into tools. Sand is just dirt until you know how to make it into glass or fiberglass. Even water needs infrastructure: pipes, pumps, treatment facilities.
- 8.40** David Deutsch, the Oxford physicist, likes to say that early humans sometimes died of cold, even though they were surrounded by wood, oxygen, and dry tinder—everything you need in order to start a fire. Why? Because they didn't know how to make fire. So, on the surface, it was the weather that killed them, but the deeper cause of why they died was ignorance, not the lack of physical stuff.
- 8.41** So, what we call a resource is really just a solution to a problem made possible by new ideas and new inventions and innovations. It is humans who create resources by discovering how to use these bits of matter and energy in new ways. Unfortunately, we keep on underestimating technological solutions. Our ancestors have evolved mainly for social fixes, not technological gadgets. When crises arose—hunger, conflict, bad weather—these were handled by sharing food, by enforcing norms of behavior, by punishing cheaters, or by cutting consumption, not by inventing new tools.
- 8.42** And the groups that cooperated and enforced these rules of survival, survived. So we have evolved to trust rules and restraints, not technological solutions. That makes us instinctively prefer social solutions today. When something goes wrong, the first thing we reach for are laws, bans, rationing, "switch off that light, use less water," and moral

pressure. Building of new reactors, carbon capture plans, or better fertilizers feels abstract and risky compared to visible social sacrifices, like using less energy or taxing emissions.

- 8.43 Moral framing of the problem then reinforces that response. Fossil fuel use, for example, is treated as a sin, and climate action is treated as atonement. Solving problems with technology feels like cheating instead of progress, but history shows us the opposite. Crises about wood, food, water, and many resources were solved by new energy sources and better technologies, not by permanent austerity. Problems are real, yes, but the rational response is more and better innovation, abundant clean energy, cleaner industry, new materials, not permanent limits on prosperity.
- 8.44 There is a new fear that we are running out of ideas. Some economists argue that the biggest, most transformative inventions such as electricity, cars, antibiotics, were exceptional. But future progress must be slower. One paper found that GDP growth per person has grown steadily, but the research effort, scientists and finances needed for those scientists, has surged. So yes, we grow at a steady rate, but the amount of effort that we have to put into that steady rate of progress has increased.
- 8.45 The argument says that we have caught the big fish in the pond; the big fish of innovation have been caught. And now we must go deeper into the pond of ideas and inventions and innovations, and we are catching smaller fish. But the counterargument is that every innovation happens only once. The question is: can new ideas keep solving new problems? Maybe the idea pond expands. Maybe there isn't just one pond. Maybe there are many other ponds. Maybe there is the AI pond. Maybe there is the biotech pond. Maybe there is the materials pond. Maybe there is an energy pond.
- 8.46 Also, our discovery tools for new innovations are improving. Computing, simulations, lab automation, AI as a research assistant, increase the productivity of research and development. With better institutions and better incentives, societies can scale both the quality and the quantity of innovation. So the real question is: can we keep generating enough valuable ideas across enough domains to keep living standards rising?
- 8.47 In one of the previous lectures, we talked about Paul Romer, the Nobel Prize-winning economist, who talked about ideas being non-rival. If I share an idea with you, I still have it. You now have it, but I still have it. We can both use it at the same time. So, instead of the Malthusian idea—more people means there is less to go around—we can have more people, more brains, more ideas, more innovations, and higher standards of living. And that's because ideas have a special property. The more ideas you consume, the more you have.
- 8.48 That's not true with a Snickers bar. Once you have eaten it, it's gone. But with ideas, the more ideas you consume, the more you have. So the real problem may not be overpopulation, as Malthus worried, but underpopulation. Low fertility rates and aging

population may be a bigger problem because, ultimately, fewer people means fewer researchers, fewer entrepreneurs, and that can slow the pace of new ideas.

8.49 We don't yet know what we can do about the falling TFRs, or total fertility rates. Maybe culture needs to change. Maybe there is a technological solution. But the fundamental key to future progress is that we must stop seeing people as primarily mouths to feed. We must start seeing people as a pair of hands and a brain capable of producing more wealth and more knowledge for the rest of us. We must stop seeing people as a problem. We must start seeing people as problem-solvers.

8.50 Now, saying that we will find new solutions that we cannot yet see may sound like blind optimism, but history suggests otherwise. Time and time again, serious predictions warned that trends could not continue and disaster was inevitable. It was around the corner. Time and time again, humans responded with new ideas and new innovations. In terms of materials and machines, we moved from stone tools to steam engines and microchips.

8.51 In terms of agriculture and chemicals, we have moved to a world filled with fertilizer and pesticides, improved seeds and food storage. In terms of health, we now have better sanitation, clean water, vaccines, and antibiotics, none of which were available to our ancestors. In terms of infrastructure and safety, we have earthquake-resistant buildings, bridges, weather forecasting. In terms of organization and communication, we have global supply chains, GPS. We used to have telegraph, now we have the internet.

8.52 There are a lot of problems that we have overcome in the past. We are not just lucky in one area. We have a general capacity to understand systems, problems, and to improve upon them. We are a problem-solving animal. But who are these problem-solving animals? Who in human society innovates? Homo sapiens, as I've noted before, survived via social cooperation. Sociality, or sociality, depending on how you want to pronounce it, is the innate human tendency to live, interact, and form relationships within groups. We share food, we divide labor, we coordinate defense from animals that want to eat us or attack us. So survival depends on social cooperation.

8.53 But flourishing depends on innovation. Survival depends on cooperation. Flourishing depends on innovation, on moving from stone tools to steam power to fusion reactors. Most people don't innovate. They solve problems socially. If I want to go to the beach, I will ask you to put an SPF on my back. That's a social solution to a problem. So innovators prefer to work things out by themselves, and they appear to be a very small minority. Why would that be the case? Maybe there is human evolution at play here. What do I mean by that?

8.54 As I've noted, for survival, humans need social cooperation, and social cooperation requires conformity. In order to survive, humans need to cooperate. Maybe there's a bunch of lions running toward us. You need to be able to unite and ward off the attack

by a group of lions with coordinated bombardment with stones, for example. So, survival requires social cooperation, and social cooperation requires conformity. You cannot have one member of the group suddenly saying, "You know what? I'm just going to wait this out. I'm not going to throw that stone. Or maybe I'm just going to recite poetry." No, you need everybody to cooperate. For survival, you need cooperation, and for cooperation, you need conformity.

- 8.55** But human flourishing requires innovation, and innovation requires nonconformity. An innovator is the person who says, "Society has been solving problems in a certain way. There's a status quo way of doing things, and I disagree; I fundamentally disagree. I can do it a better way. I'm going to upset the status quo." But to be able to upset the status quo, to be able to have that kind of self-confidence, that kind of disregard for conformity, you have to be a nonconformist.
- 8.56** So it happens that these individuals who are innovators and nonconformists are also less social. People who are focused more on systems and things, rather than people who are more interested in their fellow human beings. There's some evidence that innovators in the STEM fields, especially, are people who have somewhat elevated autistic traits. We have people here who are nonconformists, who are less social, somewhat more autistic. It will not surprise you that a disproportionate number of them are men who are interested in systems, not people. These nonconformist antisocial individuals can be very difficult.
- 8.57** The question is, whether in our society today, we are still willing to be tolerant of nonconformity, of antisocial and slightly autistic individuals. Whether, in fact, the society is becoming more intolerant of behaviors and personality and psychological characteristics that make the innovators who they are. One thing we do know for sure, it seems to me, and that is that we cannot identify innovators ahead of time. We don't know who these individuals are and where they will emerge from, which is why it is so difficult to stimulate innovation, and why no country out there has a blueprint for doing so.
- 8.58** So what I advocate for is solutionism. Solutionism is an attitude toward problems in a world of progress. It starts by taking problems seriously, not minimizing or denying them. It insists that we should understand problems scientifically, measure them, categorize them, and test possible causes of those problems. It pushes us to prevent, mitigate, and solve problems using all the tools at our disposal, technology and engineering, law, regulation and institutions, culture, norms, and education. That's what I mean by solutionism, by problem-solving animals.
- 8.59** Solutionism is not a naive faith that technology magically fixes everything by itself. There will always be problems. Solutionism does not deny risks or harms. It does not say that every problem has a purely technical fix. Sometimes we need institutional or cultural change to fix problems ahead of us. Solutionism also rejects two unhelpful extremes—blind optimism. Blind optimists who say, "Everything is fine, stop worrying."

But also it rejects blind pessimism, which says that "nothing can be done, so stop trying." Both extremes lead to doing nothing, which is exactly the wrong move.

- 8.60** Solutionism is the middle stance of clear-eyed concern plus active effort. What could the future of solutionism look like? Well, one question which I'm often asked when giving talks is: aren't we wealthy enough? Aren't we already rich enough? I would suggest to you that comfort is relative. Comfort and what it means to have enough are moving targets, and they change with technology and with expectations.
- 8.61** People in the 1800s often felt that they were doing OK, but they didn't have electricity, plumbing, refrigeration, or modern medicine. Yet not having those conditions today would seem like extreme hardship. If growth continues, future generations may view today's living standards as modest by their benchmarks. What counts as decent life tracks what is widely available, not what is biologically necessary. Modern public goals focus on preventing disasters— climate change, for example, pandemics, or poverty. They do not focus on expanding that which is possible. And big transformation often arrives through compounding small improvements, whereby yesterday's science fiction becomes today's baseline.
- 8.62** So human history has moved through three broad economic eras— the Stone Age, the Agricultural Age, and the Industrial Age. Each age raised living standards. And maybe we are on the cusp of the fourth age, the Intelligence Age. Each past age was built on a fundamental technology—tools, farming engines. The leading candidate for a new age is artificial intelligence, possibly combined with advanced robotics. Engines multiplied human muscle power. AI could multiply human mental power, help us think, plan, design, and solve problems.
- 8.63** If this new age arrives, and our economy grows faster than today, the world economy could double, not every several decades, but perhaps every few years. Imagine how much more wealthy we would be, not just here in the West, but around the world. AI could turn intelligence into a utility like electricity—cheap, flexible, cognitive labor. Human teams need hiring, training, motivation. People vary in ability, get tired, quit, and make unpredictable mistakes. AI systems, once trained, could be copied endlessly, work 24/7, scale up or down, fix a bug, and every copy improves.
- 8.64** The biggest impact of AI could be in research and development. AI can read and summarize huge literatures, suggest hypotheses, run simulations, and control robotic labs. In economic terms, agriculture was limited by land. The Industrial Age was limited by labor and capital. But in an Intelligence Age, AI plus robotics could replace most mental and physical labor, so output would depend mainly on technology and capital, rather than human workers. And that means that the future of economic growth would no longer be tightly bound to population size. As long as we keep on improving technology, production could grow very rapidly, perhaps all the way up to the limits set by physics.

- 8.65** What about the future of work in this Intelligence Age? Well, AI will probably not make humans useless. It will change what we do—less busywork, less going down into the mine shafts. Instead, choosing goals and making sure that AI systems deliver on those goals—management, if you will. There's a worry that AI will take all jobs, make us unemployed, and therefore, none of us will have any income, and we will all be dirt poor. But that story doesn't really work. Even if AI makes the economy much more productive, humans will still need to purchase the goods produced by AI or access them in some way. Otherwise, what's the point of producing if nobody can purchase what the AI is producing?
- 8.66** We will need to find new ways of spreading the purchasing power, of spreading wealth. Maybe we'll have new kinds of jobs. Maybe the prices will be incredibly low because all the production will be mechanized. Maybe we'll come up with different broader schemes of ownership of AI and robotic companies, and different schemes for profit-sharing, or maybe some mix of all of those things. That is how technology usually works. Some jobs vanish, new ones appear, and the work shifts from one place to another.
- 8.67** Another reason why I don't think that, in the world of AIs, there's going to be massive unemployment is because human needs are infinite. If human needs are infinite, then by definition, there will always have to be jobs which are performed not just by AIs and robots, but human beings. Let's look at some of the ways in which the AI revolution, the intelligence era that I talked about, could help us solve many of the problems that remain.
- 8.68** Let's start with human mastery over biology— curing diseases, advanced early cancer detection through blood tests and imaging, mRNA vaccines, DNA editing, improved treatment of heart disease with advanced surgery, valve replacements, GLP-1 agonists, deploying gene editing to tackle muscular dystrophy, cystic fibrosis, sickle cell anemia and other genetic diseases, develop artificial organs grown from stem cells, and therapies that prevent neurodegenerative diseases, understanding metabolic health well enough to end obesity and design optimal nutrition formulas. All of that we could do in the Intelligence Age.
- 8.69** We can attempt to end pandemics, use UVC light to kill airborne pathogens and sanitize indoor air as reliably as chlorine sanitizes water. Monitor wastewater and airports for early warnings of outbreaks of new pandemics. Apply CRISPR gene drives to eliminate disease-carrying mosquitoes. Augment immunity with wearable devices that synthesize mRNA vaccines and coordinate instant global distribution. Mastery over aging and human enhancement.
- 8.70** What about curing aging? Treating aging as a solvable biological process, rather than an inevitable fate of humanity? Target muscle loss, bone weakness, cancer risks, sensory decline, and energy loss with new drugs and gene therapies. Use transcription factor

cocktails and other interventions to reset or refresh human tissue. Aim to grant everyone as many healthy and vigorous years as they choose.

- 8.71** What about human enhancement? We could try to raise physical abilities such as strength, endurance, flexibility, energy, and even attractiveness. We could improve cognitive traits, including intelligence, creativity, memory, focus, willpower, and resilience. We could create safe ways for people to thrive on very short sleep, or potentially eliminate the need for sleep altogether.
- 8.72** What about intelligence-driven human mastery over agriculture and food production? Optimize crops for efficiency, nutrition, taste, hardiness, and disease resistance. Scale lab-grown meat that uses resources more efficiently than traditional livestock. Develop chemical food synthesis to bypass the need to grow entire organisms. Invent entirely new types of food with flavors and textures that are not found in nature. Engineer super-strong fibers such as spider-silk-like materials for clothing and industry.
- 8.73** What about using AI in order to create human mastery over information? Shrink the cost of building software from years and millions of dollars to months and tens of thousands of dollars. Use virtual AI workers for engineering, product management, design, marketing, and sales. Make it feasible to build custom applications for small communities such as reading groups or local clubs. Deliver first-rate legal, medical, accounting, and therapeutic services to everyone.
- 8.74** Personalize education so that every student gets adaptive tutoring, flexible pacing, and motivation-aligned curricula. Let individual creators produce feature-length films and other content at a fraction of today's cost. Use AI to match people for relationships, jobs, business deals, and collaborations, while removing language barriers with universal translation. Complete major open problems in mathematics using formal proofs that can be checked independently of AI intuition. Integrate information into daily life with advanced robotics, wearable devices, and brain-computer interfaces.
- 8.75** What about using AI in order to achieve human mastery of space? Faster travel on Earth and in orbit. Revive supersonic flights, so that routes like New York to London fall below 4 hours and trans-Pacific trips shrink accordingly. Develop practical flying cars that combine the flexibility of personal vehicles with the speed of air travel. Roll out autonomous vehicles and trucks that cut freight costs, reduce accidents, and raise convenience and privacy.
- 8.76** Drive down launch costs into space from thousands of dollars per kilogram to tens of dollars per kilogram. Experiment with alternatives such as high-velocity launch systems that throw cargo into orbit. Expand the use of space beyond tourism and science to manufacturing, resource extraction, and planetary defense. Recognize that exploration often begins for adventure before the economics are clear.
- 8.77** What about AI-powered human mastery over energy? We should treat energy as a positive enabler of progress, rather than a necessary evil. We should use energy to

desalinate and pump water, making drought largely a choice rather than fate. We can extract key materials such as lithium, nickel, copper, and uranium from seawater. We can move towards solar, nuclear fission and fusion, and geothermal sources that can power a super abundant future for humanity.

- 8.78** We can also use AI in order to develop human mastery over matter. Continue the historical march of humanity toward finer tolerances until machines are built with atomic precision. Design molecular-scale gears, bearings, pumps, and engines assembled into cell-sized robots. Create objects that change shape, move, and display visual effects through dense embedded mechanisms and optics. Use household synthesizers, like those in Star Trek, to compile matter into goods on demand and recycle them back into raw atoms after use.
- 8.79** We can rebuild large portions of infrastructure extremely quickly and cheaply, once manufacturing becomes nearly instantaneous. We can apply medical nanotechnology to build new tissues, destroy cancer, clear toxins, and provide organs that outperform natural ones. We can use AI in order to develop human mastery of the environment. Develop systems that control not just carbon dioxide, but the full range of atmospheric variables.
- 8.80** Engineer weather to optimize agriculture and to reduce destructive storms. Undertake geographic transformations far beyond historical projects such as canals and dams. Remove toxins and pollutants from air, water, and soil, while restoring and creating new ecosystems. Prevent extinctions, rescue vulnerable species, and de-extinct, which is to say, reintroduce long-lost animals. If we are worried about rising sea levels, we could pump water into depressions such as the Qattara Depression in North Africa. We can terraform other worlds to support ecosystems and habitable environments across the solar system.
- 8.81** So what I'm advocating for is not a static utopia, but a protopia. Traditional utopias imagine a final, unchanging state where all problems are solved, but in reality, everything in the universe is in motion, and humans thrive on growth and challenge. So protopia frames success as being better today than yesterday. That essentially is the fundament of human progress. We are not claiming that we are living in a utopia where everything is working out for everybody, everywhere, at all times. We are saying that today is better than yesterday. And what we are hoping to do is to create a world where tomorrow is better than today.
- 8.82** So as you can see, the Intelligence Age, if it should ever arrive, will have a lot of work to do. To accomplish human mastery over nature, with all of its components, will be an extremely difficult task. But given the many ways in which humans have mastered previous problems, I think there is a reasonable chance that we will be able to accomplish even more in the future.

- 8.83** What sort of a future do I have in mind? What is the endpoint of human progress? I have no idea. What I do know is that it is not going to be a utopia. There is never going to be a point in time at which everything will be working out for everyone, everywhere, at all times. If utopia means a world without problems, I very much doubt we will ever reach that place, and perhaps we are not meant to. Because in the past, when humans attempted to build a utopia, be it during the French Revolution or in the Soviet Union, a lot of people died. And utopia, at the end of all that slaughter, was as far away as it ever was.
- 8.84** No, to me, human progress is a path of continued improvement. It means that I can confidently say that today is better than yesterday. I can work towards making tomorrow better than today. In order for human progress to continue, the drivers of human progress, the institutions, the economic settings, the technological and scientific and epistemic advancements, the moral and ethical precepts that underpin modernity, all of those must be preserved, because only if those drivers of human progress are preserved, can we be sure that modernity continues into the future, rather than becoming just another efflorescence that is snuffed out.
- 8.85** How do you think about existential risk in the context of innovation and human progress? So some innovations solve major problems while adding new total annihilation scenarios. Do you think, for example, that we're living with more or less existential risk today than we were 2,000 years ago?
- 8.86** I say that's a difficult question. In some ways, yes, in some ways, no. If there was a large space rock flying toward the Earth, 2,000 years ago, people didn't have any way to stop it. Today we do. But the same technologies that will enable us to prevent that from happening, from a space rock destroying the planet, can also destroy our species through, for example, nuclear conflict. So, every technology is a double-edged sword.
- 8.87** But technological stagnation, in my view, is really not an option because eventually there will come a point at which humanity will be facing problems that today we cannot even contemplate. In my mind, the worst possible thing that could happen would be a space rock or would be some sort of a pathogen. But no matter what comes our way in the future, I would prefer us to be the most technologically advanced civilization and the richest civilization possible, so that we have the resources in order to be able to tackle those existential challenges.
- 8.88** So, that would be my pitch for why I think that continued technological scientific research and economic growth are good, because of these unknown unknowns out there in the future. But fundamentally, in some ways, we are more at risk; existentially, in some ways, we are less so.
- 8.89** You had spoken about innovators going through— how do I want to say this? Innovators escaping their current situation through kings and princes and going through these different spaces to get to different regions. But now, your opinion on

passports, immigration—human progress would suggest that we've gone back versus forward, when it comes to people going to different countries. Who could say that there's innovators hiding in India or in the fields of Africa that have just not been recognized because the borders are closed now? And what part of history did we decide that that was the best thing to do?

8.90 Undoubtedly, there are innovators in Latin America, for example, or Africa, or parts of Asia, who are, for whatever reason, unable to participate in the global innovation race. Ideally, what we would want to do is to have an immigration system that encourages people who are high-skilled STEM immigrants to come and work in the United States, rather than working for our competitors. Hopefully, access to the Internet for example, and the ease with which people can communicate across borders, will enable truly gifted individuals to make themselves known, so that ultimately they can be hired in the United States and contribute to the tremendous technological prowess of the United States, whether it be in Silicon Valley or New York or other places. Thank you very much.